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Volumes of Nullhomotopies in Nilpotent Spaces

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by

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May 2025

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Kyle Dean Hansen

For Mom, Dad, Kristian, and Connor.

For Nana, and in loving memory of Grandpa Hansen, Granny Rose, and Grandpa Brown.

Soli Deo Gloria.

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Abstract

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The Shadowing Principle of Manin has proved a valuable tool for addressing questions of quantitative topology raised by Gromov in the late 1900s. The principle informally provides a way for bounded algebraic maps between differential graded algebras to be translated into nearby genuine maps between their geometric realizations. We extend this principle to finite towers of principal $K(G, n)$ fibrations, and in particular apply this construction to nilpotent spaces. As a specific application of the extended principle, we provide upper bounds on the asymptotic behavior of volumes of nullhomotopies of Lipschitz maps into nilpotent spaces, producing bounds for coformal spaces which are demonstrated to be nearly sharp.

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Chapter 1

Introduction

In [1], Gromov demonstrates for reasonable spaces that there is a polynomial in L bounding the number elements of a homotopy class that admit L -Lipschitz representatives. This provides one of the first quantitative links between the algebraic and quantitative geometric data of a topological space which Gromov proposes later in the 1990s (see [2]) as a valuable direction of study. That is, he suggested that topological invariants be studied not only algebraically, but quantitatively as well. Whether two manifolds are cobordant, or two maps are homotopic, are questions which in theory carry some geometric data along with them. Historically such equivalences have been dealt with in mostly algebraic ways, ignoring issues of computability or the potential sizes of their geometric realizations. However, the word problem for finitely generated groups convinces us that even algebraically simple objects are not *computably* simple in any reasonable amount of time. As Gromov has said in [2] regarding the corresponding story in topology,

...we can organize an innocuous enough metric on $\mathbf{S}^3$ so that it takes more than 10^{30} years to contract certain loops in the sphere and in the course of contraction we need to stretch the loop to something like 10^{30} light years in size. So if 10^{30} years is all the time you have, you conclude that the loop is

not contractible and whether or not $\pi_1(\mathbf{S}^3) = 0$ becomes a matter of opinion.

Sullivan’s localization methods in homotopy theory have repeatedly proved to be an invaluable tool for understanding algebraic invariants of topological spaces, and proves useful in Gromov’s suggested program as well. Together with Gromov, the work of Nabutovsky and Weinberger (such as [3,4]) paved the way for further quantitative study of topological spaces. The past decade of work (e.g., [5–10]) by Chambers, Guth, Manin, and Weinberger, among others have developed Gromov’s initial inquiries into a fruitful field of study, turning **quantitative topology** from an idea into a practical theory. As long as we only care about the rational homotopy type of a space (i.e., if we start by rationalizing our space, or localizing our space at $\{0\}$), standard algebraic invariants for a large class of spaces can be completely encoded by Sullivan’s model. This includes **simply connected spaces**, which are spaces for which the fundamental group is trivial. The **rationalization** of a simply connected space is one whose invariants are identical to the original, up to tensoring with $\mathbb{Q}$.

While one may hope to define the rationalization of spaces more generally, one may run into issues with rationalization as soon as a nontrivial fundamental group is allowed. If the fundamental group is abelian one can easily make sense of tensoring with $\mathbb{Q}$, but the natural Whitehead action on higher homotopy groups starts to cause other problems, even in spaces as innocuous as $\mathbb{RP}^2$ or $\mathbf{S}^1 \vee \mathbf{S}^2$ whose localizations fail to behave as expected. Here, the theory of localization does not seem to provide as much valuable insight as it does in the simply connected setting. An obvious remedy to this situation is to require the fundamental group to be abelian, and to act *trivially* on higher homotopy groups. Such a space is called **abelian** or **simple**. Any H-space, such as the torus $\mathbb{T}^n$, is simple. For instance, given an H-space X and $\alpha \in \pi_k(X)$, and a loop $\gamma : [0, 1] \rightarrow X$ based at $e \in X$, one can define $\Gamma(x, t) = \gamma(t) \star \alpha(x)$, where $\star : X \times X \rightarrow X$ is

multiplication. Clearly $\Gamma(x, 0) = \Gamma(x, 1) = e \star \alpha(x)$ so that Γ is a self-homotopy; hence $[\gamma, \alpha] = 0 \in \pi_k(X)$ where $[\cdot, \cdot]$ is the Whitehead bracket (see [Chapter 2](#)), and the action is trivial. Rationalization works in this setting, providing a class of spaces which naturally generalizes simply connected ones from the perspective of rational homotopy theory.

However, this generalization is incomplete in some sense. One can rationalize a space in meaningful ways even when the fundamental group is nonabelian. Allowing a further detour into the world of **nilpotence**, we find a satisfying stew of nontriviality mixed together with powerful structure. One may think of a **nilpotent group** as being one which is finitely far from an abelian one, and a **nilpotent action** as being finitely far from a trivial one. For example, if G is a finitely generated nilpotent group, then the Eilenberg-MacLane space $K(G, 1)$ can be rationalized by passing through the Malcev completion of G . Of course, the action of the fundamental group on higher homotopy groups in this setting is (trivially) trivial, but if we allow for spaces where the Whitehead action is a nilpotent action, rationalization still provides the invariants one desires. It is somewhat unsurprising—but gratifying nonetheless—that **c -step nilpotent spaces** (ones with π_1 a c -step nilpotent group, and $\pi_1 \curvearrowright \pi_n$ a c -step nilpotent action for $n \geq 2$) produce a natural generalization of abelian ones. Since Sullivan models completely encode the rational homotopy invariants of nilpotent spaces, such spaces make ideal objects of study in the story of quantitative topology.

Nilpotent spaces show up naturally in the study of almost nonnegatively curved manifolds. A closed smooth manifold is said to be **almost nonnegatively curved** if it can Gromov-Hausdorff converge to a single point under a lower curvature bound, generalizing “almost flat” and nonnegatively curved spaces. It is shown in [\[11\]](#) that all such spaces are nilpotent, up to finite covers. In particular the fundamental group is nilpotent up to finite index, with uniform control on the index given by the dimension of the space.

In the specific setting of $Y = K(G, 1)$, we uncover a connection between the story

of quantitative topology and that of geometric group theory. Instead of studying the geometry of Y , one may pass to the universal cover $\tilde{Y}$ which, though topologically Euclidean, is seen geometrically as a connected, simply connected, nilpotent Lie group $\tilde{Y} = \mathbb{G}$ equipped with a left-invariant metric. Recall that two metric spaces are **quasi-isometric** if (heuristically) they cannot be distinguished while squinting at them from far away. Since G sits inside of $\mathbb{G}$ as a lattice, one can study the coarse geometry of the group G by studying $\mathbb{G}$ instead.

For example, recall that the **Dehn function** δ_G of a finitely presented group G measures the size of a disk needed to fill a loop of length ℓ (i.e., the geometric realization of a word w of wordlength ℓ) in the Cayley graph of G . Since the asymptotic behavior of δ_G is invariant under coarse geometry, one is allowed to study this invariant in the Lie group $\mathbb{G}$ instead. The additional structure here allows us to define a number of natural high-dimensional analogues of Dehn functions, all of which are isoperimetric in spirit. That is, we ask how large an n -dimensional filling must be among all $(n-1)$ -dimensional objects of a given size which admit a filling, where a “filling” may be defined (for example, and not exclusively) by the existence of a nullhomotopy. One does not always need to measure both the filling and the boundary object in the same manner. One may, for instance, measure the filling object according to volume while measuring the boundary object in terms of the Lipschitz constant. Such variety of measures, and variety of definitions of “filling” produce a number of typically inequivalent notions of high-dimensional Dehn functions. The asymptotic behaviors of the corresponding higher-order Dehn functions need not agree; see [12, 13] for example, as well as [14–16].

Nilpotent groups have enjoyed a decent amount of study in the story of Dehn functions, and geometric group theory more generally. An increase in nilpotence step, for example, corresponds to a potential increase in asymptotic complexity of the Dehn function. Gromov shows (see [17]) that the 3-dimensional Heisenberg group has cubic Dehn

function $\delta(L) = \Theta(L^3)$. In general, if G is c -step nilpotent, then $\delta_G(L) = O(L^{c+1})$.

However, the story is often more complicated than this, especially when one starts considering higher order Dehn functions. In [18], for example, Pittet demonstrates that the 5-dimensional Heisenberg group has purely quadratic Dehn function and hence that the bound $O(L^{c+1})$ is not necessarily sharp. It has been further shown (see [19]) that $\delta_G(L)$ can exhibit strictly non-polynomial growth. The isoperimetric results of Pansu and Varopoulos (see [20, 21]) serve as some motivation for studying higher Dehn functions in the context of nilpotent groups. Recent work by Young in [13, 22] characterizes the asymptotic behavior of certain higher order Dehn functions in each dimension for Heisenberg groups, for instance.

In this paper, we are primarily interested in a question of quantitative topology which can also be seen as an investigation of higher order Dehn functions.

Question 1.1. *Let Y be a closed Riemannian manifold. If $f : \mathbf{S}^n \rightarrow Y$ is a nullhomotopic, L -Lipschitz map, what is the volume of an efficient nullhomotopy in terms of L ? How does this depend on the topology of Y ?*

In [8] the answer to the above question is shown to depend in general on the rational homotopy type of the target space Y , and Manin shows the following in [23]:

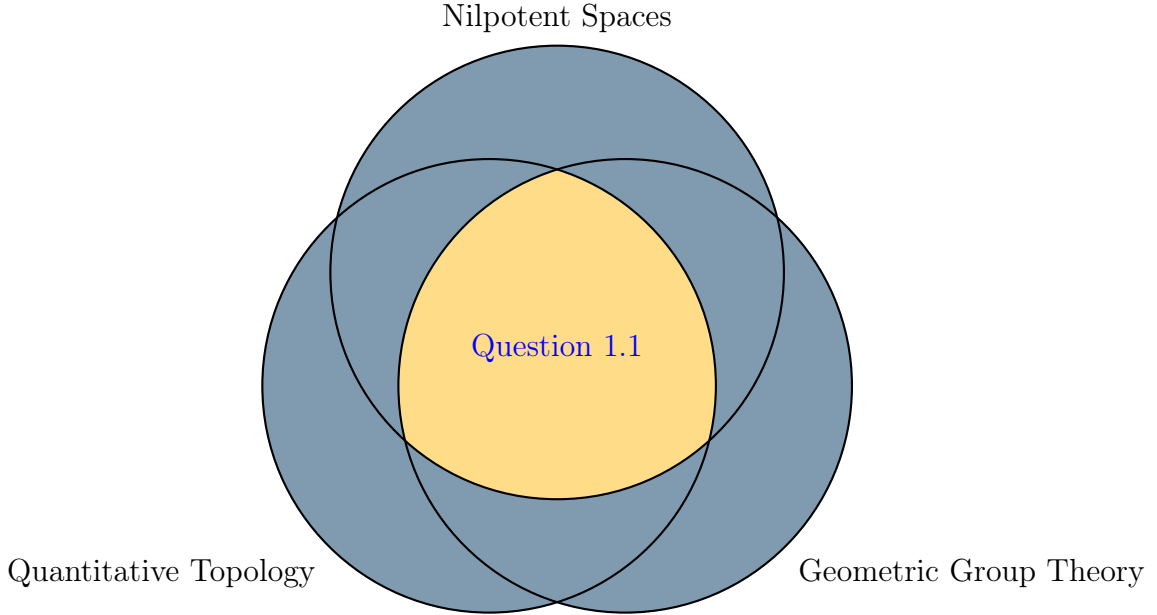
Theorem 1.2 ([23, Theorem A(ii)]). *If Y is simply connected and $f : \mathbf{S}^n \rightarrow Y$ is L -Lipschitz, then one can always find a nullhomotopy whose volume is at most $O(L^{2n})$.*

Moreover, a family of examples in [8] shows that this bound is, in some sense, sharp.

Theorem 1.3 ([9, §7]). *Let $n \geq 2$. There is a simply connected space Y_n and an L -Lipschitz map $f : \mathbf{S}^n \rightarrow Y_n$ such that any nullhomotopy of f has volume $\Omega(L^{2(n-1)})$.*

In the case $Y = K(G, 1)$ with G a finitely generated nilpotent group, this question can be seen as asking for the volume of an optimal (homotopic) filling of an L -Lipschitz

map $F : \mathbf{S}^n \rightarrow \tilde{Y} = \mathbb{G}$, which in turn can be seen as a higher order Dehn function of G . The current paper seeks to answer this question for Y a nilpotent space more generally, and is therefore a natural progression of both prongs of this story, in a very natural setting for both.



One of the significant breakthroughs in quantitative topology has been the development of the **Shadowing Principle** of Manin in [23]. Heuristically, the principle provides a method for homotoping a map which is algebraically bounded into a bounded geometric one. Somewhat more concretely, suppose a genuine map $f : X \rightarrow Y$ into a simply connected space has an algebraic model f^*m_Y which lies in the same homotopy class as a map whose dilatation is bounded by L . Then f itself lies in the homotopy class of a genuine map whose Lipschitz constant is bounded (asymptotically) by L . This principle is used, for example, in the proof of Theorem 1.2.

The guiding heuristic of the principle is that it is often easier to create bounded maps on the level of algebraic information, and that this is nearly enough to produce bounded geometric maps; if you have an (unbounded) geometric map which has a bounded al-

gebraic counterpart, then there is a bounded geometric map homotopic to the original one. Moreover, the algebraic counterpart to this new map is not too far away from the bounded algebraic counterpart of the original. We encourage the reader to refer to the original paper [23] for further exposition and intuition for this principle, as well as its applications.

One limitation of this principle is that one must manage to find bounded algebraic maps in the first place, to which one may then apply the principle. This task may not be any easier than finding their geometric counterparts. For instance, in [23, §5.3], a naive application of the principle produces nullhomotopies of volume $O(L^{n+2})$ for maps $\mathbf{S}^m \rightarrow \mathbf{S}^{2n}$ as long as $m \geq 2n - 1$. However, [24] shows that it is possible to find linear nullhomotopies for maps $\mathbf{S}^3 \rightarrow \mathbf{S}^2$, and [25, Theorem A(iii)] implies that this linear bound is always possible for maps into spheres.

1.1 Summary of Results

The Shadowing Principle in [23] requires the target spaces to be simply connected. The main contribution of this paper is the extension of the Shadowing Principle to certain (towers of) principal fibrations (Theorem 4.22). Concretely, this results in a generalization to nilpotent spaces (Theorem 4.23). As an application, we produce upper bounds on the volumes of nullhomotopies in general nilpotent spaces (Theorem 5.3). The upper bounds we attain by this method are not yet known to be sharp, and we expect they can be refined. A more careful analysis on simple spaces, for instance, reduces the expected upper bound somewhat significantly (Corollary 5.6). Thus far we have been unable to find a family of example spaces which come close to the general upper bounds, and we explore this failure in the first part of Chapter 3.

Restricting our attention to a class of nilpotent spaces called **coformal** spaces, we

can again refine the upper bounds on volumes of nullhomotopies ([Corollary 5.8](#)). In this setting, the upper bounds attained are (nearly) sharp. Indeed, we produce a family of spaces ([Theorem 3.39](#)) which both demonstrate the dependence of such volumes on the nilpotence class of the space, and generalize [Theorem 1.3](#) in a natural way.

To be more concrete, we briefly introduce some notation. First, following [\[26\]](#), we introduce the idea of the volume of a Lipschitz map $g : X \rightarrow Y$ between compact (piecewise) smooth Riemannian manifolds with boundary. After embedding $Y \hookrightarrow \mathbb{R}^N$ in a (piecewise) smooth manner, Rademacher's Theorem implies that g is differentiable almost everywhere (since g is Lipschitz), and hence one may define the **volume** of g by

$$\text{vol } g := \int_X |\text{Jac}(g)| \, \text{dvol}.$$

Given $f : X \rightarrow Y$ an L -Lipschitz, nullhomotopic function we define the **(homotopic) filling volume of f** to be

$$\text{FV}(f) := \inf \{ \text{vol } H \mid H : X \times I \rightarrow Y, H|_{X \times \{0\}} = f, H|_{X \times \{1\}} = * \}$$

where $*$ is the constant map to the basepoint of Y and $I = [0, 1]$ is the unit interval, and where the infimum is taken with respect to Lipschitz maps H so that the notion of volume is well-defined. The term “filling volume” comes from when $X = \mathbf{S}^n$ and we view the nullhomotopy H as a map $H : \mathbf{D}^{n+1} \rightarrow Y$. We further define

$$\text{V}\delta_Y^X(L) := \sup \{ \text{FV}(f) \mid f : X \rightarrow Y, \text{Lip } f \leq L, [f] = [*] \in [X, Y] \}.$$

In the event $X = \mathbf{S}^n$ we write $\text{V}\delta_Y^n(L) := \text{V}\delta_Y^{\mathbf{S}^n}(L)$. We summarize our resulting bounds together with the bounds for simply connected spaces in [Table 1.1](#) below for comparison.

<i>Type of Space Y</i>	<i>Simply Connected</i>	<i>Simple</i>	<i>c-Step Nilpotent</i>	<i>Coformal + c-Step Nilpotent</i>
Lower Bounds on $V\delta_Y^n(L)$	$\Omega(L^{2(n-1)})$	?	?	$\Omega(L^{(c+1)(n-1)})$
Reference	[9, §7.1]	See Chapter 3	See Chapter 3	Theorem 3.39
Upper Bounds on $V\delta_Y^n(L)$	$O(L^{2n})$	$O(L^{2n+1})$	$O(L^{(4c-1)n})$	$O(L^{(c+1)n})$
Reference	[23, Theorem A]	Corollary 5.6	Corollary 5.4	Corollary 5.8

Table 1.1: Bounds on $V\delta(L)$

We expect the general nilpotent bounds can be refined, and explore the following:

Conjecture 1.4. *If Y is a c -step nilpotent space then $V\delta_Y^n(L) = O(L^{3nc})$.*

1.2 Structure of Paper

In [Chapter 2](#) we orient the reader with the notation, conventions, and background concerning fibrations, nilpotent spaces, minimal Sullivan extensions/models, and the category equivalences induced by the localization and realization functors. After explaining our attempt (and failure) to fill in the top row of [Table 1.1](#) for general simple and c -step nilpotent spaces, we construct a family of nullhomotopic maps into coformal nilpotent spaces in [Chapter 3](#) for which any nullhomotopy must have large volume. In [Chapter 4](#) we generalize the Shadowing Principle to principal $K(G, n)$ fibrations, which specifically results in a corresponding principle for nilpotent spaces. As an application of the principle, in [Chapter 5](#) we produce upper bounds on the volumes of nullhomotopies filling in the second row of [Table 1.1](#).

Chapter 2

Background and Notation

We provide here our notation and conventions for nilpotent spaces, as well as Sullivan DGAs. All spaces throughout the paper, unless otherwise stated, are assumed to be compact Riemannian manifolds or finite simplicial complexes. Let Y be such a space, and recall that there is an action $\pi_1(Y) \curvearrowright \pi_n(Y)$ for all $n \geq 1$ induced by the Whitehead bracket. Namely, we have $\gamma \bullet \beta := [\gamma, \beta] + \beta$ where $[\cdot, \cdot] : \pi_1(Y) \otimes \pi_n(Y) \rightarrow \pi_n(Y)$ is the Whitehead product. This product is generalized in [Section 3.2](#); the details of this action are unnecessary at the moment, and so we invite the reader unfamiliar with the Whitehead product to continue reading this chapter first. The action itself can be identified with the action of deck transformations $\pi_1(Y) \curvearrowright \pi_n(\tilde{Y})$ in the universal cover for $n \geq 2$ (see [\[27, §4.1\]](#)).

Before providing all of the relevant definitions, we note here that we are primarily interested in spaces up to **rational equivalence** which we define a little later on. Suffice to say, we only focus on the non-torsion aspects of algebraic invariants when possible. For instance we might identify $\mathbf{S}^{2n+1}$ and $K(\mathbb{Z}, 2n+1)$. We will be primarily interested in spaces which admit triangulations, and the algebra of their differential forms. We conflate the notation in this setting, writing $\Omega^*(Y)$ to denote the smooth differential

forms when Y is a Riemannian manifold, as well as to denote the algebra of piecewise smooth forms when Y is a simplicial complex.

2.1 Nilpotent Spaces

We start by fixing notation and conventions for nilpotent groups and actions, and immediately contextualize our notation to the setting of homotopy groups of topological spaces. We then provide a method for constructing nilpotent spaces out of Postnikov towers. Along the way, we provide a number of examples and non-examples of nilpotent spaces.

2.1.1 Definitions and Notation

Let G be a group. Recall that for two subgroups $A, B \leq G$ their commutator is defined by $[A, B] = \langle [a, b] \mid a \in A, b \in B \rangle$. Define the **j th term of the lower central series of G** inductively by

$$G_j := \begin{cases} G & j = 1, \\ [G, G_{j-1}] & j > 1. \end{cases}$$

A group G is **c -step nilpotent** if $G_{c+1} \cong \{e\}$ and $G_c \not\cong \{e\}$. If G is c -step nilpotent for some unspecified c , we say that G is **nilpotent**. If G is 1-step nilpotent, we call G **abelian**. When $G = \pi_1(Y)$, we write $\pi_1^{(j)}(Y) := G_j$. If the space Y is clear from context we write simply $\pi_1^{(j)} = \pi_1^{(j)}(Y)$.

We now extend these definitions to group actions. Suppose now that $(G, \cdot)$ is a nilpotent group and $(M, +)$ is an abelian group, and that $G \curvearrowright M$. Write this action as $g \bullet m$. The **j th term of the lower central series of the action** is defined inductively

by

$$M_j^G := \begin{cases} M & j = 1, \\ \langle g \bullet m - m \mid g \in G, m \in M_{j-1}^G \rangle & j > 1. \end{cases}$$

If G is clear from context, we may write $M_j = M_j^G$. We call the action of G a (c -**step**) **nilpotent action** if $M_{c+1} \cong \{0\}$ and $M_c \not\cong \{0\}$. If $c = 1$, we call the action **simple** or **abelian**. When $G = \pi_1(Y)$ and $M = \pi_n(Y)$ for $n \geq 2$ with $G \curvearrowright M$ by the Whitehead product, we write $\pi_n^{(j)} = \pi_n^{(j)}(Y) := M_j$.

Definition 2.1. If $\pi_1(Y)$ is nilpotent and acts nilpotently on $\pi_n(Y)$ for all n we say that Y is **nilpotent**. If $\pi_1(Y)$ is c -step nilpotent and the action is c -step nilpotent for all n (respectively, for all $n \leq N$ for some N), we say that Y is c -**step nilpotent** (resp. up through degree N). If $c = 1$, then we say that Y is **simple** or **abelian** (resp. up through degree N).

Example 2.2. Simply connected spaces are simple, and simple spaces are nilpotent. $\blacktriangle$

Throughout the remainder of this chapter, we assume that Y is a c -step nilpotent space. We introduce some notation now which will later help describe the Postnikov tower of Y . As a warning, our notation does not necessarily agree with that of [28, 29]. Given a nilpotent group G we define

$$\overline{G_j} := \frac{G_j}{G_{j+1}}.$$

Similarly, given a nilpotent action $G \curvearrowright M$ we define

$$\overline{M_j^G} := \frac{M_j^G}{M_{j+1}^G}.$$

If G is clear from context, we write $\overline{M_j} = \overline{M_j^G}$. Note that if the action is c -step nilpotent,

then $j = c + 1$ is the smallest index for which $\overline{M_{c+1}} \cong \{0\}$. Contextualizing to the case $G = \pi_1(Y)$ for $n = 1$ and $M = \pi_n(Y)$ when $n \geq 2$, we define

$$\overline{\pi_n}^{(j)} = \overline{\pi_n}^{(j)}(Y) := \begin{cases} \overline{G_j} & n = 1, \\ \overline{M_j} & n \geq 2. \end{cases}$$

We first start with some non-examples of nilpotent spaces, showcasing some of the ways in which we must be careful when dealing with them.

Example 2.3. Note that $\pi_1(\mathbb{RP}^{2n}) = \mathbb{Z}/2\mathbb{Z}$ is torsion abelian. Since $\mathbb{RP}^{2n}$ has $\mathbf{S}^{2n}$ as a double cover, we have $\pi_k(\mathbb{RP}^{2n}) \cong \pi_k(\mathbf{S}^{2n})$ for all $k > 1$. In particular, $\pi_{2n}(\mathbb{RP}^{2n}) = \mathbb{Z}$. However, $\pi_1(\mathbb{RP}^{2n})$ acts by involution so that if $a \in \pi_1(\mathbb{RP}^{2n})$ is a generator and $x \in \pi_{2n}(\mathbb{RP}^{2n})$, then $a \bullet x = -x$. Therefore

$$\pi_2^{(2)}(\mathbb{RP}^{2n}) = \langle a \bullet x - x \mid x \in \mathbb{Z} \rangle = \langle 2x \mid x \in \mathbb{Z} \rangle = 2\mathbb{Z}.$$

More generally,

$$\pi_2^{(j)}(\mathbb{RP}^{2n}) = 2^{j-1}\mathbb{Z}$$

so that $\mathbb{RP}^{2n}$ is not nilpotent. ▲

In contrast to simply connected spaces, the next example shows that nilpotence is not closed under wedge sums.

Example 2.4. Let $X = \mathbf{S}^1 \vee \mathbf{S}^2$. Then $\pi_1(X) = \mathbb{Z}$, and passing to the universal cover $\tilde{X}$ one finds that $\pi_2(X) = \pi_2(\tilde{X}) = \prod_{\mathbb{Z}} \mathbb{Z}$. Write $\pi_2(X) \cong \langle e_i \mid i \in \mathbb{Z} \rangle$, where e_i is the generator of the i th copy of $\mathbf{S}^2$ in $\tilde{X}$. Let $a \in \pi_1(X)$ be a generator, and observe that $a \bullet e_i = e_{i+1}$. Then $\pi_2^{(2)}(X) = \langle e_{i+1} - e_i \rangle \cong \mathbb{Z}[\mathbb{Z}]$ and more generally $\pi_2^{(j)}(X) \cong \mathbb{Z}[\mathbb{Z}]$. Therefore, $\mathbf{S}^1 \vee \mathbf{S}^2$ is not nilpotent, even though both $\mathbf{S}^1$ and $\mathbf{S}^2$ are nilpotent. ▲

Of course, this may be seen even with $\mathbf{S}^1 \vee \mathbf{S}^1$ as $\pi_1(\mathbf{S}^1 \vee \mathbf{S}^1)$ is free abelian (i.e., not nilpotent), but the above demonstrates the failure for the action on higher homotopy groups more generally. On the other hand, nilpotence is preserved under taking products of spaces:

Example 2.5. Let X and Y be compact. Recall that $\pi_k(X \times Y) \cong \pi_k(X) \times \pi_k(Y)$ for all k . In particular, $\pi_1(X \times Y)$ is nilpotent if $\pi_1(X)$ and $\pi_1(Y)$ each are. Similarly, the Whitehead action $\pi_1(X \times Y) \curvearrowright \pi_k(X \times Y)$ is nilpotent if $\pi_1(X) \curvearrowright \pi_k(X)$ and $\pi_1(Y) \curvearrowright \pi_k(Y)$ are each nilpotent. Therefore, if X and Y are nilpotent, so is $X \times Y$. $\blacktriangle$

Example 2.6. Let G be a finitely generated nilpotent group. Then the Eilenberg-MacLane space $K(G, 1)$ is a nilpotent space since higher homotopy actions are trivial. $\blacktriangle$

Later, we will build more examples of nilpotent spaces through Postnikov towers.

2.1.2 Principal Fibrations and k-invariants

Fix G a finitely generated abelian group and $n \in \mathbb{N}$ throughout this section.

Proposition 2.7. *Let PX denote the pathspace of the space X . Then PX is contractible.*

Proof. Recall that PX is the set of paths in X based at $*$ $\in X$. Therefore a point $\gamma \in PX$ is a path $\gamma : X \times I \rightarrow X$ for which $\gamma|_{s=0} = *$. Define a retract $r : PX \times I \rightarrow PX$ of PX by truncation of paths. That is, define

$$r : (\gamma(x, s), t) \mapsto \gamma(x, (1 - t)s).$$

Clearly $r|_{t=0} = \mathbb{1}$ is the identity and $r|_{t=1} = *$ is constant. $\square$

Definition 2.8. Let $K = K(G, n)$ be an Eilenberg-MacLane space. Define the space $K^+ = K(G, n+1)$ with fixed basepoint $*$ $\in K^+$. The **pathspace fibration** is the obvious fibration $PK^+ \rightarrow K^+$ with fiber $p^{-1}(*) = \Omega K^+ \simeq K$.

Now let $k: B \rightarrow K^+$ for some topological space B . The **pullback of PK^+ by k** is the fibration $Y = B \times_k PK^+ \rightarrow B$ (unique up to weak homotopy equivalence) under the pullback

$$\begin{array}{ccc} Y & \longrightarrow & PK^+ \\ \downarrow p & \lrcorner & \downarrow \\ B & \xrightarrow{k} & K^+ \end{array}$$

Such a fibration $p: Y \rightarrow B$ has fiber K and is called a **principal $K(G, n)$ fibration**. In particular, a fibration $p: Y \rightarrow B$ is a principal $K(G, n)$ fibration if there is some map $k: B \rightarrow K^+$ for which p is the pullback.

Proposition 2.9. *Given a principal $K(G, n)$ -fibration $Y \rightarrow B$ induced by a map $k: B \rightarrow K(G, n+1)$, there is a class $[k] \in H^{n+1}(B; G)$ which classifies $p: Y \rightarrow B$ up to weak homotopy equivalence. That is there are bijections*

$$\left\{ \begin{array}{c} \text{principal} \\ K(G, n) \text{ fibrations} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{homotopy classes of} \\ \text{maps } B \rightarrow K(G, n) \end{array} \right\} \longleftrightarrow H^{n+1}(B; G)$$

Proof. Y is up to weak equivalence the homotopy fiber of k , and therefore it is enough to show that the choice of map k is equivalent to a class $[k] \in H^{n+1}(B; G)$. But by the universal coefficient theorem (or rather, by an interpretation of the theorem) the homotopy class of $k: B \rightarrow K(G, n+1)$ is precisely determined by $[k] \in H^{n+1}(B; G)$. $\square$

Definition 2.10. The class $[k] \in H^{n+1}(B; G)$ is called the **k -invariant** of the fibration. We call $k: B \rightarrow K(G, n+1)$ itself the **classifying map** of the fibration.

2.1.3 Obstruction Theory

Fix G a finitely generated abelian group and fix a principal $K(G, n)$ fibration $Y \xrightarrow{p} B$ of simplicial complexes Y and B .

Definition 2.11. Let the following diagram commute:

$$\begin{array}{ccc} A & \xrightarrow{\alpha} & Y \\ \iota \downarrow & & \downarrow p \\ X & \xrightarrow{\beta} & B \end{array}$$

Let $k : B \rightarrow K(G, n+1)$ be the classifying map of the fibration. Let CX and CA denote the cones of X and A respectively. The map α induces a nullhomotopic map $A \rightarrow K(G, n+1)$, and the **obstruction class** for this diagram is the class $\mathcal{O} \in H^{n+1}(X, A; G)$ induced by the composition

$$X \cup CA \rightarrow B \xrightarrow{k} K(G, n+1).$$

Proposition 2.12 ([27, Proposition 4.72]). *A map $X \rightarrow Y$ fitting into the diagonal of the above diagram exists if and only if the obstruction of the diagram vanishes.*

Proof. We claim that a lift exists if and only if the composition $X \cup CA \rightarrow B \xrightarrow{k} K^+$ extends to a map $CX \rightarrow K^+$. Indeed if a lift exists, then the map $X \rightarrow Y \rightarrow K^+$ factors through $PK^+ \simeq *$, and in particular means that there is an extension to CX as desired. On the other hand, such an extension defines a nullhomotopy into K^+ which in turn lifts (by the universal property of pullbacks) to a homotopy in Y . The end of this homotopy fits into the diagonal of the diagram.

On the other end of the problem, $\mathcal{O}$ is trivial if and only if the composition $X \cup CA \xrightarrow{\tilde{\beta}} K^+$ is nullhomotopic. If an extension $CX \rightarrow K^+$ exists, then there is a nullhomotopy which restricts to $X \cup CA$, and thus it suffices to show if $X \cup CA$ is nullhomotopic, then we can find the desired extension.

Indeed, if h_t is a nullhomotopy of $\tilde{\beta}$, the constant map h_1 clearly extends to a constant $CX \rightarrow K^+$. Let $\tilde{h}_t = h_{1-t}$ be the reverse of h . The homotopy extension property applied to $(CX, X \cup CA)$ allows us to extend $\tilde{h}$ to a homotopy $\tilde{H}$, and $\tilde{H}_0 : CX \rightarrow K^+$ defines the desired extension. $\square$

In a similar manner (see [Proposition 2.15](#) below), the obstruction to a relative homotopy between two potential fillings defines an element of $H^n(X, A; G)$. Here we introduce some terminology which we will use later on, and which will be relevant to the statement of this idea. Let $\mathcal{D} = \mathcal{D}(X, A, Y, B)$ be the set of all diagrams of the above form whose obstruction class vanishes.

Definition 2.13. The **external diagram** of a map $f : X \rightarrow Y$ is the diagram $\mathcal{D}_f$ given by

$$\begin{array}{ccc} A & \xrightarrow{f \circ \iota} & Y \\ \iota \downarrow & & \downarrow p \\ X & \xrightarrow{p \circ f} & B \end{array}$$

By [Proposition 2.12](#) we have $\mathcal{D}_f \in \mathcal{D}$ since f itself provides the diagonal map of interest. If $\mathcal{D}_f = \mathcal{D}_g$ for two maps $f, g : X \rightarrow Y$ we call f and g **externally equivalent**.

Clearly external equivalence in $\mathcal{D}$ defines an equivalence relation on maps $X \rightarrow Y$. This notion—although weaker than what we will be primarily interested in—has some value.

Definition 2.14. Suppose that f and g are externally equivalent via a common diagram

$$\begin{array}{ccc} A & \xrightarrow{\alpha} & Y \\ \iota \downarrow & \nearrow f & \downarrow p \\ X & \xrightarrow{\beta} & B \end{array}$$

g

Suppose that there is a homotopy H from f to g for which $H|_t$ is externally equivalent to f (hence to g) for all $t \in I$. Then f and g are said to be **internally homotopic**

through H and we call H an **internal homotopy** from f to g . If f and g are internally homotopic through some (unspecified) homotopy, we say that f and g are **internally homotopic** or **internally equivalent**.

Clearly internal equivalence is an equivalence relation. We will often be interested in the situation $(X, A) = (\Delta, \partial\Delta)$, the standard simplex. When $X = \Delta$, we will assume that $A = \partial\Delta$ without specifying so, and we write $\mathcal{D}_\Delta = \mathcal{D}(\Delta, \partial\Delta, Y, B)$.

Proposition 2.15 ([27, Exercise 4.24]). *Let $f, g : X \rightarrow Y$ be externally equivalent maps. The obstruction to an internal homotopy from f to g is a class in $H^n(X, A; G)$.*

Proof. Because f, g are externally equivalent with the same diagram, we view them each as being the extension $CX \rightarrow K^+$ of $X \cup CA \rightarrow K^+$ as in the previous proposition. View the suspension of X as $\Sigma X = CX \cup X \times I \cup CX$, two copies of CX connected by a cylinder along X . The existence of a homotopy H from f to g is equivalent to the existence of a map $\Sigma X \rightarrow Y$ which is f in the first instance of CX , which is H on $X \times I$, and which is g on the second CX . Hence, the existence of a homotopy is equivalent to a map fitting in the diagonal of the following diagram:

$$\begin{array}{ccc} \Sigma A & \longrightarrow & Y \\ \downarrow & \nearrow \text{dashed} & \downarrow \\ \Sigma X & \longrightarrow & B \end{array}$$

By [Proposition 2.12](#), the existence of such a homotopy is determined by an obstruction class in $H^{n+1}(\Sigma X, \Sigma A; G) \approx H^n(X, A; G)$. $\square$

Note that this proposition details the obstruction to the existence of a homotopy between two maps. Thus,

Corollary 2.16. *Internal homotopy classes form a torsor for $H^n(X, A; G)$. In particular, the collection of internal homotopy classes for $(\Delta^n, \partial\Delta^n)$ is a torsor for $G \cong H^n(\Delta^n, \partial\Delta^n; G)$.*

2.1.4 Postnikov Towers

Recall that a Postnikov system for Y is an inverse system

$$\cdots \rightarrow Y_n \xrightarrow{p_n} Y_{n-1} \rightarrow \cdots$$

where each p_n is a $K(\pi_n(Y), n)$ fibration. Moreover, the inclusion $Y \rightarrow Y_n$ induces an isomorphism $\pi_k(Y) \cong \pi_k(Y_n)$ for all $k \leq n$. In general this fibration is *not* principal, but when Y is nilpotent, p_n decomposes into a **principal refinement** of principal $K(G, n)$ fibrations. That is, if the action of $\pi_1(Y)$ on $\pi_n(Y)$ is c -step nilpotent, [30, Thm. I.2.9] shows that there is a sequence of principal $K(\overline{\pi}_n^{(j)}, n)$ fibrations $p_n^{(j)} : Y_n^j \rightarrow Y_n^{j-1}$ such that

$$p_n = p_n^{(1)} \circ p_n^{(2)} \circ \cdots \circ p_n^{(c)}.$$

In particular, we have the following commutative diagram:

$$\begin{array}{ccccc}
 Y_{n+1}^0 & \xlongequal{\quad} & Y_n & \xlongequal{\quad} & Y_n^c \\
 & \searrow \text{dashed} & \downarrow p_n^{(c)} & & \\
 & & Y_n^{c-1} & & \\
 & & \downarrow & & \\
 & & \vdots & & \\
 & & \downarrow & & \\
 & & Y_n^1 & & \\
 & & \downarrow p_n^{(1)} & & \\
 & & Y_n^0 & \xlongequal{\quad} & Y_{n-1} & \xlongequal{\quad} & Y_{n-1}^c
 \end{array}
 \tag{2.17}$$

(2.17)

In the event that Y is a Riemannian manifold, the construction of the Postnikov tower in general creates a fibration of CW complexes whose fibration maps $p_n^{(j)}$ are *a priori* known only to be continuous. More specifically, there is no guarantee we have any

reasonable notion of “local trivializations” around each fiber as one might hope. However, one may replace Y with a rationally equivalent simplicial complex whose geometric properties are not far from those of Y . Working in the simplicial category instead allows one to produce fibrations in the simplicial category; in particular, the maps $p_n^{(j)}$ may be assumed to be simplicial maps (see for example [31, Theorem 11.4] and [29, §1.6]).

2.1.5 Examples

Here, we build examples of Postnikov towers with principal refinements, giving us a few examples of nilpotent spaces. These constructions will show up later when we turn to build Sullivan models.

Example 2.18. Let $\mathbb{T}^{n-1}$ be the $(n-1)$ -torus. One can build $\mathbb{T}^n$ as a trivial principal $K(\mathbb{Z}, 1)$ fibration over $\mathbb{T}^{n-1}$ as

$$\begin{array}{ccc} \mathbf{S}^1 = K(\mathbb{Z}, 1) & \longrightarrow & \mathbb{T}^n \\ & & \downarrow \\ & & \mathbb{T}^{n-1} \end{array}$$

Here, the classifying map $k_0 : \mathbb{T}^{n-1} \rightarrow K(\mathbb{Z}, 2)$ is one for which $[k_0] \in H^2(\mathbb{T}^{n-1}; \mathbb{Z})$ is trivial. Of course, one may also build $\mathbb{T}^n$ as a single principal $K(\mathbb{Z}^n, 1)$ fibration over $\{*\}$. This is related to the fact that $\mathbb{Z}^n$ is 1-step nilpotent (i.e., abelian). $\blacktriangle$

Note that in the above example, $H^2(\mathbb{T}^{n-1}; \mathbb{Z}) \approx \mathbb{Z}^{\binom{n}{2}}$ so that a number of non-trivial k -invariants are possible.

Example 2.19. Let $\mathbb{T}^2$ be the 2-torus with $H^1(\mathbb{T}^2; \mathbb{Z})$ generated by classes $[x]$ and $[y]$, so that $[\mathbb{T}^2] = [x] \smile [y]$. Consider a principal $K(\mathbb{Z}, 1)$ fibration over $\mathbb{T}^2$ as

$$\begin{array}{ccc} K(\mathbb{Z}, 1) & \longrightarrow & M \\ & & \downarrow \\ & & \mathbb{T}^2 \end{array}$$

In this situation, either $[k] = 0$ is trivial (in which case $M = \mathbb{T}^3$ as above) or else $[k]$ is a nonzero multiple of $[\mathbb{T}^2]$. In the case $[k] = [\mathbb{T}^2]$ we call M the **3-dimensional Heisenberg Manifold** which we denote Heis^3 . We end up with a 1-cochain $z \in C^1(\text{Heis}^3)$ whose coboundary is nontrivial. In particular, $\delta z = x \smile y$. The cohomology groups are therefore given by

$$H^k(\text{Heis}^3; \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & k = 0, 3 \\ \mathbb{Z}^2 & k = 1, 2, \end{cases}$$

where H^1 is generated by $[x]$ and $[y]$ while H^2 is generated by $[z \smile x]$ and $[z \smile y]$.

For the situation $[k] = r[\mathbb{T}^2]$ for $r \neq 1$ more broadly, we get a $K(G, 1)$ for which $G^{ab} \approx \mathbb{Z}^2 \times \mathbb{Z}/r\mathbb{Z}$ whose universal cover is $\mathbb{H}^3$ the Heisenberg Lie group. In this setting, $K(G, 1)$ is still rationally equivalent to Heis^3 .

More generally, suppose that $\mathbb{T}^{2n}$ is the $2n$ torus with $H^2(\mathbb{T}^2; \mathbb{Z})$ generated by classes $[x_1], \dots, [x_n]$ and $[y_1], \dots, [y_n]$. Consider a principal $K(\mathbb{Z}, 1)$ fibration over $\mathbb{T}^{2n}$ as

$$\begin{array}{ccc} K(\mathbb{Z}, 1) & \longrightarrow & M' \\ & & \downarrow \\ & & \mathbb{T}^{2n} \end{array}$$

There are clearly a number of classifying maps one could choose, though many of these end up providing (rationally) equivalent fibrations. When $[k] = 0$, the resulting space is $\mathbb{T}^{2n+1}$. When $[k] = \sum_{i=1}^n [x_i \smile y_i]$ the manifold M' is called the **$(2n+1)$ -dimensional Heisenberg manifold** which we denote Heis^{2n+1} . ▲

Example 2.20. Return to the 3-Heisenberg manifold as a principal $K(\mathbb{Z}, 1)$ fibration over $\mathbb{T}^2$. Rename $z_0 = x$, and $z_1 = y$, and $z_2 = z$. Inductively, suppose that $\mathbb{F}^{c-1}$ is a c -dimensional space built out of a principal $K(\mathbb{Z}, 1)$ fibration over $\mathbb{F}^{c-2}$, and that $C^1(\mathbb{F}^{c-1})$ is generated by c cocycles $z_0, z_1, z_2, \dots, z_{c-1}$ for which $\delta z_i = z_{i-1} \smile z_0$ for $i \geq 1$. In particular, note that $\delta z_0 = \delta z_1 = 0$.

Define a principal $K(\mathbb{Z}, 1)$ fibration

$$\begin{array}{ccc}
K(\mathbb{Z}, 1) & \longrightarrow & \mathbb{F}^c \\
& & \downarrow \\
& & \mathbb{F}^{c-1}
\end{array}$$

for which the k -invariant is given by a class $[k] = [z_{c-1} \smile z_0] \in H^2(\mathbb{F}^c; \mathbb{Z})$. The space $\mathbb{F}^{c+1}$ is called the **c -step Filiform space**. Overall, this produces a $K(G, 1)$ -fibration over $*$ which is *not* principal (and where G is non-abelian); however, this iterative process provides a decomposition into a principal refinement. $\blacktriangle$

So far, we have only dealt with principal $K(\mathbb{Z}, 1)$ fibrations. In general, we will only ever be concerned with $K(\mathbb{Z}^p, n)$ fibrations (ignoring torsion, in particular) since we are primarily interested in spaces only up to rational equivalence. Spaces where the fundamental group is nilpotent will be of special importance, but for now we provide an example of a nontrivial principal $K(\mathbb{Z}, 3)$ -fibration where the fundamental group does not cause extra interference.

Example 2.21. Recall that $\mathbb{CP}^\infty$ is a $K(\mathbb{Z}, 2)$ Eilenberg-MacLane space, and that cohomologically $H^*(\mathbb{CP}^\infty) \cong \langle \alpha^{(2)} \rangle$ is generated by a single generator in degree 2. In particular, $H^4(\mathbb{CP}^\infty; \mathbb{Z}) \cong \mathbb{Z}$ as a group. Suppose we have a principal $K(\mathbb{Z}, 3)$ fibration:

$$\begin{array}{ccc}
K(\mathbb{Z}, 3) & \longrightarrow & M \\
& & \downarrow \\
& & \mathbb{CP}^\infty
\end{array}$$

The k -invariant is a class $[k] \in H^4(\mathbb{CP}^\infty; \mathbb{Z}) \cong \mathbb{Z}$ so that we are again left with (fundamentally) two choices: either $[k]$ is trivial or $[k]$ is a nonzero multiple of the generator α^2 . In the event that $[k] = 0$ we are left with $M = K(\mathbb{Z}, 2) \times K(\mathbb{Z}, 3)$ which is rationally equivalent to $\mathbb{CP}^\infty \times \mathbf{S}^3$. On the other hand, $[k] = \alpha^2$ produces the Hopf fibration, in which case M is rationally equivalent to $\mathbf{S}^2$. $\blacktriangle$

2.2 Sullivan Models

We now turn to producing algebraic models of the above ideas. This will allow use to produce a more broad collection of examples of nilpotent spaces fairly easily, without needing to describe the Postnikov tower of a space quite so tediously.

Much of this chapter is a conglomeration of definitions and results from [28, 29, 32]. We again start by providing the relevant definitions and notation for Sullivan models, and then describe when it is possible to build up such models as elementary extensions. We conclude with a few examples of such models.

2.2.1 Definitions and Notation

For our purposes, a **commutative differential graded algebra** (a **DGA**) is a cochain complex $(\mathcal{A}, d)$ of vector spaces $\mathcal{A} = \{\mathcal{A}^n\}$ over $\mathbb{Q}$ or $\mathbb{R}$, together with a graded commutative multiplication which satisfies the (graded) Leibniz rule. We assume from here on out that every DGA $\mathcal{A}$ is **connected**, meaning that $H^0(\mathcal{A}) = \mathbb{Q}$. We define $\mathcal{A}^+ := \{\mathcal{A}^n\}_{n>0}$.

The alternating algebra $\wedge Z$ on a graded vector space $Z = \{Z_n\}_{n \geq 1}$ is a graded algebra, which can be made into a DGA with an appropriate choice of differential (a trivial example being $d = 0$). We define

$$\wedge^k Z = \text{span} \{z_1 \wedge \cdots \wedge z_k \mid z_i \in Z\}$$

and define $\wedge^{\geq k} Z$ and $\wedge^{\leq k} Z$ and $\wedge^{>k} Z$ and $\wedge^{<k} Z$ in the obvious ways.

When $\mathcal{A} = \wedge Z$ the **space of indecomposables** is the graded vector space

$$Q\mathcal{A} = \frac{\mathcal{A}^+}{\mathcal{A}^+ \wedge \mathcal{A}^+} \cong Z$$

and in degree n the map $Z \rightarrow \wedge Z$ induces an identification

$$(Q\mathcal{A})^n \cong Z_n.$$

In this setting the differential splits as $d = d_1 + d_2 + \cdots$ where $\text{im}(d_k) \subseteq \wedge^{k+1} Z$. The **quadratic part** of the differential is d_1 . It turns out that $(\mathcal{A}, d_1)$ is itself a DGA, which we call the **associated quadratic** DGA. The remaining pieces $d - d_1$ are a **deformation**, studied for instance by [33].

Example 2.22. Let $\mathcal{K}_n = \wedge(x^{(n)})$ with $d = 0$. We use this notation to indicate that $\mathcal{K} = \wedge V$ where $V = V_n$ is generated by a single element x of degree n . If n is odd, then $x^2 = -x^2 = 0$. Then $H^*(\mathcal{K}_n) = H^n(\mathcal{K}_n) \cong \mathbb{Z}$. On the other hand, if n is even then $x^2 \neq 0$ so that $H^{p \cdot n}(\mathcal{K}_n) = \mathbb{Z}$ for any multiple $p \in \mathbb{N}$ of n . That is, $H^*(\mathcal{K}_n)$ is a free polynomial ring on a single generator in degree n . ▲

2.2.2 Cofibrations and Sullivan Extensions

Now, let $(\mathcal{A}, d) \rightarrow (\mathcal{A} \otimes \wedge Z, d)$ be a commutative cochain algebra morphism, such that $a \mapsto a \otimes 1$. We identify $\mathcal{A} = \mathcal{A} \otimes 1$ and $Z = 1 \otimes Z$. This morphism is a **Sullivan extension** if

- (1) $H^0(\mathcal{A}, d) = \mathbb{Q}$ (that is, $\mathcal{A}$ is connected),
- (2) $Z = \{Z_n\}_{n \geq 1}$ is concentrated in positive degrees, and
- (3) Z satisfies the **nilpotence condition**. That is, Z is the union of an increasing family of subspaces

$$0 = Z(0) \subseteq \cdots \subseteq Z(r) \subseteq \cdots$$

where the differential satisfies

$$\mathrm{im}(d)|_{Z(r)} \subseteq \begin{cases} \mathcal{A} & r = 1 \\ \mathcal{A} \otimes \wedge Z(r-1) & r > 1. \end{cases}$$

The extension is **minimal** if in addition

$$(4) \quad \mathrm{im}(d) \subseteq \mathcal{A}^+ \otimes \wedge Z + \mathcal{A} \otimes \wedge^{\geq 2} Z.$$

The **fiber** of the extension is the DGA $(\wedge Z, \bar{d})$ where $\bar{d}$ is induced by the quotient

$$\wedge Z \cong \frac{\mathcal{A} \otimes \wedge Z}{\mathcal{A}^+ \otimes \wedge Z}.$$

We will drop the adjective ‘‘Sullivan’’ from here on out, except if the extension in question is not minimal. Abusing terminology, we may call $\mathcal{A} \otimes \wedge Z$ itself a minimal extension of $\mathcal{A}$. We may write $\mathcal{A} \langle Z \rangle$ for $\mathcal{A} \otimes \wedge Z$ in the event that $\mathrm{im}(d|_Z) \subseteq \mathcal{A}$ (i.e., under the nilpotence condition $Z = Z(1)$), and we call such a minimal extension **elementary**.

We are primarily interested in studying connected DGAs of the form $\mathcal{A} = \wedge V$ for some graded vector space $V = \{V_n\}_{n \in \mathbb{N}}$ that can be built by minimal extensions of the form $\wedge V_{\leq n} \otimes \wedge V_{n+1}$. Here, $V_{\leq n}$ denotes $\{V_k\}_{k \leq n}$.

Suppose for instance that the extension $\wedge V_{\leq n} \otimes \wedge V_{n+1}$ is minimal. If $V_{n+1} = V_{n+1}(c)$, then this extension decomposes into a sequence of c elementary extensions. If this is true for every n , we say that $\wedge V$ is **c -step nilpotent**. If $c = 1$ then every extension is elementary; hence for every n the differential satisfies the more strict requirement

$$\mathrm{im}(d|_{V_n}) \subseteq \wedge^{\geq 2} V_{< n} \tag{2.23}$$

and we call $\wedge V$ **simple** or **abelian**. In the event $V_1 = 0$ the above expression is auto-

matically satisfied, and we call $\wedge V$ **simply connected**.

Here, we provide some examples which we revisit shortly at the end of this section.

Example 2.24. Let n be even and let $\mathcal{K}_n = \wedge(\alpha^{(n)})$ with $d = 0$. The extension $\mathcal{S}_n = \wedge(\alpha) \otimes \wedge(\beta^{(2n-1)})$ with $d\beta = \alpha^2$ is a minimal extension, and in fact is an elementary extension, of $\mathcal{K}_n$. In this setting, we have $H^n(\mathcal{S}_n) = \mathbb{Z}$ and all other cohomology groups are trivial. ▲

Example 2.25. Let $\mathcal{T} = \wedge(\alpha^{(1)}, \beta^{(1)})$ with $d = 0$. The extension $\mathcal{H} = \wedge(\alpha, \beta) \otimes \wedge(\gamma^{(1)})$ with $d\gamma = \alpha \wedge \beta$ is an elementary extension. Here we have

$$H^k(\mathcal{H}) \cong \begin{cases} \mathbb{Z} & k = 0, 3 \\ \mathbb{Z}^2 & k = 1, 2. \end{cases}$$

More generally, let $\mathcal{T}_{2n} = \wedge(\alpha_1^{(1)}, \dots, \alpha_n^{(1)}, \beta_1^{(1)}, \dots, \beta_n^{(1)})$ with $d = 0$. The extension $\mathcal{H}_{2n+1} = \mathcal{T}_{2n} \otimes \wedge(\gamma^{(1)})$ with $d\gamma = \sum_{i=1}^n \alpha_i \wedge \beta_i$ is an elementary extension. ▲

Example 2.26. Inductively (with $c = 1$ as a trivial base case) suppose that $\mathcal{F}_c = \wedge V$ can be built out of $c - 1$ elementary extensions, where $V = V_1$ is generated by degree 1 generators $\zeta_0, \zeta_1, \dots, \zeta_{c-1}$ with $d\zeta_i = \zeta_{i-1} \wedge \zeta_0$ for $i \geq 1$. In particular, $d\zeta_0 = d\zeta_1 = 0$. Define $\mathcal{F}_{c+1}$ to be the extension $\mathcal{F}_c \otimes \wedge(\zeta_c^{(1)})$ with $d\zeta_c = \zeta_{c-1} \wedge \zeta_0$. It is easy to check that this extension is again minimal, and in fact elementary. Hence, $\mathcal{F}_{c+1}$ is c -step nilpotent.

On the other hand, one may consider the extension $\mathbb{Q} \rightarrow \mathbb{Q} \otimes \mathcal{F}_c \cong \mathcal{F}_c$ on its own. This extension is a minimal Sullivan extension, but is not elementary. ▲

2.2.3 Minimal Models

Recall that a map of DGAs is a **quasi-isomorphism** if it induces an isomorphism on cohomology. A **minimal (Sullivan) model** of a DGA morphism $\varphi: (\mathcal{A}, d) \rightarrow (\mathcal{B}, d)$

is a quasi-isomorphism $m_\varphi: (\mathcal{A} \otimes \wedge Z, d) \rightarrow (\mathcal{B}, d)$ where $\mathcal{A} \otimes \wedge Z$ is a minimal extension of $\mathcal{A}$, and where $m_\varphi(a \otimes 1) = \varphi(a)$ for all $a \in \mathcal{A}$. Again, we will drop the adjective “Sullivan” and simply call such models **minimal**.

Note. Our terminology disagrees slightly with that in [29]. We require that $Z = \{Z_n\}_{n \geq 1}$, but they relax this condition to allow for nontrivial subspaces of degree $n = 0$ in the fiber. In this way, their definition of a minimal Sullivan model is slightly more broad; however, we will soon restrict our attention to a context where these definitions agree. (See [29, §3.7].)

The existence of minimal models of fibrations is a well-established fact:

Theorem 2.27 ([28, Thm. 14.12]). *Let $\varphi: (\mathcal{A}, d) \rightarrow (\mathcal{B}, d)$ be a morphism of connected DGAs for which $H^1(\varphi)$ is an injection. Then φ has a (unique) minimal model.*

In particular, the minimal model of a connected DGA always exists:

Proposition 2.28 ([29, Prop. 1.8]). *If $(\mathcal{B}, d)$ is a connected DGA, there is a minimal Sullivan algebra $\wedge V$ and a quasi-isomorphism $\varphi: (\wedge V, d) \rightarrow (\mathcal{B}, d)$.*

Abusing terminology, we call $\wedge V$ itself the **minimal Sullivan model** of $\mathcal{B}$ in this situation. In particular, we can now define the minimal model of a topological space. As before, we will frequently drop the adjective “Sullivan.”

Definition 2.29. If Y is a Riemannian manifold or simplicial complex, the **minimal model of Y** is $\mathcal{M}_Y^*$, the minimal model of $\Omega^*(Y)$. If $p: Y \rightarrow B$ is a principal $K(G, n)$ fibration for G abelian, the **minimal model of p** is the minimal model of $p^*: \Omega^*(B) \rightarrow \Omega^*(Y)$.

2.2.4 Examples

We now connect the examples we have been producing. The duality between their constructions will be addressed in the following section.

Example 2.30. The Sullivan model $\mathcal{K}_n$ from [Example 2.22](#) is the minimal model of $K(\mathbb{Z}, n)$. ▲

Example 2.31. When n is even, the Sullivan model $\mathcal{S}_n$ from [Example 2.24](#) is the minimal model of $\mathbf{S}^n$. When n is odd, the minimal model of $\mathbf{S}^n$ is $\mathcal{K}_n$ since $\mathbf{S}^n$ is rationally equivalent to $K(\mathbb{Z}, n)$ in this setting. ▲

Example 2.32. The Sullivan model $\mathcal{T}_{2n}$ from [Example 2.25](#) is the minimal model of $\mathbb{T}^{2n}$. More generally, the minimal model of $\mathbb{T}^m$ is $\wedge(x_1, \dots, x_m)$ with $d = 0$. ▲

Example 2.33. The Sullivan model $\mathcal{H}_{2n+1}$ from [Example 2.25](#) is the minimal model of Heis^{2n+1} . In particular, $\mathcal{H}$ is the minimal model of Heis^3 . ▲

Example 2.34. The Sullivan model $\mathcal{F}_c$ from [Example 2.26](#) is the minimal model of $\mathbb{F}^c$. ▲

2.3 Realization and Rationalization

2.3.1 Rational Equivalence

In [\[30\]](#) it is shown that if G is finitely generated nilpotent, there is a **rationalization** (equivalently, $\{0\}$ -localization) of G , by which we mean that there is a group denoted $G \otimes \mathbb{Q}$ in which $g \mapsto g^p$ is a bijection for all $p \neq 0$. Moreover, the rationalization functor $G \mapsto G \otimes \mathbb{Q}$, thought of as the Malcev completion of G , is exact in the category of (c -step) nilpotent groups. When M is abelian, this rationalization $M \otimes \mathbb{Q}$ agrees with the standard tensor product with $\mathbb{Q}$.

Definition 2.35. Let X and Y be nilpotent spaces. If $f : X \rightarrow Y$ is such that $f_k : \pi_k(X) \otimes \mathbb{Q} \rightarrow \pi_k(Y) \otimes \mathbb{Q}$ is an isomorphism for all k , we call f a **rational (homotopy) equivalence**. Two spaces X and X' are **rationally equivalent** if there is a chain of rational equivalences connecting them:

$$X \rightarrow Y_0 \leftarrow Y_1 \rightarrow \cdots \leftarrow Y_N \rightarrow X'$$

Minimal models provide a functor from topological spaces to differential graded algebras $\mathcal{M}_{\square}^* : \mathbf{Top} \rightarrow \mathbf{DGA}$. Dually there is a “realization” functor $\langle \cdot \rangle : \mathbf{DGA} \rightarrow \mathbf{Top}$ for producing a topological space out of the given algebraic data.

Definition 2.36. Given a space X the **rationalization of X** is $X_{\mathbb{Q}} = \langle \mathcal{M}_X^* \rangle$.

A central power of rational homotopy theory is that for nilpotent spaces, these functors are inverses up to rational equivalence. That is

Theorem 2.37 ([29, Proposition 8.2]). *Given a nilpotent space X , the geometric realization of the minimal model $m_X : \mathcal{M}_X^* \rightarrow \Omega^*(X)$ induces a rational equivalence*

$$X \rightarrow \langle \Omega^*(X) \rangle \xrightarrow{\langle m_X \rangle} \langle \mathcal{M}_X^* \rangle = X_{\mathbb{Q}}.$$

Since we are interested in nilpotent spaces up to rationalization, we may assume that we are working with any rationally equivalent space. Because of this it will help to condense our notation when talking about nilpotent spaces. For instance, we will want to talk about the rationalization of each term of the lower central series of the action of the fundamental group on a higher homotopy group. Our notation in this setting is already quite cumbersome, but we are now at a point where we change our notation somewhat drastically, since duality between principal $K(G, n)$ fibrations and elementary extensions allows us to make natural identifications between certain rational invariants.

When $M = \pi_n(Y)$ for some $n \geq 2$ we define

$$V_n = V_n(Y) := \text{Hom}(M, \mathbb{Q}) = \text{Hom}(\pi_n(Y), \mathbb{Q}).$$

In the situation $G = \pi_1(Y)$, it is fitting to be slightly more delicate. If we assume that G is finitely generated, there is a Lie group $\widehat{G}$ called the **Malcev completion** of G for which $G \subseteq \widehat{G}$ is a cocompact lattice. In fact, $\widehat{G}$ has rational structure constants, meaning that there is a basis $\{X_1, \dots, X_m\}$ of the Lie algebra $\mathfrak{g} = \text{lie}(\widehat{G})$ for which all bracket relations have rational coefficients:

$$[X_i, X_j] \in \text{span}_{\mathbb{Q}} \{X_1, \dots, X_m\},$$

and one views $\widehat{G}$ as a rational Lie group. Following [34, §3.2.1] one may define $V_1 = \mathfrak{g}^* = \text{Hom}(\mathfrak{g}, \mathbb{Q})$. In [32] they denote $G \otimes \mathbb{Q} = \mathfrak{g}$, and we will often adopt this notation as well. We say that Y is of **finite type** if V_n is a finite dimensional $\mathbb{Q}$ vector space for every n . Unless otherwise stated, we assume that all nilpotent spaces are of finite type. Moreover define

$$\Lambda_n^{(j)} = \Lambda_n^{(j)}(Y) := \begin{cases} \text{Hom}(\overline{G_j}, \mathbb{Q}) & n = 1, \\ \text{Hom}(\overline{M_j}, \mathbb{Q}) & n \geq 2. \end{cases}$$

That is, $\Lambda_n^{(j)} = \text{Hom}(\overline{\pi_n^{(j)}}(Y), \mathbb{Q})$ for all $n \geq 1$. It will also help to define

$$V_n^{\leq J} := \bigoplus_{j=1}^J \Lambda_n^{(j)}$$

and note moreover that $V_n^{\leq c} \cong V_n$.

In particular, observe that $V_n^{\leq J} \cong \pi_n(Y_n^J) \otimes \mathbb{Q}$, and that $V_n \cong \pi_n(Y_n) \otimes \mathbb{Q}$. A key consequence in this context is that the minimal model of Y can be described by $\mathcal{M}_Y^* = \wedge V$

where $V = \{V_n\}_{n \geq 1}$ with $V_n \cong \text{Hom}(\pi_n(Y), \mathbb{Q}) \cong \pi_n(Y) \otimes \mathbb{Q}$ for all n . We make this identification without confusion. In fact, the principal refinement of the Postnikov tower of Y induces a sequence of minimal extensions whose limit is $\mathcal{M}_Y^*$. That is, the principal $K(\overline{\pi}_n^{(j)}, n)$ fibration

$$p_n^{(j)}: Y_n^j \rightarrow Y_n^{j-1}$$

is precisely dual to an elementary extension

$$V_n(j-1) \langle \Lambda_n^{(j)} \rangle = V_n(j).$$

There is a powerful theorem for fibrations which connects some of the subjects we have been discussing in a bit more generality. In particular, the minimal models of the fibrations from [Example 2.25](#) through [Example 2.24](#) can all be described by this theorem.

Theorem 2.38 ([29, eq. 3.8, and Thm. 5.1]). *Let $F \rightarrow Y \rightarrow B$ be a fibration of path connected Riemannian manifolds of finite type. Then there is a minimal extension $\mathcal{M}_B^* \rightarrow \mathcal{M}_B^* \otimes \wedge V$ such that the following diagram of DGAs commutes*

$$\begin{array}{ccccc} (\mathcal{M}_B^*, d) & \longrightarrow & (\mathcal{M}_B^* \otimes \wedge V, d) & \longrightarrow & (\wedge V, \bar{d}) \\ \simeq \downarrow m_B & & \simeq \downarrow m_Y & & \downarrow m_F \\ \Omega^*(B) & \longrightarrow & \Omega^*(Y) & \longrightarrow & \Omega^*(F) \end{array}$$

Moreover, if $\pi_1(B)$ acts locally nilpotently on $H^*(F; \mathbb{Q})$ then $m_F: \wedge V \xrightarrow{\simeq} \Omega^*(F)$ is a quasi-isomorphism, and hence a minimal model for F .

If the fibration $F \rightarrow Y \rightarrow B$ is a principal $K(G, n)$ fibration, it is shown in [32, §12.2] that the k -invariant $[k] \in H^n(B; G)$ is dual to the differential d of $\mathcal{M}_B^* \otimes \wedge V$ in the sense that $[d] \in H^n(\mathcal{M}_B^*; V)$ corresponds to $[k] \otimes 1 \in H^n(B; G) \otimes \mathbb{Q}$ under the following:

Lemma 2.39. $H^n(\mathcal{M}_B^*; V) \cong H^n(B; G) \otimes \mathbb{Q}$.

Proof. Under applications of the Universal Coefficient Theorem, we have

$$\begin{aligned}
H^n(\mathcal{M}_B^*; V) &= H^n(\text{Hom}(V, \mathcal{M}_B^*)) \\
&\cong \text{Hom}(V, H^n(\mathcal{M}_B^*)) \\
&\cong \text{Hom}(\text{Hom}(G, \mathbb{Q}), \text{Hom}(H_n(B), \mathbb{Q})) \\
&\cong \text{Hom}(H_n(B), G \otimes \mathbb{Q}) \\
&\cong H^n(B; G \otimes \mathbb{Q}) \\
&\cong H^n(B; G) \otimes \mathbb{Q}.
\end{aligned}
\tag*{$\square$}$$

In particular, then, as in the proof of [32, Proposition 15.3], we have that the following diagram commutes for any (finite) complex X :

$$\begin{array}{ccc}
[X, B] & \xrightarrow[\text{to lifting}]{\text{obstruction}} & H^n(X; G) \\
\downarrow & & \downarrow \otimes \mathbb{Q} \\
& & H^n(X; V) \\
& & \downarrow \cong \\
[\mathcal{M}_B^*, \mathcal{M}_X^*] & \xrightarrow[\text{to extending}]{\text{obstruction}} & H^n(\mathcal{M}_X^*; V)
\end{array}
\tag{2.40}$$

2.4 Algebraic Obstruction Theory

Following now [32, §11], we present some homotopy theory for DGAs, with a quantitative focus as in [23, §3.2]. Let $\mathcal{A}$ and $\mathcal{B}$ be DGAs with morphisms $\varphi, \psi : \mathcal{A} \rightarrow \mathcal{B}$. An **(algebraic) homotopy** from φ to ψ is then a morphism

$$\Phi : \mathcal{A} \rightarrow \mathcal{B} \otimes \mathbb{R} \langle t, dt \rangle$$

for which $\Phi|_{\frac{t=0}{dt=0}} = \varphi$ and $\Phi|_{\frac{t=1}{dt=0}} = \psi$. For such an algebraic homotopy Φ , we will write Φ_j^i for the $t^i(dt)^j$ term of Φ . Moreover, we think of $\mathbb{R}\langle t, dt \rangle := \mathbb{R}\langle t^{(0)}, dt^{(1)} \rangle$ as an algebraic model for the unit interval $I := [0, 1]$. In the event that $\mathcal{B} = \Omega^*(X)$ for a sufficiently nice space X , we may realize this algebraic model as a genuine interval by a morphism

$$\rho : \Omega^*(X) \otimes \mathbb{R}\langle t, dt \rangle \rightarrow \Omega^*(X \times I).$$

We may further utilize this algebraic model of the unit interval as a tool for integration of homotopies. That is, we define a linear operator $\int_0^1 : \mathcal{B} \otimes \mathbb{R}\langle t, dt \rangle \rightarrow \mathcal{B}$ by

$$\int_0^1 b \otimes t^i = 0 \quad \text{and} \quad \int_0^1 b \otimes t^i dt = (-1)^{\deg b} \frac{1}{i+1} b,$$

as well as an operator $\int_0^t : \mathcal{B} \otimes \mathbb{R}\langle t, dt \rangle \rightarrow \mathcal{B} \otimes \mathbb{R}\langle t, dt \rangle$ by

$$\int_0^t b \otimes t^i = 0 \quad \text{and} \quad \int_0^t b \otimes t^i dt = (-1)^{\deg b} \frac{1}{i+1} b \otimes t^{i+1}.$$

These operators will allow us to discuss obstructions to extending homotopies, and serve as formal analogues of fiberwise integration (to be defined in [Chapter 4](#)). Before this discussion, however, we state the following facts which are easily proved from the given definitions.

Proposition 2.41. *These operators satisfy the conditions*

$$d \left(\int_0^1 u \right) + \int_0^1 du = u|_{\frac{t=1}{dt=0}} - u|_{\frac{t=0}{dt=0}} \tag{2.42}$$

$$d \left(\int_0^t u \right) + \int_0^t du = u - u|_{\frac{t=0}{dt=0}} \otimes 1. \tag{2.43}$$

Corollary 2.44. *If $\Phi : \mathcal{A} \rightarrow \mathcal{B} \otimes \mathbb{R} \langle t, dt \rangle$ is a homotopy from φ to ψ then*

$$d \left(\int_0^1 \Phi(a) \right) + \int_0^1 d\Phi(a) = \psi(a) - \varphi(a).$$

We now turn to algebraic obstruction theory. The point of this theory is to provide computable obstructions which determine whether a partial algebraic homotopy (a map which is a homotopy when restricted to a sub-DGA) may be extended to the entire DGA.

That is, suppose that $\varphi, \psi : \mathcal{A} \rightarrow \mathcal{B}$ are homotopic by a homotopy $\Phi : \mathcal{A} \rightarrow \mathcal{B} \otimes \mathbb{R} \langle t, dt \rangle$. Suppose moreover that $\iota : \mathcal{A} \hookrightarrow \tilde{\mathcal{A}}$, and that there are lifts $\tilde{\varphi}, \tilde{\psi} : \tilde{\mathcal{A}} \rightarrow \mathcal{B}$. When is there a homotopy $\tilde{\Phi} : \tilde{\mathcal{A}} \rightarrow \mathcal{B} \otimes \mathbb{R} \langle t, dt \rangle$ from $\tilde{\varphi}$ to $\tilde{\psi}$ for which $\tilde{\Phi} \circ \iota = \Phi$? There are of course a variety of refinements and generalizations of this question, some of which we address below. In general, however, this question is hard to answer unless we have a bit more structure on the inclusion $\iota : \mathcal{A} \hookrightarrow \tilde{\mathcal{A}}$.

To describe these obstructions, we need access to certain cohomology classes. Given a morphism $\varphi : \mathcal{A} \rightarrow \mathcal{B}$ we may define the **relative cohomology** of φ as the cohomology of the complex $(C^n(\varphi), d) = (\mathcal{A}^n \oplus \mathcal{B}^{n-1}, d)$ where $d(a, b) = (da, \varphi(a) - db)$. We denote this cohomology ring by any of the following means, sometimes with vector valued coefficients

$$H^*(\varphi : \mathcal{A} \rightarrow \mathcal{B}) = H^*(\varphi) = H^*(\mathcal{A} \rightarrow \mathcal{B}).$$

Moreover (see [32, §10.3]) there is an exact sequence

$$\cdots \rightarrow H^{n-1}(\mathcal{A}) \rightarrow H^{n-1}(\mathcal{B}) \rightarrow H^n(\varphi) \rightarrow H^n(\mathcal{A}) \rightarrow \cdots$$

Proposition 2.45 ([32, Proposition 11.1]). *Let $Z = Z_n$ and let $\iota : \mathcal{A} \rightarrow \mathcal{A} \langle Z \rangle$ be an elementary Sullivan extension of a DGA $\mathcal{A}$. Suppose we are given the following (non-commutative) DGA diagram*

$$\begin{array}{ccc}
\mathcal{A} & \xrightarrow{f} & \mathcal{B} \\
\downarrow \iota & & \downarrow \eta \\
\mathcal{A}\langle Z \rangle & \xrightarrow{g} & \mathcal{C}
\end{array}$$

as well as a homotopy Φ from $\eta \circ f$ to $g|_{\mathcal{A}} := g \circ \iota$.

Then the obstruction class $[\mathcal{O}] \in H^{n+1}(\eta; Z)$ to producing a map $\tilde{f} : \mathcal{A}\langle Z \rangle \rightarrow \mathcal{B}$ such that $\tilde{f} \circ \iota = f$, as well as a homotopy $\tilde{\Phi}$ from $\eta \circ \tilde{f}$ to g for which $\tilde{\Phi} \circ \iota = \Phi$, is represented by

$$\begin{aligned}
\mathcal{O} : Z &\rightarrow \mathcal{B}^{n+1} \oplus \mathcal{C}^n \\
z &\mapsto \left(f(dz), g(z) + \int_0^1 \Phi(dz) \right).
\end{aligned}$$

Moreover, in the event that the obstruction vanishes, there are maps $(b, c) : Z \rightarrow \mathcal{B}^n \oplus \mathcal{C}^{n-1}$ such that $d(b, c) = \mathcal{O}$, which means

$$\begin{aligned}
db(z) &= f(dz), \\
dc(z) &= \eta \circ b(z) - g(z) - \int_0^1 H(dz).
\end{aligned}$$

The map $\tilde{f}$ can then be defined by $\tilde{f}(z) = b(z)$ and the homotopy extension $\tilde{\Phi}$ can be defined by

$$\tilde{\Phi}(z) = g(z) + d(c(z) \otimes t) + \int_0^t \Phi(dz).$$

As a relative version of this, we have

Proposition 2.46 ([32, Proposition 11.2]). *Let $Z = Z_n$ and let $\iota : \mathcal{A} \rightarrow \mathcal{A}\langle Z \rangle$ be an elementary Sullivan extension of a DGA $\mathcal{A}$. Suppose we are given the following (non-commutative) DGA diagram*

$$\begin{array}{ccccc}
\mathcal{A} & \xrightarrow{f} & \mathcal{B} & \xrightarrow{\mu} & \mathcal{D} \\
\downarrow \iota & & \downarrow \eta & \nearrow \nu & \\
\mathcal{A}\langle Z \rangle & \xrightarrow{g} & \mathcal{C} & &
\end{array}$$

such that

(1) $g|_{\mathcal{A}} \simeq \eta f$ by a homotopy $\Phi : \mathcal{A} \rightarrow \mathcal{C} \otimes \mathbb{R}\langle t, dt \rangle$ with $\nu \circ \Phi$ constant

(2) μ is surjective

(3) the righthand triangle commutes exactly (i.e., $\nu \circ \eta = \mu$ exactly)

(4) the outer portion of the diagram commutes exactly (i.e., $\mu \circ f = \nu \circ g|_{\mathcal{A}}$)

Then the obstruction class $[\mathcal{O}] \in H^{n+1}(\eta; Z)$ to producing a map $\tilde{f} : \mathcal{A}\langle Z \rangle \rightarrow \mathcal{B}$ such that $\tilde{f} \circ \iota = f$, as well as a homotopy $\tilde{\Phi}$ from $\eta \circ \tilde{f}$ to g for which $\tilde{\Phi} \circ \iota = \Phi$, and for which $(\nu \circ \tilde{\Phi} = (\mu \circ \tilde{f}) \otimes 1)$ is constant, is represented by a map

$$\begin{aligned}
\mathcal{O} : Z &\rightarrow \mathcal{B}^{n+1} \oplus \mathcal{C}^n \\
z &\mapsto \left(f(dz), g(z) + \int_0^1 \Phi(dz) \right).
\end{aligned}$$

We will often make use of the following specific relative version of this fact:

Proposition 2.47. *Let $\mathcal{A}\langle Z \rangle$ be an elementary extension of a DGA $\mathcal{A}$. Suppose there are maps*

$$\mathcal{A}\langle Z \rangle \xrightarrow{\varphi, \psi} \mathcal{B} \xrightarrow{\mu} \mathcal{C}$$

with μ surjective, for which $\varphi|_{\mathcal{A}} \simeq \psi|_{\mathcal{A}}$ by a homotopy Φ , and such that $\mu \circ \varphi \simeq \mu \circ \psi$ by a homotopy $\chi : \mathcal{A}\langle Z \rangle \rightarrow \mathcal{C} \otimes \mathbb{R}\langle t, dt \rangle$ extending $\mu \circ \Phi$. The obstruction $[\mathcal{O}] \in H^n(\mu; Z)$ to extending Φ to a homotopy $\tilde{\Phi} : \mathcal{A} \rightarrow \mathcal{C} \otimes \mathbb{R}\langle t, dt \rangle$ which simultaneously lifts χ and

extends Φ is given by

$$\mathcal{O}(z) = \left(\psi(z) - \varphi(z) - \int_0^1 \Phi(dz); \chi(z) \right).$$

The above proposition is a corollary of a more general fact, the proofs of which can be found in [23, §3.1]. We now turn to the quantization of the above ideas. We fix X a finite piecewise Riemannian simplicial complex, and a principal $K(G, n)$ fibration $Y \rightarrow B$ in which G is abelian and $V = V_n = \text{Hom}(G, \mathbb{Q})$ is a finite dimensional vector space. As in Theorem 2.38, there is a minimal model $\mathcal{M}_Y^* := \mathcal{M}_B^* \otimes \wedge V \xrightarrow{m_Y} \Omega^*(Y)$ relative to the fibration.

We would like to talk about the size of a morphism $\varphi : \mathcal{M}_Y^* \rightarrow \Omega^*(X)$. We first equip $\Omega^*(X)$ with the L^∞ norm. Fix some norms on $\mathcal{M}_B^*$ and on $V = V_n$. Assume moreover that $n \leq \dim X$. Since our spaces are of finite type, any dependence on this choice of norm for V will be scaled only up to a constant. Assume that there is some fixed definition of $\text{Dil}(\varphi|_{\mathcal{M}_Y^*}) \in \mathbb{R}^{\geq 0}$. For instance, if $\mathcal{M}_B^* = \wedge W$ is a minimal model (say for a nilpotent space) we set

$$\text{Dil}(\varphi|_{\mathcal{M}_B^*}) := \max_{1 \leq k \leq \dim X} \left\{ \|\varphi|_{W_k}\|_{op}^{1/k} \right\}.$$

Having fixed a definition for $\text{Dil}(\varphi|_{\mathcal{M}_B^*})$ define the **dilatation** of φ to be

$$\text{Dil}(\varphi) := \max \left\{ \text{Dil}(\varphi|_{\mathcal{M}_B^*}), \|\varphi|_V\|_{op}^{1/n} \right\}.$$

For example, suppose that $\bar{f} : X \rightarrow B$ is L -Lipschitz. Then $\bar{f}^* : \Omega^k(B) \rightarrow \Omega^k(X)$ satisfies

$$\|\bar{f}^* \omega\|_\infty \leq L^k \|\omega\|_\infty.$$

Then under an appropriate definition we have $\text{Dil}(\bar{f}^* m_B) \leq CL$. Suppose further that $f : X \rightarrow Y$ is a lift of $\bar{f}$ for which $\|f^* m_Y|_V\|_{op}^{1/n} \leq L$. Since $(f^* m_Y)|_{\mathcal{M}_B^*} = \bar{f}^* m_B$, we have $\text{Dil}(f^* m_Y) \leq C'L$ as well, where C' depends only on m_B and the norms of $\Omega^*(B)$ and V . More specifically if $f : X \rightarrow Y$ is itself L -Lipschitz, then $\text{Dil}(f^* m_Y) \leq C'L$.

Given a DGA homotopy $\Phi : \mathcal{M}_Y^* \rightarrow \Omega^*(X) \otimes \mathbb{R}\langle t, dt \rangle$, we can define the dilatation once we fix a particular realization of the algebraic interval into its geometric counterpart. For each $T \in \mathbb{R}^{>0}$ we define a realization $\rho_T : \Omega^*(X) \otimes \mathbb{R}\langle t, dt \rangle \rightarrow \Omega^*(X \times I)$ by $t \mapsto t/T$, using T as a heuristic for the “formal length” of the formal interval. With this, we may define a family of dilatations for Φ by

$$\text{Dil}_T(\Phi) := \text{Dil}(\rho_T \Phi).$$

We now state a useful proposition about the quantitative bounds on extending homotopies between maps. Namely, if we have a partially-defined homotopy between φ and ψ , both of which have bounded dilatation, we would like to bound the dilatation of a homotopy between them. This quantizes the algebraic information from [Proposition 2.47](#).

Proposition 2.48 ([23, Proposition 3.9]). *Let $\varphi, \psi : \mathcal{M}_Y^* \rightarrow \Omega^*(X)$, and suppose there is a homotopy $\Phi : \mathcal{M}_B^* \rightarrow \Omega^*(X) \otimes \mathbb{R}\langle t, dt \rangle$ from $\varphi|_{\mathcal{M}_B^*}$ to $\psi|_{\mathcal{M}_B^*}$.*

- (1) *The obstruction to extending Φ to a homotopy $\tilde{\Phi} : \mathcal{M}_Y^* \rightarrow \Omega^*(X) \otimes \mathbb{R}\langle t, dt \rangle$ from φ to ψ is a class in $H^n(X; V)$ represented by a cochain σ for which (for any $\tau > 0$) we have*

$$\|\sigma\|_{op} \leq \tau C(n, d|_V) \text{Dil}_\tau(\Phi)^{n+1} + \text{Dil}(\varphi)^n + \text{Dil}(\psi)^n.$$

- (2) *In the event the obstruction vanishes, we can choose $\tilde{\Phi}$ so that*

$$\|\tilde{\Phi}_i^j|_V\|_{op} \leq (C_{IP} + 2) (\tau C(n, d|_V) \text{Dil}_\tau(\Phi)^{n+1} + \text{Dil}(\varphi)^n + \text{Dil}(\psi)^n)$$

where C_{IP} is the isoperimetric constant for $(n + 1)$ -forms in X , and $\tau > 0$ is arbitrary.

Furthermore, consider the relative situation (i.e., there is a subcomplex $A \subseteq X$ for which we have an existing homotopy $\chi : \mathcal{M}_Y^* \rightarrow \Omega^*(A) \otimes \mathbb{R} \langle t, dt \rangle$ between $\varphi|_A$ and $\psi|_A$). If the obstruction from [Proposition 2.47](#) vanishes, we can get similar bounds using a relative isoperimetric constant together with an additional $O(\tau \text{Dil}_\tau(\chi))$ term.

In [\[23, §3.2\]](#), it is shown that if $\text{Dil}(\varphi), \text{Dil}(\psi) \leq L$, and a homotopy Φ between them can be built formally up through degree n without encountering any nonzero obstructions, then as long as $\mathcal{M}_Y^*$ is simply connected with minimal model $\mathcal{M}_Y^* = \wedge V$ for $V = \{V_n\}_{n \geq 2}$, we can create more specific bounds on the dilatation of Φ . Namely, it is shown that for $j \in \{0, 1\}$ we have

$$\|\Phi_i^j|_{V_k}\|_{op} \leq CL^{2k-2}$$

for some constant $C = C(k, X, Y)$ depending additionally on the choice of norms for each V_n . We revisit this in [Chapter 5](#) to produce upper bounds on dilatation when Y is a nilpotent space.

Chapter 3

High Volume Nullhomotopies

The main goal of this chapter is to produce nilpotent spaces and Lipschitz maps into those spaces for which any nullhomotopy has large volume. We build a promising candidate space through constructing their algebraic duals, and then taking the realization of this model. Attempting to generalize the process of [9], our maps will be constructed through the help of higher order Whitehead products. This approach leads to some difficulty in verifying the existence of certain Lipschitz maps.

Analysis of this failure leads us to consider spaces and maps for which higher order Whitehead products do not play a role. Such spaces are called **coformal**. This leads us with a collection of examples which do not approach the expected general upper bounds of Theorem 5.3, and it remains open whether these upper bounds can be reduced, or whether spaces exist which approach these bounds. On the other hand, these examples produce bounds which are (nearly) sharp for the class of coformal spaces.

This chapter may be read as a cautionary tale that geometric complexity of a space is not *a priori* indicated by its algebraic complexity, and is organized as follows. We begin in Section 3.1 by first introducing our candidate space through its minimal DGA, and examine why the algebraic structure might naturally lead us to conclude that it admits

maps of high degree (and hence, after gluing in appropriate cells, nullhomotopies of high volume). In order to understand the geometric properties of the candidate maps into our space, we are naturally lead to consider higher order Whitehead products, which we do in [Section 3.2](#). The maps we construct in this way are meant to be L -Lipschitz, and it is natural to ask whether the maps we construct factor through scalable spaces, which we investigate in [Section 3.3](#). This investigation finally brings us to a point where the error of our ways is revealed, and in [Section 3.4](#) we uncover a topological obstruction to creating L -Lipschitz maps in the manner that we desired. We conclude in [Section 3.5](#) by defining coformal spaces and producing a generalization of the family of examples from [\[9, §7\]](#) for which the previous issues no longer present themselves, allowing us to prove [Theorem 3.39](#).

3.1 A Worthy Candidate

From now on, unless otherwise stated, $[\cdot, \cdot]$ will indicate the Whitehead bracket. Recall that $[\cdot, \cdot]: \pi_{j+1}(Y) \otimes \pi_{k+1}(Y) \rightarrow \pi_{j+k+1}(Y)$ is defined as follows. If f and g are representatives of α and β in $\pi_{j+1}(Y)$ and $\pi_{k+1}(Y)$ respectively, they define a map

$$f \vee g: \mathbf{S}^{j+1} \vee \mathbf{S}^{k+1} \rightarrow Y.$$

Furthermore, one has

$$\mathbf{S}^{j+1} \times \mathbf{S}^{k+1} \simeq (\mathbf{S}^{j+1} \vee \mathbf{S}^{k+1}) \cup_{\tau} \mathbf{D}^{j+k+2}$$

where

$$\tau: \mathbf{S}^{j+k+1} \rightarrow \mathbf{S}^{j+1} \vee \mathbf{S}^{k+1}$$

is the **attaching map**. Then the class $[f, g] \in \pi_{j+k+1}(Y)$ is the homotopy class of the composition

$$\mathbf{S}^{j+k+1} \xrightarrow{\tau} \mathbf{S}^{j+1} \vee \mathbf{S}^{k+1} \xrightarrow{f \vee g} Y.$$

The Whitehead bracket induces a graded Lie algebra structure (see [35]) on $\pi_{*+1}(Y)$.

Recall from [9, §7] that, for all $n \geq 2$ one constructs a simply connected space Y_n admitting $O(L)$ -Lipschitz maps with high-volume nullhomotopies as follows. Let $b = b_2$ be one of the generators of $\pi_2(\mathbf{S}^2 \vee \mathbf{S}^2) \cong \mathbb{Z}^2$. For any L , one can find $b_{n,L} = L^{2(n-1)}b \in \pi_n(\mathbf{S}^2 \vee \mathbf{S}^2)$ which has an $O(L)$ -Lipschitz representative (also denoted $b_{n,L}$). One defines Y_n to be the space formed by killing $\pi_n(\mathbf{S}^2 \vee \mathbf{S}^2)$ by attaching $(n+1)$ -cells, in which case $b_{n,L}$ then becomes nullhomotopic. In fact, any nullhomotopy of $b_{n,L}$ in Y_n must have degree $L^{2(n-1)}$ over one of these new $(n+1)$ -cells, hence will have volume $\Omega(L^{2(n-1)})$.

The homotopy Lie algebra $\pi_{*+1}(\mathbf{S}^2 \vee \mathbf{S}^2)$ is a free Lie algebra on two generators, making this space a natural candidate for attaining high-volume nullhomotopies. The Whitehead bracket is generated by two maps $a, b_2: \mathbf{S}^2 \rightarrow \mathbf{S}^2 \vee \mathbf{S}^2$ representing the identity maps on each sphere. In particular, there is an element $b_n \in \pi_n(\mathbf{S}^2 \vee \mathbf{S}^2)$ which decomposes inductively as $n-2$ iterated Whitehead brackets

$$b_n = [a, b_{n-1}] = [a, [a, b_{n-2}]] = \cdots = [a, [a, \cdots [a, b_2] \cdots]].$$

We attain an $O(L)$ -Lipschitz, high-degree map through a representative of

$$L^{2(n-1)}b_n = [L^2a, L^{2(n-2)}b_{n-1}] = [L^2a, [L^2a, \cdots [L^2a, L^2b_2] \cdots]].$$

Dually, from the perspective of minimal models, we see that the minimal model of $\mathbf{S}^2 \vee \mathbf{S}^2$ has generators α, β_2 in degree 2, and in degree $n > 2$ there is always a generator β_n for which $d\beta_n = \alpha \wedge \beta_{n-1}$. Heuristically, view each generator γ as the attaching map of a cell

Γ , and the differential $d\gamma$ as a product of cochains that prevent the attaching map from representing an element of homology. If these cochains can be made to have high degree relative to L -Lipschitz attaching maps, then one might hope to find an L -Lipschitz, high-degree representative of γ by use of these high-degree cochains. In [Chapter 5](#) we define the weight of an indecomposable ω to capture the increasing complexity of ω through its differential $d\omega$. Heuristically, ω has high weight if $d\omega$ contains a summand with an indecomposable factor ω' of high weight. The above ideas suggest that iterated applications of Whitehead products might be closely tied to the weight of an indecomposable—and indeed, they are related. We might then expect a Sullivan DGA with long “threads” of differentials to produce a nilpotent space admitting high-volume nullhomotopies.

The diagram below provides a visual understanding of the relationships we are interested in for the DGA of $\mathbf{S}^2 \vee \mathbf{S}^2$. We present this here, since we are looking for a generalization of this picture in the setting of nilpotent spaces. In this diagram, a harpoon $b_q \rightharpoonup b_{q+1}$ indicates that $db_{q+1} = a \wedge b_q$. Alternatively, viewing each b_q as a map $\mathbf{S}^q \rightarrow \mathbf{S}^2 \vee \mathbf{S}^2$, the notation $b_q \rightharpoonup b_{q+1}$ indicates that $b_{q+1} = [a, b_q]$. With this notation, we attain the sequence

$$b_2 \rightharpoonup b_3 \rightharpoonup b_4 \rightharpoonup \cdots \rightharpoonup b_q \rightharpoonup b_{q+1} \rightharpoonup \cdots$$

When we instead consider a map corresponding to L^2a and begin the chain with L^2b_2 , we get a sequence of $n - 1$ products of L^2 , all culminating in a map of degree $L^{2(n-1)}$ corresponding to $L^{2(n-1)}b_n$. We would like somehow to incorporate higher-step nilpotence into this picture.

With this in mind, fix n and c .

Definition 3.1. Throughout the rest of this chapter, let $\mathcal{N} = \mathcal{N}_c$ denote the c -step nilpotent space called the **candidate space** whose minimal model is generated in degree

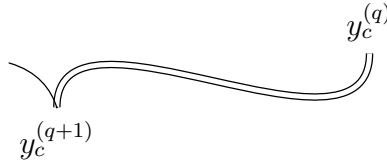
$1 \leq q \leq n$ by

$$\begin{aligned} \left\{ dt, x_1, \dots, x_c, y_1^{(1)}, \dots, y_c^{(1)} \right\} & \quad q = 1 \\ \left\{ y_1^{(q)}, \dots, y_c^{(q)} \right\} & \quad n \geq q > 1 \end{aligned}$$

with differential given by

$$\begin{aligned} d(dt) &= dx_1 = dy_1^{(1)} = 0 \\ dx_{j+1} &= dt \wedge x_j \\ dy_1^{(q)} &= dt \wedge x_c \wedge y_c^{(q-1)} \quad (q > 1) \\ dy_{j+1}^{(q)} &= dt \wedge y_j^{(q)}. \quad (q \geq 1) \end{aligned}$$

We produce a similar diagram to that above, providing a visual understanding of these differentials. Here, a harpoon $A_j \rightarrow A_{j+1}$ indicates $dA_{j+1} = dt \wedge A_j$ for some symbol A , and an arrow such as



indicates the fact that $dy_c^{(q+1)} = dt \wedge x_c \wedge y_c^{(q)}$. The differential then induces a chain as below:

$$(3.2)$$

When we replace dt with Ldt , and begin these chains with $Ly_1^{(1)}$ and Lx_1 , we see that the horizontal arrows correspond to multiplication by L , and one might hope the “thick” arrows correspond to multiplication by L^{c+1} . In the two-term case, this happens by duality with the usual Whitehead product, while in the 3-term case, this will happen by certain duality with higher order Whitehead products to be examined later. We hence hope we can attain a high-degree, L -Lipschitz map representing $L^{(2q-2)c+j}y_j^{(q)}$ through these “threads” of differentials. For example, when $j = c$ and $q = n$ we expect to achieve an L -Lipschitz map $\eta_{L,n,c}$ representing $L^{(2n-1)c}y_c^{(n)}$.

If one were to attach an $(n+1)$ -cell along the map $\eta_{L,n,c}$ dual to y_c^n , the map $\eta_{L,n,c}$ would be nullhomotopic via a nullhomotopy which must pass through this new $(n+1)$ -cell. By construction, this nullhomotopy would have to have volume $\Omega(L^{(2n-1)c})$ over this cell, by the same argument as in [9]. We now turn to investigation of this minimal

model, and its geometric realization.

3.2 Higher Order Whitehead Products

In order to investigate this example more thoroughly, we must understand how the structure of the DGA (in particular, the differential d itself) allows us to construct the maps we want. We focus our efforts on $\mathcal{N} = \mathcal{N}_1$, since the failure we experience here will be indicative of failure for $c > 1$. In the case of $\mathbf{S}^2 \vee \mathbf{S}^2$, the differential was quadratic, and hence the fact that $db_{n+1} = a \wedge b_n$ corresponds directly to the fact that $\beta_{q+1} = [a, \beta_q]$, where β_q is a fixed representative of b_q for each q . Given a Sullivan DGA $\mathcal{A} = \wedge V$ with a differential d , the quadratic part $d_1 : V \rightarrow V \wedge V$ of d preserves only the two-term summands of the differential on any indecomposable. Recall that $(\wedge V, d_1)$ is a Sullivan algebra in its own right, called the **associated quadratic** DGA. In this associated quadratic DGA, d_1 reflects the Whitehead product of the geometric realization of $\wedge V$. That is,

Theorem 3.3 ([36, Theorem B.(iii)]. See also [28, Prop. 13.16]). *The standard 2-term Whitehead products on homotopy for a rational space are described by d_1 . That is, for a rational space X the Whitehead product on $\pi_{*+1}(X)$ is dual under the Sullivan pairing (see below) to d_1 in the associated quadratic DGA of $\mathcal{M}_X^*$.*

Definition 3.4. A space for which $d = d_1$ is called **coformal** (see [37, Proposition 3.3]).

It is shown in [38, Theorem 5.4] that a similar (though more restrictive result) applies when we remove the quadratic restriction on d , allowing one to infer genuine topological information from certain r -factor summands of differentials in $\mathcal{M}_X^*$. This requires the use of **higher order Whitehead products**. We do not provide here the technical definition of these Whitehead products, instead referring the interested reader to [38]

where the definitions are clearly and carefully explained. Instead, we focus here on the relevant qualities of the product. For this section, fix X a rational space with minimal model $\wedge V$. (It will not hurt to think of $X = \mathcal{N}_c$.) In particular, the indecomposables of $\mathcal{M}_X^*$ are elements of $V = \{V_n\}_{n \in \mathbb{N}}$.

We borrow the following notation from [38]:

Definition 3.5. The **Sullivan pairing** is the natural pairing $\langle\langle \cdot, \cdot \rangle\rangle : V \otimes \pi_{*+1}(X) \rightarrow \mathbb{Q}$.

Given homotopy elements $\{\beta_i\} \subseteq \pi_*(X)$ and indecomposables $\{b_i\} \subseteq V$ for which $V = \text{span}\{b_i\}$, we will say that these collections are **dual** to one another provided $\langle\langle a_i, \alpha_j \rangle\rangle = \delta_{ij}$ where δ is the Kronecker delta. Moreover, given such dual collections, if $\deg(b_i) \leq \deg(b_{i+1})$ (hence, $\deg(\beta_i) \leq \deg(\beta_{i+1})$) for all i , we call the collections **monotone increasing in degree**.

Note that under the identification

$$\mathbf{S}^{j+1} \times \mathbf{S}^{k+1} \simeq (\mathbf{S}^{j+1} \vee \mathbf{S}^{k+1}) \cup_{\tau} \mathbf{D}^{j+k+2}$$

we may view the wedge $\mathbf{S}^{j+1} \vee \mathbf{S}^{k+1}$ as $\mathbf{S}^{j+1} \times \mathbf{S}^{k+1}$ where the top $(j+k+2)$ cell has been removed. This is part of our motivation for the following definition.

Definition 3.6. Let $X_1, \dots, X_r$ be a sequence of spaces with basepoints $x_1, \dots, x_r$ respectively. For a point $p = (a_1, \dots, a_r) \in X_1 \times \dots \times X_r$, let $\mathcal{B}(p) := \#\{i \mid a_i \neq x_i\}$ be the number of coordinates which do not correspond to basepoints. The **fat wedge** is defined by

$$\mathbb{V}(X_1, \dots, X_r) := \left\{ p \in \prod_{i=1}^r X_i \mid \mathcal{B}(p) \neq r \right\}.$$

In other words, this is the subcomplex of $X_1 \times \dots \times X_r$ based at $(x_1, \dots, x_r)$ for which at least one coordinate must keep track of the basepoint.

Notation. Let A_i be some spaces for $i = k + 1, \dots, r$. We will often write

$$\mathbb{V}(n_1, \dots, n_k, A_{k+1}, \dots, A_r) := \mathbb{V}(\mathbf{S}^{n_1}, \dots, \mathbf{S}^{n_k}, A_{k+1}, \dots, A_r)$$

Example 3.7. In general, $\mathbb{V}(n_1, \dots, n_r)$ is a copy of $\prod_{i=1}^r \mathbf{S}^{n_i}$ whose top $(\sum_{i=1}^r n_i)$ -cell has been removed. For example, $\mathbb{V}(1, 1, 1) = (\mathbb{T}^3)^{(2)}$ is the 2-skeleton of the 3-torus. $\blacktriangle$

Note that if $n_i > 1$ for all i , then $\mathbb{V}(n_1, \dots, n_r)$ is in fact simply connected. On the other hand, since wedges of nilpotent spaces are rarely nilpotent, we in general will not expect fat wedges of nilpotent spaces to be nilpotent.

Example 3.8. $\mathbf{S}^1 \vee \mathbf{S}^1$ is not nilpotent, since the fundamental group is the free group on two generators. Similarly, $\mathbb{V}(1, 1, 1)$ is not nilpotent. From an algebraic perspective, these claims are fairly obvious, but we would like to understand them a bit more geometrically. Note that $\mathbb{V}(1, 1, 1)$ is comprised of three 2-tori, glued along certain copies of $\mathbf{S}^1$. We refer to these as the (xy) -torus, the (xz) -torus, and the (yz) -torus. A cover of $\mathbb{V}(1, 1, 1)$ is given by taking all integer shifts of the (xy) -, (xz) -, and (yz) -planes in $\mathbb{R}^3$, each plane generated by the corresponding subtorus of $\mathbb{V}(1, 1, 1)$. This divides $\mathbb{R}^3$ into hollow cubes, each of which are distinct elements of $\pi_2 \cong \mathbb{Z}^\infty$, but related via the action of π_1 . For example, the cubes $Q_{(0,0,0)}$ and $Q_{(1,0,0)}$ centered at $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ and $(\frac{3}{2}, \frac{1}{2}, \frac{1}{2})$ respectively, share a common (yz) -face. Basing this cover at $(0, 0, 0)$, there is an obvious (based) map $\mathbf{S}^2 \rightarrow Q_{(0,0,0)}$ and a corresponding map $\mathbf{S}^2 \rightarrow Q_{(1,0,0)}$ which becomes based at $(0, 0, 0)$ after attaching a tail along the edge from $(0, 0, 0)$ to $(1, 0, 0)$. Each of these maps generate a nontrivial element of π_2 which cannot be identified, because of the hole centered at $(\frac{3}{2}, \frac{1}{2}, \frac{1}{2})$.

However, one might remedy this by gluing cells to kill higher homotopy so that these cubes become identified, and so that there is only one generator of π_2 as a π_1 -module. In fact, if we preserve one generator of rational homotopy we kill the fundamental group action in this way, producing a simple space, and for fixed n , we may then produce a

space whose minimal model is given by

$$\langle dt, dx, dy, y^{(2)}, y^{(3)}, \dots, y^{(n)} \mid dy^{(q+1)} = dt \wedge dx \wedge y^{(q)} \rangle \quad (3.9)$$

where we identify $dy = y^{(1)}$. Compare this with [Definition 3.1](#) to see that (3.9) describes the space candidate space $\mathcal{N}_1$ for $c = 1$. ▲

With this generalization of wedges at our disposal, we now generalize Whitehead products. Note that in what follows, we will often conflate a specific map with its homotopy class for convenience and ease of notation. Though the following is stated as a proposition, we will essentially treat it as the definition of the r th order Whitehead product. We also note that we will actually be concerned with maps from the localization of spheres (rather than genuine spheres), which we do without further mention.

Proposition 3.10. *Let $n_1, \dots, n_r \in \mathbb{N} \setminus \{0\}$, and let $N := n_1 + \dots + n_r$. The **Whitehead bracket of order r** is a map*

$$\pi_{n_1}(X) \times \dots \times \pi_{n_r}(X) \rightarrow \mathcal{P}(\pi_{N-1}(X))$$

where $\mathcal{P}(A)$ denotes the power set of A . An element $x \in [f_1, \dots, f_r]$ is such that a representative of x factors through

$$\mathbf{S}^{N-1} \xrightarrow{\tau_N} \mathbb{V}(n_1, \dots, n_r) \xrightarrow{\tilde{x}} X$$

where $\tilde{x}|_{\mathbf{S}^{n_i}} = f_i$ and where τ is a fixed attaching map.

We write $\tau = \tau_N$ as the attaching map, though in actuality this map depends on $(n_1, \dots, n_r)$ rather than N itself.

In general, higher order Whitehead products are *sets* of homotopy classes, rather

than individual homotopy classes. In particular, this set may be empty or have multiple elements.

Example 3.11. Let $\iota : \mathbf{S}^2 \rightarrow \mathbf{S}^2$ be the identity. Consider $0 \vee \iota \vee \iota : \mathbf{S}^k \vee \mathbf{S}^2 \vee \mathbf{S}^2 \rightarrow \mathbf{S}^2$. Then $[0, \iota, \iota] = \emptyset$. Indeed, the fact that this is empty is equivalent to saying that there are no maps $G : \mathbf{S}^2 \times \mathbf{S}^2 \rightarrow \mathbf{S}^2$ for which the composition

$$\mathbf{S}^2 \vee \mathbf{S}^2 \hookrightarrow \mathbf{S}^2 \times \mathbf{S}^2 \rightarrow \mathbf{S}^2$$

is homotopically trivial on each component simultaneously. One way to see that no such G exists is that if γ is the fundamental class of the target space $\mathbf{S}^2$, and α, β are the fundamental classes of each copy of $\mathbf{S}^2$ in the wedge/product, then $0 = \gamma \smile \gamma \xrightarrow{(\iota \vee \iota)^*} \alpha \smile \beta = 0$ factors through

$$\gamma \smile \gamma \xrightarrow{G^*} \alpha \smile \beta = [\mathbf{S}^2 \times \mathbf{S}^2] \neq 0,$$

since $\iota^* : \alpha \mapsto \alpha$ and similarly for β . ▲

See [39, Proposition 4] for an example of a 3rd order Whitehead product with multiple elements. Note, though, that their example is concerned with Whitehead products from genuine spheres rather than their rationalizations.

Although higher order Whitehead products are generally difficult to compute, certain methods make computations possible. For instance, the following result provides a method for detecting when the higher order Whitehead bracket is nonempty.

Theorem 3.12 ([40, Thm. 2.7] and [39, p.8]). *For any homotopy elements $f_i \in \pi_{n_i}(X)$, if $0 \in [f_{i_1}, \dots, f_{i_k}]$ for all $1 \leq i_1 \leq \dots \leq i_k \leq r$ with $k \leq r - 1$ then $[f_1, \dots, f_r] \neq \emptyset$.*

Moreover, in the event that the product is nonempty, there is a certain duality with the differential in the minimal model. This is explored in detail in [38], and we present

the relevant information for our purposes below. For X a rational space, fix a set of indecomposables $\{\eta_1, \eta_2, \dots\}$ of $\mathcal{M}_X^*$, and define $\mathcal{F}^r(\mathcal{M}_X^*)$ to be the graded vector space generated by $\{\eta_{i_1}\eta_{i_2}\cdots\eta_{i_s} \mid s \geq r\}$.

Theorem 3.13 ([38, Corollary 6.5]). *If $d\mu \in \mathcal{F}^r$ for all μ , then all r th order Whitehead brackets exist and consist of a single element.*

Example 3.14. Let $\mathcal{N}_1$ be the space whose minimal model is given by (3.9). Then all 3rd order Whitehead brackets are uniquely determined and nonempty. A representative $\eta_2 = \tilde{\eta}_2 \circ \tau_2$ of $y^{(2)}$ must make the following diagram commute for fixed representatives T , ξ , and η of dt , dx , and dy respectively:

$$\begin{array}{ccc} \mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbf{S}^1 & \xrightarrow{T \vee \xi \vee \eta} & \mathcal{N}_1 \\ \downarrow & \nearrow \tilde{\eta}_2 & \\ \mathbf{S}^2 & \xrightarrow{\tau_2} & \mathbb{V}(1, 1, 1) \end{array}$$

Hence, a representative of $y^{(3)}$ is a map $\eta_3 = \tilde{\eta}_3 \circ \tau_3$ which factors in such a way that the following diagram commutes

$$\begin{array}{ccc} \mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbf{S}^2 & \xrightarrow{1 \vee 1 \vee \tau_2} & \mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbb{V}(1, 1, 1) \xrightarrow{T \vee \xi \vee \tilde{\eta}_2} \mathcal{N}_1 \\ \downarrow & & \nearrow \tilde{\eta}_3 \\ \mathbf{S}^3 & \xrightarrow{\tau_3} & \mathbb{V}(1, 1, 2) \end{array} \quad (3.15)$$

Inductively, interpreting $y^{(1)} = dy$, we have $\eta_q = \tilde{\eta}_q \circ \tau_q$ representing $y^{(q)} = [T, \xi, \eta_{q-1}]$ as an extension of a map that factors through $\mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbb{V}(1, 1, q-1)$. That is, the following diagram commutes for all q :

$$\begin{array}{ccc} \mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbf{S}^{q-1} & \xrightarrow{1 \vee 1 \vee \tau_{q-1}} & \mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbb{V}(1, 1, q-2) \xrightarrow{T \vee \xi \vee \tilde{\eta}_{q-1}} \mathcal{N}_1 \\ \downarrow & & \nearrow \tilde{\eta}_q \\ \mathbf{S}^q & \xrightarrow{\tau_q} & \mathbb{V}(1, 1, q-1) \end{array} \quad (3.16) \quad \blacktriangle$$

The hope is that repeated iteration of these higher order Whitehead products will allow us to find $O(L)$ -Lipschitz maps of high degree. In order to do this, we must have some notion of linearity, or at least homogeneity. In usual Whitehead products, this is a standard fact required for making π_{*+1} into a Lie algebra under the Whitehead bracket. Indeed, higher order Whitehead products enjoy certain results analogous to those of an alternating multilinear mapping. The following ensures, for example, that we have homogeneity in some sense. We encourage the reader to see [39, §2] for more details.

Proposition 3.17 ([39, Corollary 2.10]). *Let $f_i: \mathbf{S}^{n_i} \rightarrow X$ for $i = 1, \dots, r$, and let $L \in \mathbb{N}$. Then*

$$[f_1, \dots, Lf_i, \dots, f_r] \subseteq L[f_1, \dots, f_i, \dots, f_r].$$

Example 3.18. Let $\mathcal{N} = \mathcal{N}_1$ as before. Fix T_L , ξ_L , and η_L as L -Lipschitz maps $\mathbf{S}^1 \rightarrow \mathcal{N}$ representing Ldt , Ldx , and Ldy respectively. Such representatives may be found, for example, by precomposition with a self-map $\mathbf{S}^1 \xrightarrow{\times L} \mathbf{S}^1$. Because 3rd order Whitehead products are unique and nonempty in this space, we inductively see that

$$L^{2q+1}y^{(q+1)} = L^{2q+1}[T, dx, y^{(q)}] = [LT, Ldx, L^{2q-1}y^{(q)}].$$

That is, a representative $\eta_{L,q}$ of $[T_L, \xi_L, \eta_{L,q-1}]$ is a representative $L^{2q-1}y^{(q)}$ for all q . $\blacktriangle$

We see that as long as the differential really does correspond to a way for producing a genuine map in the first place, this makes precise the notion that our high-degree “threads” of differentials from Section 3.1 produces genuine high-degree maps. It remains to see whether we can ensure that these maps (a well-selected representative $\eta_{L,q}$ in the above example) can be made L -Lipschitz.

3.3 Scalable Spaces

We continue to analyze the situation $\mathcal{N} = \mathcal{N}_1$. Consider the diagram (3.16) where we attain a representative $\eta_q = \tilde{\eta}_q \circ \tau_q$ of $y^{(q)}$ for each q such that, inductively, $\tilde{\eta}_q$ extends the composition

$$\mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbf{S}^{q-1} \xrightarrow{1 \vee 1 \vee \tau_{q-1}} \mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbb{V}(1, 1, q-2) \xrightarrow{T \vee \xi \vee \tilde{\eta}_{q-1}} \mathcal{N}_1.$$

We seek L -Lipschitz representatives of $L^{2q-1}y^{(q)} = [Ldt, Ldx, L^{2(q-1)-1}y^{(q-1)}]$ for each q . First, fix T_L , ξ_L , and $\eta_{L,1}$ as L -Lipschitz representatives of Ldt , Ldx and Ldy respectively by precomposition of the relevant maps with a self-map $\mathbf{S}^1 \xrightarrow{\times L} \mathbf{S}^1$.

Proposition 3.19. *There is an $O(L)$ -Lipschitz representative of $L^3y^{(2)}$.*

Proof. A representative of $L^3y^{(2)}$ is a representative of

$$L^3y^{(2)} = [Ldt, Ldx, Ldy] = [T_L, \xi_L, \eta_{L,1}]$$

and hence any map $\tilde{\eta}_{L,2}$ which fits into the following diagram will produce the desired map:

$$\begin{array}{ccccc} \mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbf{S}^1 & \xrightarrow{\times L} & \mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbf{S}^1 & \xrightarrow{T \vee \xi \vee \eta_1} & \mathcal{N}_1 \\ \downarrow & & & \nearrow \tilde{\eta}_{L,2} & \\ \mathbf{S}^3 & \xrightarrow{\tau_2} & \mathbb{V}(1, 1, 1) & & \end{array}$$

Here the map $\times L$ is multiplication by L on each component. Indeed, viewing $\mathbb{V}(1, 1, 1)$ as a subcomplex of $\mathbb{T}^3$, there is an L -Lipschitz self-map $\sigma_L: \mathbb{T}^3 \rightarrow \mathbb{T}^3$ such that $(\sigma_L)|_{\mathbf{S}^1}$ is multiplication by L on each $\mathbf{S}^1$ factor. In particular, $(\sigma_L)|_{\mathbb{V}(1,1,1)}$ is a self-map of $\mathbb{V}(1, 1, 1)$.

Define $\tilde{\eta}_{L,2} := \tilde{\eta}_2 \circ (\sigma_L)|_{\mathbb{V}(1,1,1)}$ which makes the following diagram commute:

$$\begin{array}{ccccc}
\mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbf{S}^1 & \xrightarrow{\times L} & \mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbf{S}^1 & \xrightarrow{T\vee\xi\vee\eta_1} & \mathcal{N}_1 \\
\downarrow & & \downarrow & \nearrow \tilde{\eta}_2 & \\
\mathbf{S}^3 & \xrightarrow{\tau_2} & \mathbb{V}(1, 1, 1) & \xrightarrow{(\sigma_L)|_{\mathbb{V}(1, 1, 1)}} & \mathbb{V}(1, 1, 1)
\end{array}$$

Then $\eta_{L,2} := \tilde{\eta}_{L,2} \circ \tau_2$ is a representative of $L^3 y^{(2)}$. Clearly we have

$$\text{Lip}(\eta_{L,2}) \leq \text{Lip}(\tau_2) \cdot \text{Lip}(\sigma_L) \cdot \text{Lip}(\tilde{\eta}_2) = O(1) \cdot O(L) \cdot O(1) = O(L). \quad \square$$

Ideally, this construction is one which we would like to generalize to $c > 1$ and $n > 1$. This high-degree, L -Lipschitz self-map of $\mathbb{V}(1, 1, 1)$, is a consequence of the fact that $\mathbb{V}(1, 1, 1)$ is an example of a scalable space. We take the following theorem statement as our (relevant) definition of scalability. The theorem itself actually provides various equivalent perspectives on scalability, but in our current context this will be the most helpful.

Theorem 3.20 ([25, Theorem A]). *A space Y is **scalable** if and only if there is a constant $C(Y)$ and infinitely many $p \in \mathbb{N}$ such that there is a $C(Y)(p+1)$ -Lipschitz self-map which induces multiplication by p^n on $H^n(Y; \mathbb{R})$. Furthermore, if Y is scalable then for all finite simplicial complexes X , nullhomotopic L -Lipschitz maps $X \rightarrow Y$ have $C(X, Y)(L+1)$ -Lipschitz nullhomotopies.*

Theorem 3.21 ([25, Theorem B]). *The following are properties of scalable spaces:*

- (1) *Scalability is invariant under rational homotopy equivalence.*
- (2) *The class of scalable spaces is closed under products and wedge sums.*
- (3) *All skeleta of simply connected, scalable complexes are scalable.*

Proposition 3.22. *Suppose Y is a scalable N -manifold (or a scalable space satisfying Poincaré Duality) and suppose that X is a space for which there is a Lipschitz inclusion of a subcomplex $f: X \rightarrow Y$ which induces an injection on homology. Then X is scalable.*

Proof. A $C(Y)(p+1)$ -Lipschitz self-map σ of Y induces multiplication by p^n on $H^n(Y; \mathbb{R})$, hence (by Poincaré duality) on $H_{N-n}(Y; \mathbb{R})$. This induces multiplication by p^n on $H_{N-n}(X)$ for all n , and thus the restriction of σ to X is a self-map with the desired properties. $\square$

Example 3.23. A finite product of spheres $\prod_{i=1}^r \mathbf{S}^{n_i}$ is scalable, and hence so is $\mathbb{V}(n_1, \dots, n_r)$. In particular, $\mathbb{T}^r$ and $\mathbb{V}(1, 1, \dots, 1)$ are scalable. $\blacktriangle$

This exhibits a slight difference in our definition of scalability compared to the original of [25]. In particular, observe that $\mathbb{V}(1, 1, 1)$ is not nilpotent, but is still scalable under this definition.

Proposition 3.24. Suppose there are maps and spaces for which the following diagram commutes, where the map along the top row is a factoring of $T \vee \xi \vee \eta_{q-1}$.

$$\begin{array}{ccccc}
 \mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbf{S}^{q-1} & \longrightarrow & \mathcal{S}_V & \longrightarrow & \mathcal{N}_1 \\
 \downarrow & & \downarrow & \nearrow & \\
 \mathbf{S}^q \xrightarrow{\tau_q} \mathbb{V}(1, 1, q-1) & \longrightarrow & \mathcal{S} & &
 \end{array}$$

Let L be given, and suppose s_L is an L -Lipschitz self-map of $\mathcal{S}$ which restricts to a self-map $s_{L,V}$ of $\mathcal{S}_V$. Suppose moreover that there are L -Lipschitz representatives T_L , ξ_L , and $\eta_{L,q-1}$ of $L_1 dt$, $L_2 dx$, and $L_3 y^{(q-1)}$ respectively (for some L_1, L_2, L_3) such that $T_L \vee \xi_L \vee \eta_{L,q-1}$ factors as

$$\mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbf{S}^{q-1} \longrightarrow \mathcal{S}_V \xrightarrow{s_{L,V}} \mathcal{S}_V \longrightarrow \mathcal{N}_1$$

Then there is a representative of $\tilde{L}y^{(q)}$ (with $\tilde{L} = L_1 L_2 L_3$) which factors through

$$\mathbf{S}^q \xrightarrow{\tau_q} \mathbb{V}(1, 1, q-1) \longrightarrow \mathcal{S} \xrightarrow{s_L} \mathcal{S} \longrightarrow \mathcal{N}_1$$

Proof. If such maps and spaces exist, then because $s_{L,V} = (s_L)|_{\mathcal{S}_V}$ the following diagram commutes, where the top row is the given factoring of $T_L \vee \xi_L \vee \eta_{L,q-1}$:

$$\begin{array}{ccccccc}
\mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbf{S}^{q-1} & \longrightarrow & \mathcal{S}_V & \xrightarrow{s_L \vee} & \mathcal{S}_V & \longrightarrow & \mathcal{N}_1 \\
\downarrow & & \downarrow & & \downarrow & & \nearrow \\
\mathbf{S}^q \xrightarrow{\tau_q} \mathbb{V}(1, 1, q-1) & \longrightarrow & \mathcal{S} & \xrightarrow{s_L} & \mathcal{S} & &
\end{array}$$

Hence by definition of higher order Whitehead products and [Proposition 3.17](#), the bottom row is a representative of

$$[T_L, \xi_L, \eta_{L,q-1}] = L_1 L_2 L_3 [dt, dx, y^{(q-1)}] = \tilde{L}y^{(q)}.$$

□

Example 3.25. [Proposition 3.19](#) is a consequence of [Proposition 3.24](#) using $\mathcal{S} = (\mathbb{T}^3)^{(2)}$ and $\mathcal{S}_V = \mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbf{S}^1$. ▲

There are three main ways we might expect to attain the desired pair of spaces $(\mathcal{S}, \mathcal{S}_V)$. First, we see that the space $\mathbb{V}(1, 1, q-1)$ is itself scalable, with a natural self-map

$$s_L: \mathbb{V}(1, 1, q-1) \rightarrow \mathbb{V}(1, 1, q-1)$$

which restricts to a self-map on $\mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbf{S}^{q-1}$. However, even for $q = 3$ there are no L -Lipschitz self-maps which will extend the desired map for our maps into $\mathcal{N}_1$, since such a self-map does not have the right degree. In particular, the map $T_L \vee \xi_L \vee \eta_{L,2}$ does not factor through the self-map on $\mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbf{S}^2$ as would be desired.

Proposition 3.26. *Let T_L , ξ_L , and $\eta_{L,2}$ be as in [Proposition 3.19](#). There is **no** Lipschitz map $\bar{\eta}_3: \mathbb{V}(1, 1, 2) \rightarrow \mathcal{N}_1$ for which following diagram commutes:*

$$\begin{array}{ccccc}
\mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbf{S}^2 & \xrightarrow{T_L \vee \xi_L \vee \eta_{L,2}} & \mathcal{N}_1 \\
\downarrow & & \nearrow \bar{\eta}_3 \\
\mathbf{S}^3 \xrightarrow{\tau_2} \mathbb{V}(1, 1, 2) & \xrightarrow{s_L} & \mathbb{V}(1, 1, 2)
\end{array}$$

Proof. The map s_L is a map of degree $L \cdot L \cdot L^2 = L^4$. As τ_3 and $\bar{\eta}_3$ are meant to be independent of L , the composition $\bar{\eta}_3 \circ s_L \circ \tau_2$ is a map of degree L^4 . But any such map

fitting into this diagram must represent $[T_L, \xi_L, \eta_{L,2}] = [Ldt, Ldx, L^3y^{(2)}] = L^5y^{(2)}$ whose degree is L^5 . $\square$

A second possibility is that $\mathcal{N}_1$ itself is scalable. Recall, however, that we constructed $\mathcal{N}_1$ by gluing in tubes in the universal cover of $\mathbb{V}(1, 1, 1)$ to identify generators of $\pi_2(\mathbb{V}(1, 1, 1))$. Rescaling this space by a factor of L induces a self-map on $H^2(\mathcal{N}_1; \mathbb{R})$ which is multiplication by L^3 rather than L^2 as would be required to be scalable. Although this hardly constitutes a proof of this fact, it gives some topological insight into the fact that $\mathcal{N}_1$ is not even formal, let alone scalable.

Thus, neither $\mathbb{V}(1, 1, q-1)$ nor $\mathcal{N}_1$ will work for the purposes of applying [Proposition 3.24](#) more broadly. We thus conduct a search for a third possibility for a scalable space through which our maps might factor. Indeed, a space $\mathbb{V}_q$ presents itself rather naturally, defined inductively by

$$\mathbb{V}_q := \begin{cases} \mathbf{S}^1 & q = 1, \\ \mathbb{V}(1, 1, \mathbb{V}_{q-1}) & q > 1. \end{cases}$$

Furthermore, inductively define $\tau_{\mathbb{V},q}: \mathbf{S}^q \rightarrow \mathbb{V}_q$ by the composition

$$\mathbf{S}^q \xrightarrow{\tau_q} \mathbb{V}(1, 1, q-1) \xrightarrow{\mathbb{V}(1, 1, \tau_{\mathbb{V},q-1})} \mathbb{V}_q.$$

Here the map $\mathbb{V}(1, 1, \tau_{\mathbb{V},q-1})$ is defined as the restriction of the obvious corresponding map $\mathbf{S}^1 \times \mathbf{S}^1 \times \mathbf{S}^{q-1} \rightarrow \mathbf{S}^1 \times \mathbf{S}^1 \times \mathbb{V}_{q-1}$.

Proposition 3.27. *The space $\mathbb{V}_q$ is scalable. Moreover, there is an L -Lipschitz self-map $s_L = s_{q,L}$ which has degree L^{2q-1} , and such that the following diagram commutes:*

$$\begin{array}{ccccc}
\mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbf{S}^{q-1} & \xrightarrow{\mathbb{1} \vee \mathbb{1} \vee \tau_{V,q-1}} & \mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbb{V}_{q-1} & \xrightarrow{(\times L) \vee (\times L) \vee s_{q-1,L}} & \mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbb{V}_{q-1} \\
\downarrow & & \downarrow & & \downarrow \\
\mathbb{V}(1, 1, q-1) & \xrightarrow{\mathbb{V}(\mathbb{1}, \mathbb{1}, \tau_{V,q})} & \mathbb{V}(1, 1, \mathbb{V}_{q-1}) = \mathbb{V}_q & \xrightarrow{s_{q,L}} & \mathbb{V}_q
\end{array}$$

Proof. Inductively one sees that $\mathbb{V}_q$ is a subcomplex of $\mathbb{T}^{2q-1}$, the inclusion of which is an injection on homology. Hence by [Proposition 3.22](#), $\mathbb{V}_q$ is scalable, and an L -Lipschitz self-map of $\mathbb{T}^{2q-1}$ induces a self-map $\mathbb{V}_q \rightarrow \mathbb{V}_q$ of degree L^{2q-1} , which we call $s_{q,L}$. On each $\mathbf{S}^1$ of $\mathbb{T}^{2q-1}$ this map is simply multiplication by L , and hence the second square commutes by induction. The commutativity of the first square of the diagram is a consequence of the definition of $\mathbb{V}(\mathbb{1}, \mathbb{1}, \tau_{V,q-1})$. $\square$

3.4 An Obstruction to Construction

[Proposition 3.27](#) appears to give us exactly what we need for success in [Proposition 3.24](#). Alas we encounter a geometric obstruction to actually producing the map $\mathbb{V}_q \rightarrow \mathcal{N}_1$. We have successfully found an L -Lipschitz $\eta_{L,2}$ representative of $L^3 y^{(2)}$ factoring as

$$\mathbf{S}^2 \longrightarrow \mathbb{V}_2 \xrightarrow{s_{2,L}} \mathbb{V}_2 \longrightarrow \mathcal{N}_1$$

We would like to apply [Proposition 3.24](#) to $q = 3$ with $\mathcal{S} = \mathbb{V}_3$ and $\mathcal{S}_V = \mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbb{V}_2$. As a consequence of [Proposition 3.27](#) we have already established the commutativity of the following diagram, in which the top row is a representative of $\eta_{L,2}$.

$$\begin{array}{ccccccc}
\mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbf{S}^2 & \longrightarrow & \mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbb{V}_2 & \xrightarrow{(\times L) \vee (\times L) \vee s_{2,L}} & \mathbf{S}^1 \vee \mathbf{S}^1 \vee \mathbb{V}_2 & \xrightarrow{T \vee \xi \vee \tilde{\eta}_2} & \mathcal{N}_1 \\
\downarrow & & \downarrow & & \downarrow & \nearrow \exists? & \\
\mathbf{S}^q & \xrightarrow{\tau_3} & \mathbb{V}(1, 1, 2) & \longrightarrow & \mathbb{V}_3 & \xrightarrow{s_L} & \mathbb{V}_3
\end{array}$$

It remains to see whether the dashed arrow exists in order to complete the diagram.

Proposition 3.28. *The obstruction to creating an extension to the following diagram is nontrivial:*

$$\begin{array}{ccc} \mathbf{S}_t^1 \vee \mathbf{S}_x^1 \vee \mathbb{V}_2 & \xrightarrow{T \vee \xi \vee \tilde{\eta}_2} & \mathcal{N}_1 \\ \downarrow & \nearrow \text{dashed} & \\ \mathbb{V}_3 & & \end{array}$$

Proof. Here we further label our $\mathbf{S}^1$ components to help distinguish them. Write

$$\mathbf{S}_t^1 \vee \mathbf{S}_x^1 \vee \mathbb{V}_2 = \mathbf{S}_{t_2}^1 \vee \mathbf{S}_{x_2}^1 \vee \mathbb{V}(\mathbf{S}_{t_1}^1 \vee \mathbf{S}_{x_1}^1 \vee \mathbf{S}_y^1).$$

The map $T \vee \xi \vee \tilde{\eta}_2$ restricts to T on $\mathbf{S}_{t_1}^1$ and $\mathbf{S}_{t_2}^1$, restricts to ξ on $\mathbf{S}_{x_1}^1$ and $\mathbf{S}_{x_2}^1$, and restricts to η_1 on $\mathbf{S}_y^1$. View $\mathbb{V}_3$ as a subcomplex of $\mathbb{T}^5$. The k -cells of $\mathbb{V}_3$ may therefore be defined according to the 1-skeleton. In particular, there is a 3-torus $z = \mathbf{S}_{t_1}^1 \times \mathbf{S}_{x_2}^1 \times \mathbf{S}_y^1 \subseteq \mathbb{V}_3$. The obstruction to extending $T \vee \xi \vee \tilde{\eta}_2$ over the top cell of z is the class of $H^3(\mathcal{N}_1; \pi_2(\mathcal{N}_1))$ representing $\eta_2 \in \pi_2(\mathcal{N}_1)$. That is, such an extension to z exists if and only if $[T, \xi, \eta_1] = 0$ is trivial. But since $[T, \xi, \eta_1] = y^{(2)} \neq 0$, no such extension exists. $\square$

The candidate space $\mathcal{N}$ even for $\mathcal{N}_c = \mathcal{N}_1$ is therefore rejected as a means of producing L -Lipschitz maps with high-volume nullhomotopies. It appears, at any rate, that the methods we had hoped to employ for creating such nullhomotopic maps do not interact well with the required passage through higher order Whitehead products and scalable spaces. We take away from this the lesson that a candidate who seems promising on paper (algebraically) may in fact not be the right fit for a job in reality (geometrically), and we turn instead to creating maps through standard Whitehead products.

3.5 Coformal Examples

Recall that a **coformal** space is one for which the differential of the minimal model is quadratic. By [37, Proposition 3.3], we may interpret this to mean that all (rational)

higher order Whitehead brackets contain zero. In this subsection we build L -Lipschitz maps into a c -step space, and maps into that space through (standard) Whitehead brackets. In [Theorem 3.39](#) we show that the resulting map has sufficiently high degree to require nullhomotopies of volume $\Omega(L^{(c+1)(n-1)})$.

Definition 3.29. Define $V_c^2 := V_{i=1}^c \mathbf{S}_i^2$. Note that $H_2(V_c^2) \cong \mathbb{Z}^c$. Define $\mathcal{T}: V_c^2 \rightarrow V_c^2$ to be any self-map inducing the $(c \times c)$ -matrix

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 1 & \cdots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & 0 & \cdots & 1 & 1 \end{bmatrix} \in \text{Aut}(\mathbb{Z}^c) \cong \text{Aut}(H_2(V_c^2)). \quad (3.30)$$

For each $q \in \{A, B\}$ Let $(\mathbb{T}_{\mathcal{T}})_q$ be a copy of the mapping torus of $\mathcal{T}$:

$$(\mathbb{T}_{\mathcal{T}})_q := \frac{(V_c^2 \times I) \cup_{\mathcal{T}} (V_c^2 \times \{1\})}{(x, 0) \sim (x, 1)}.$$

Given $*$ $\in V_c^2$ common to each copy of $\mathbf{S}^2$ define $\gamma = \{*\} \times \mathbf{S}^1 \subseteq (\mathbb{T}_{\mathcal{T}})_q$. Denote $V_{2c}^2 = (V_c^2)_A \vee (V_c^2)_B = (V_{i=1}^c \mathbf{S}_{A,i}^2) \vee (V_{i=1}^c \mathbf{S}_{B,i}^2)$. Define $\mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B}$ to be the mapping torus of $\mathcal{T}_A \vee \mathcal{T}_B: V_{2c}^2 \rightarrow V_{2c}^2$ under the obvious choices of domain and codomain for each copy of $\mathcal{T}$. Note that $\mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B} = (\mathbb{T}_{\mathcal{T}})_A \cup_{\gamma} (\mathbb{T}_{\mathcal{T}})_B$.

Lemma 3.31. $\pi_1(\mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B}) \cong \mathbb{Z}$ is generated by $t: \mathbf{S}^1 \rightarrow \gamma \hookrightarrow \mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B}$.

Proof. This is a consequence of van Kampen's Theorem. □

We state the following result for α_j generators of $\pi_2((\mathbb{T}_{\mathcal{T}})_A)$, but similar results hold for β_j generators of $\pi_2((\mathbb{T}_{\mathcal{T}})_B)$.

Proposition 3.32. *Let $\alpha_j \in \pi_2((\mathbb{T}_{\mathcal{T}})_A)$ be the homotopy class of the map $\mathbf{S}^2 \rightarrow \mathbf{S}_j^2 \hookrightarrow (\mathbb{T}_{\mathcal{T}})_A$. Then $t \bullet \alpha_j = \alpha_j + \alpha_{j+1}$, and $[t, \alpha_j] = \alpha_{j+1}$ for $j < c$ while $t \bullet \alpha_c = \alpha_c$ and $[t, \alpha_c] = 0$.*

Proof. By definition the generator $t \in \pi_1(\mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B})$ acts on $\pi_2((\mathbb{T}_{\mathcal{T}})_A) \otimes \mathbb{Q} \subseteq \pi_2(\mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B}) \otimes \mathbb{Q}$ under the basis $\{\alpha_c, \alpha_{c-1}, \dots, \alpha_1\}$ by the $(c \times c)$ matrix in (3.30) above. $\square$

Definition 3.33. Inductively define

$$\zeta_{k,j} := \begin{cases} \beta_1 & k = 2, j = 1, \\ [\alpha_1, \zeta_{k-1,c}] & k > 2, j = 1, \\ [t, \zeta_{k,j-1}] & j > 1. \end{cases}$$

Lemma 3.34. *For all $k > 2$ and $j < c$ we have $[t, [\alpha_j, \zeta_{k-1,c}]] = [\alpha_{j+1}, \zeta_{k-1,c}]$ and for all $k \geq 2$ we have $[t, \zeta_{k,c}] = 0$.*

Proof. The fact $[t, \zeta_{2,c}] = 0$ is true by Proposition 3.32 (applied, of course, to β_j instead of α_j). Let $k > 2$ and assume for induction that the result holds for all lower indices. By the Jacobi identity we have

$$\begin{aligned} [t, [\alpha_j, \zeta_{k-1,c}]] &= -[\alpha_j, [\zeta_{k-1,c}, t]] - [\zeta_{k-1,c}, [t, \alpha_j]] \\ &= -[\alpha_j, 0] - [\zeta_{k-1,c}, [t, \alpha_j]] && \text{(induction)} \\ &= [[t, \alpha_j], \zeta_{k-1,c}]. && \text{(antisymmetry)} \end{aligned}$$

In the event $j < c$, Proposition 3.32 implies then that $[t, [\alpha_j, \zeta_{k-1,c}]] = [\alpha_{j+1}, \zeta_{k-1,c}]$ while if $j = c$, the same result indicates that $[t, [\alpha_j, \zeta_{k-1,c}]] = [0, \zeta_{k-1,c}] = 0$ as required. $\square$

Corollary 3.35. $\zeta_{k,j} = [\alpha_j, \zeta_{k-1,c}]$ for all k and j . In particular $\zeta_{k,c} = [\alpha_c, \zeta_{k-1,c}]$.

Proof. Inductively by the previous lemma, $\zeta_{k,j} = [t, \zeta_{k,j-1}] = [t, [\alpha_{j-1}, \zeta_{k-1,c}]] = [\alpha_j, \zeta_{k-1,c}]$, where the base case $j = 1$ is true by definition. $\square$

Lemma 3.36. *For all k , one has $\zeta_{k,c} \neq 0$.*

Proof. Let $\mathcal{M} = \mathcal{M}_{\mathcal{T}_A \vee \mathcal{T}_B}$ denote the mapping cylinder of $\mathcal{T}_A \vee \mathcal{T}_B$. The universal cover $\widetilde{\mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B}}$ of $\mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B}$ may be viewed as $\mathbb{Z}$ copies of $\mathcal{M}$ identified in the obvious ways. Let $\mathcal{M}_p$ denote the copies of $\mathcal{M}$ in $\widetilde{\mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B}}$ over the interval $[-p, p]$ so that $\mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B} = \bigcup_{p \in \mathbb{Z}} \mathcal{M}_p$.

Let $\tilde{\alpha}_c : \mathbf{S}_{A,c}^2 \hookrightarrow \mathbb{V}_{2c}^2 \times \{0\} \hookrightarrow \widetilde{\mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B}}$ lift $\alpha_c : \mathbf{S}_{A,c}^2 \hookrightarrow \mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B}$ in the obvious way. Similarly denote $\tilde{\beta}_c$ and $\tilde{\zeta}_{k,c}$. It is not hard to see that $\mathcal{M}_p$ deformation retracts to $\mathbb{V}_{2c}^2 \times \{p\}$ through standard deformation retractions of each copy of $\mathcal{M}$ so that $\pi_n(\mathcal{M}_p) \cong \pi_n(\mathbb{V}_{2c}^2)$ for all p . Moreover, because $t \bullet \alpha_c = \alpha_c$ (and similarly for β_c) we see that this retract sends $(\mathbf{S}_{A,c}^2 \vee \mathbf{S}_{B,c}^2) \times \{0\}$ identically onto $(\mathbf{S}_{A,c}^2 \vee \mathbf{S}_{B,c}^2) \times \{p\}$. Then the retract identifies the class of $\tilde{\alpha}_c$ with that of $\mathbb{1}_{A,c} : \mathbf{S}_{A,c}^2 \hookrightarrow \mathbb{V}_{2c}^2$. The same is true for $\tilde{\beta}_c$ under the obvious changes. That is to say, the isomorphism $\pi_2(\mathcal{M}_p) \cong \pi_2(\mathbb{V}_{2c}^2)$ induced by the retraction will identify $\tilde{\alpha}_c$ and $\tilde{\beta}_c$ with $\mathbb{1}_{A,c}$ and $\mathbb{1}_{B,c}$ respectively. By induction, since $\zeta_{k,c} = [\alpha_c, \zeta_{k-1,c}]$ and since $t \bullet \zeta_{k,c} = \zeta_{k,c}$, we see that the retraction of $\mathcal{M}_p$ identifies the class of $\tilde{\zeta}_{k,c}$ in $\pi_k(\mathcal{M}_p)$ with the nonzero class $[\mathbb{1}_{A,c}, [\mathbb{1}_{A,c}, \dots [\mathbb{1}_{A,c}, \mathbb{1}_{B,c}] \dots]] \in \pi_k(\mathbb{V}_{2c}^2)$. Hence for any p , the map $\tilde{\zeta}_{k,c}$ is not nullhomotopic in $\mathcal{M}_p$.

If to the contrary $\zeta_{k,c}$ were nullhomotopic by some $H : \mathbf{S}^k \times I \rightarrow \mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B}$, there would be a lift $\tilde{H} : \mathbf{S}^k \times I \rightarrow \widetilde{\mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B}}$ nullhomotoping $\zeta_{k,c}$ inside $\widetilde{\mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B}}$. By compactness there is some p for which this nullhomotopy is contained entirely within $\mathcal{M}_p$, a contradiction. $\square$

Lemma 3.37. *Let Y be a space for which $f : \mathbf{S}^{n_1} \rightarrow Y$ and $g : \mathbf{S}^{n_2} \rightarrow Y$ are $O(L)$ -Lipschitz maps. Then $[f, g] \in \pi_{n_1+n_2-1}(Y)$ has an $O(L)$ -Lipschitz representative.*

Proof. The map $[f, g]$ is formed by the composition

$$\mathbf{S}^{n_1+n_2-1} \rightarrow \mathbf{S}^{n_1} \vee \mathbf{S}^{n_2} \xrightarrow{f \vee g} Y.$$

The first map is $O(1)$ -Lipschitz, and the second is $O(L)$ -Lipschitz, and hence the composition is $O(L)$ -Lipschitz. $\square$

Proposition 3.38. *For all k there is an $O(L)$ -Lipschitz representative $\zeta_{k,c}^{(L)}$ of $L^{(c+1)(k-1)}\zeta_{k,c}$.*

Proof. Let $\alpha_1^{(L)}$ be the map $\mathbf{S}^2 \xrightarrow{\times L} \mathbf{S}^2 \xrightarrow{\alpha_1} \mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B}$ be the $O(L)$ -Lipschitz map formed by precomposing α_1 with an L -Lipschitz self-map of $\mathbf{S}^2$ of degree L^2 . That is, $\alpha_1^{(L)}$ is an L -Lipschitz representative of $L^2\alpha_1$. Similarly, define $t^{(L)}$ and $\beta_1^{(L)}$ to be L -Lipschitz representatives of Lt and $L^2\beta_1$ respectively. Inductively, we attain L -Lipschitz representatives $\alpha_j^{(L)}$ and $\beta_j^{(L)}$ of $L^{j+1}\alpha_j$ and $L^{j+1}\beta_j$ respectively by

$$\alpha_j^{(L)} := [t^{(L)}, \alpha_{j-1}^{(L)}] = [Lt, L^j\alpha_{j-1}] = L^{j+1}\alpha_j$$

and similarly for $\beta_j^{(L)}$. In particular there is an $O(L)$ -Lipschitz representative $\zeta_{2,c}^{(L)} := \beta_c^{(L)}$ of $L^{c+1}\zeta_{2,c} = L^{c+1}\beta_2$. This establishes the base case $k = 2$.

Assume the result inductively for $k - 1$ for some $k > 2$. By [Lemma 3.37](#) there is an $O(L)$ -Lipschitz representative $\zeta_{k,c}^{(L)}$ of

$$\begin{aligned} [\alpha_c^{(L)}, \zeta_{k-1,c}^{(L)}] &= [L^{c+1}\alpha_c, L^{(c+1)(k-2)}\zeta_{k-1,c}] && \text{(induction)} \\ &= L^{(c+1)+(c+1)(k-2)}[\alpha_c, \zeta_{k-1,c}] && \text{(homogeneity)} \\ &= L^{(c+1)(k-1)}\zeta_{k,c}, && \text{(Corollary 3.35)} \end{aligned}$$

which completes the proof. $\square$

Theorem 3.39. *There is a space Y which is c -step nilpotent and coformal up through degree n , together with a nullhomotopic, L -Lipschitz map $f : \mathbf{S}^n \rightarrow Y$ such that any nullhomotopy of f has volume $\Omega(L^{(c+1)(n-1)})$.*

Proof. The argument here is identical to that of [\[9, §7\]](#). Let $f = \zeta_{n,c}^{(L)}$, and let Y be

the space formed by attaching $(n + 1)$ -cells to $\mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B}$, whose attaching maps form a basis for $\pi_n(\mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B}) \otimes \mathbb{Q}$. Ensure that Y is coformal through degree n by attaching higher cells to kill any nontrivial higher order Whitehead products. In particular, fix a standard nullhomotopy $N_{n,c}$ of $\zeta_{n,c}$ in Y . Such a nullhomotopy has degree $C \neq 0$ (in the sense of relative homology) over some $(n + 1)$ cell q of Y because (by [Lemma 3.36](#)), $\zeta_{n,c}$ is not nullhomotopic in $\mathbb{T}_{\mathcal{T}_A \vee \mathcal{T}_B}$. This yields a closed $(n + 1)$ -form ω on Y such that $\int_{\mathbf{S}^n \times I} N_{n,c}^* \omega = C \neq 0$. Let $f' = \zeta_{n,c} \circ \varphi_{n,L}$ where $\mathbf{S}^n \rightarrow \mathbf{S}^n$ is a self-map of degree $L^{(c+1)(n-1)}$. Note that $f' \simeq f$ as both are representatives of $L^{(c+1)(n-1)}\zeta_{n,c}$ by definition. Hence a nullhomotopy $N_{n,c,L}$ of f may be defined by first homotoping f to f' , followed by the standard nullhomotopy $N_{n,c}$. This nullhomotopy has volume bounded below by $\int_{\mathbf{S}^n \times I} N_{n,c,L}^* \omega = L^{(c+1)(n-1)}C = \Omega(L^{(c+1)(n-1)})$.

Now suppose that N' is some other nullhomotopy of f , and let $\tilde{N}$ be the map $\mathbf{S}^{n+1} \rightarrow Y$ formed by performing $N_{n,c,L}$ along the northern hemisphere and N' along the southern hemisphere, connected along the $\mathbf{S}^n$ equator along the map f itself. The (rational) Heurewicz map sends $\pi_{n+1}(Y)$ to 0 and so the total degree of $\tilde{N}$ must be 0, and hence $\tilde{N}$ has zero degree on any $(n + 1)$ -cell. In particular, this means that $\int_{\mathbf{S}^n \times I} (N')^* \omega = \int_{\mathbf{S}^n \times I} N_{n,c,L}^* \omega = \Omega(L^{(c+1)(n-1)})$. $\square$

Chapter 4

The Shadowing Principles

We come now to what may be considered the heart of the paper, where we generalize the Shadowing Principle of [23] to targets which appear as principal $K(G, n)$ fibrations, and hence to nilpotent spaces. The previous chapter provides some motivation for this; we would ultimately like to get an upper bound on the volume of nullhomotopies in nilpotent spaces, and the original principle provided a useful tool for such analysis in the simply connected setting. By the help of [Theorem 4.22](#), we attain our goal in [Chapter 5](#).

Before turning to our versions of the Shadowing Principle, we provide some intuition for interpreting the principle outside of the fibered setting. Given a target Y and an n -dimensional space X , we assume that we have an algebraic map $\varphi : \mathcal{M}_Y^* \rightarrow \Omega^*(X)$ which has dilatation bounded by a quantity L . We might wonder how far away we must look to find a genuine map which has similar bounds. That is, we put some reasonable metric on $\text{Hom}(\mathcal{M}_Y^*, \Omega^*(X))$, such as

$$d(\varphi, \psi) = \inf \left\{ \text{length}(\Phi) \mid \varphi \stackrel{\Phi}{\simeq} \psi \right\}$$

where the **formal length** of such a homotopy Φ is

$$\text{length}(\Phi) := \text{Dil} \left(\int_0^1 \Phi \right).$$

This gives constant homotopies trivial length, is nondegenerate according to applications of (2.43), and satisfies the triangle inequality by [23, Prop. 3.11]. With this metric in mind, the Shadowing Principle states that if φ is itself in the homotopy class of some genuine map and $\text{Dil}(\varphi) \leq L$, then there is a pullback of a genuine $O(L)$ -Lipschitz map at most $O(L)$ far from φ , homotopic to the original genuine map. If it is too hard to construct a bounded geometric map from the outset, it may be easier to construct one whose pullback is bounded. The principle then produces a genuine map homotopic to the original whose Lipschitz constant is bounded.

In Section 4.1 we introduce establish necessary quantitative results in simplicial geometry, which are crucial to the proof of the fibered and nilpotent versions of the principle. In Section 4.2 we prove the relevant principles.

4.1 Simplicial Geometry

In the following, we work primarily in the simplicial category as our model for homotopy theory. See [31] for details on this setting, and its equivalence to the model category of CW complexes. In particular, fibrations will all be (geometric realizations of) fibrations of finite simplicial sets. A key ingredient in this work (as in [8, 9, 23]) is that of simplicial approximation. This serves as a tool for standardizing local maps in such a way as to control Lipschitz constants. Given a simplicial complex X with standard metric, one of the most common subdivision schemes of X is barycentric subdivision. This scheme has certain geometric disadvantages though; simplices may become arbitrarily

skinny, causing massive distortion in comparison to the standard simplex.

We would like to prove lemmas on the standard simplex which may then be used locally to prove results about more general complexes, and it is therefore helpful to have subdivision schemes where our simplices are uniformly bilipschitz to the standard simplex. To be more specific, we say that a simplicial subdivision of X is **L -regular** if there is a constant $r > 0$ (depending on the dimension of X) for which simplices are r -bilipschitz to a standard simplex with edge length $\frac{1}{L}$. Although barycentric subdivision is not regular, there are a handful of schemes (see [41, 42] for examples) which are. We assume from here on out that all subdivisions will be regular.

Lipschitz maps between simplicial complexes can be approximated by simplicial maps without increasing the Lipschitz constant too much. First, recall the following definitions:

Definition 4.1. Let $H : X \times I \rightarrow Y$ be a Lipschitz homotopy. After embedding $Y \hookrightarrow \mathbb{R}^N$ in a (piecewise) smooth manner, Rademacher's Theorem implies that H is differentiable almost everywhere (since H is Lipschitz), and hence one may define the **volume** of H by

$$\text{vol } H := \int_{X \times I} |\text{Jac}(H)| \, \text{dvol}.$$

The **length** of H is

$$\text{length}(H) := \max \left\{ \text{Lip}(H|_{\{x\} \times I}) \mid x \in X \right\}.$$

The **thickness** of H is

$$\text{thickness}(H) := \max \left\{ \text{Lip}(H|_{X \times \{t\}}) \mid t \in I \right\}.$$

Lemma 4.2. For X an n -dimensional space, $\text{vol}(H) = O(\text{length}(H)) \cdot O(\text{thickness}(H)^n)$.

Proof. Clearly for any $x \in X$

$$\int_{\{x\} \times I} |\text{Jac } H| \, \text{dvol} \lesssim \int_{\{x\} \times I} |\text{Lip}(H|_{\{x\} \times I})| \, \text{dvol} \lesssim \text{length}(H),$$

and similarly for any $t \in I$

$$\int_{X \times \{t\}} |\text{Jac } H| \, \text{dvol} \lesssim \int_{X \times \{t\}} |\text{Lip}(H|_{X \times \{t\}})|^n \, \text{dvol} \lesssim \text{thickness}(H)^n. \quad \square$$

Proposition 4.3 (Quantitative Simplicial Approximation Theorem [8, Prop. 2.1]). *Let X be a finite-dimensional simplicial complex with standard simplices, and let Y be a finite simplicial complex with piecewise linear metric. There is a constant $C = C(X, Y)$ such that any L -Lipschitz map $f : X \rightarrow Y$ has a simplicial approximation f_Δ with $\text{Lip } f_\Delta \leq C(L + 1)$. Moreover, f is homotopic to f_Δ by a C -length homotopy H of thickness $C(L + 1)$. In particular, $\text{Lip } H \leq C'(L + 1)$ for some $C' = C'(X, Y)$.*

Sketch of proof. Paying quantitative attention to the proof of the (non-quantitative) simplicial approximation theorem from [27], one takes a regular subdivision of X and uses the Lebesgue covering lemma to carefully produce nearly-linear homotopies between the uncontrolled map and its simplicial approximation. By induction up the skeleta of X , this process is made to be globally well-controlled. $\square$

We will need to borrow some of these ideas in what follows. Specifically, we will want to talk about different notions of Lipschitz bounds in the diagrammatic setting. Recalling notation from Chapter 2, we are primarily interested in diagrams $\mathcal{D}_f$ of the form

$$\begin{array}{ccc} A & \xrightarrow{\alpha} & Y \\ \downarrow \iota & \nearrow f & \downarrow p \\ X & \xrightarrow{\beta} & B \end{array}$$

where $A \hookrightarrow X$ is a simplicial pair and $Y \twoheadrightarrow B$ is a principal $K(G, n)$ fibration. We will often write $f|_A = \alpha$ and $\bar{f} = \beta$.

Definition 4.4. Let $f: X \rightarrow Y$. If each of the maps $\bar{f}$ and $f|_A$ in $\mathcal{D}_f$ are L -Lipschitz, we say that $\mathcal{D}_f$ (or f) is **externally L -Lipschitz**. If f is externally equivalent to some L -Lipschitz map, we say that $\mathcal{D}_f$ (or f) is **L -Lipschitz at heart**. If f is internally equivalent to some L -Lipschitz map, we say that f is **internally L -Lipschitz**.

If we have a map $f: X \rightarrow Y$ which is externally L -Lipschitz, we would like to bound the best-behaved fillings of the corresponding diagram $\mathcal{D}_f$. In the setting of maps from the standard simplex, we can be a bit more precise. Let $f: \Delta \rightarrow Y$. We would like to come up with a map f_0 such that, for each $\xi \in G$ there is a map f_ξ representing ξ in the G -torsor relative to f_0 . Moreover, we would like uniform bounds on the Lipschitz constants of these maps.

One useful tool for quantifying the behavior of the maps we build will be the language of **shard complexes** and **mosaic maps**.

Definition 4.5. For each k , let $\mathcal{F}_k$ be a finite set of maps $\Delta^k \rightarrow Y$. The collection $\mathcal{F} = \bigcup_k \mathcal{F}_k$ is called a **shard complex** on Y , and is naturally a finite subcomplex of the singular simplicial complex on Y . In particular, $\mathcal{F}$ is a semi-simplicial set whose k -simplices are $\mathcal{F}_k$. We call a map $f: X \rightarrow Y$ from a simplicial complex X an **$\mathcal{F}$ -mosaic map** if $f|_{\Delta^k} \in \mathcal{F}_k$ for all k -simplices $\Delta^k \subseteq X$, for all k .

One feature of this definition is that being mosaic is preserved under postcomposition. For example, if $\varphi: Y \rightarrow Y'$ is a homotopy equivalence, then $\varphi \circ f$ will be mosaic with respect to a shard complex $\mathcal{F}'$ on Y' if f is mosaic with respect to some $\mathcal{F}$ on Y .

For the sake of brevity and clarity, it will be convenient to prove results for 1-Lipschitz maps. However, such results will often times generalize to L -Lipschitz maps. In particular, if $X \rightarrow Y$ is L -Lipschitz, one subdivides X at scale $\frac{1}{L}$ and rescales so that simplices are unit size. The same map then becomes 1-Lipschitz, and when convenient we will assume without loss of generality that $L = 1$.

Fix a principal $K(G, n)$ -fibration of finite simplicial sets $p: Y \rightarrow B$. A (minimal) fibration of simplicial sets has a geometric realization which is locally trivial over simplices in the base. In other words, if the fiber of $p: Y \rightarrow B$ is F , then for any simplex $\sigma \subseteq B$ we may identify $p^{-1}(\sigma)$ with $F \times \sigma$. (See, for example, the proof of [31, Theorem I.10.9].) We make this identification freely in what transpires and after sufficient subdivision we may assume that for $b \in B^{(0)}$ we have $p^{-1}(\star(b)) \cong \star(b) \times F$, where $\star(b)$ denotes the star of b . By rescaling Y and B as necessary, we assume that if $s: \Delta \rightarrow B$ is 1-Lipschitz, then $\text{im}(s) \subseteq \star(b)$ for some $b \in B^{(0)}$.

Proposition 4.6. *Let $L > 0$ be given. There is an $L_0 = L_0(L, p)$ such that the following holds: Suppose the diagram*

$$\begin{array}{ccc} A & \xrightarrow{\alpha} & Y \\ \downarrow & \nearrow f & \downarrow \\ X & \xrightarrow{\beta} & B \end{array}$$

is externally L -Lipschitz. Assume moreover that α is mosaic with respect to some fixed shard complex $\mathcal{Z}$ on Y for which $\mathcal{Z}^{(0)}$ consists of basepoints over fibers of vertices in B . Then f is L_0 -Lipschitz at heart through some f_0 . Moreover, for each $\xi \in H^n(X, A; G)$, there is an L_ξ such that f is L_ξ -Lipschitz at heart through some f_ξ for which the obstruction to performing an internal homotopy from f_ξ to f_0 is ξ .

Proof. Let $\mathcal{D}_\Delta$ be the collection of diagrams

$$\begin{array}{ccc} A & \xrightarrow{\alpha} & Y \\ \downarrow & \dashrightarrow & \downarrow \\ X & \xrightarrow{\beta} & B \end{array}$$

for which β is simplicial, for which α is mosaic with respect to $\mathcal{Z}$, and whose obstruction to finding a lift in the diagonal vanishes. Given such a diagram $\mathcal{D} \in \mathcal{D}_\Delta$, let $f_{\mathcal{D},0}$ be a filling of minimal Lipschitz constant. Because α is assumed to be mosaic and β is assumed

to be simplicial, the set $\mathcal{D}_\Delta$ is finite, and hence the quantity $C_{\Delta,0} := \max_{\mathcal{D} \in \mathcal{D}_\Delta} \{f_{\mathcal{D},0}\}$ is finite.

Furthermore, given $\xi \in H^n(X, A; G)$, let $f_{\Delta,\xi}$ be of minimal Lipschitz constant such that $[f_{\Delta,\xi}] = \xi$ in the torsor relative to $f_{\Delta,0}$. (That is, the obstruction to performing an internal homotopy from $f_{\Delta,\xi}$ to $f_{\Delta,0}$ is ξ .) Again, the quantity $C_{\Delta,\xi} = \max_{D_\Delta \in \mathcal{D}_\Delta} \{\text{Lip}(f_{D_\Delta,\xi})\}$ is finite. Therefore, if we had first assumed that β were simplicial, we would be done.

It suffices then to show that, given an arbitrary L -Lipschitz diagram $\mathcal{D}$, one can find a “nearby” diagram of the above form whose fillings $f_{\Delta,0}$ and $f_{\Delta,\xi}$ induce L_0 -Lipschitz fillings of $\mathcal{D}$. Indeed, this is easily attained by similar methods to [Proposition 4.3](#). At the risk of increasing the Lipschitz constant slightly we attain a simplicial approximation β' of β , as well as a linear homotopy H from β to β' . By the homotopy lifting property, we get the following commutative diagram

$$\begin{array}{ccc} A \times I & \xrightarrow{H_\alpha} & Y \\ \downarrow & & \downarrow p \\ X \times I & \xrightarrow{H_\beta} & B \end{array}$$

where at $t = 1$ the diagram is filled as

$$\begin{array}{ccc} A & \xrightarrow{\alpha'} & Y \\ \downarrow & & \downarrow p \\ X & \xrightarrow{\beta'} & B \end{array}$$

If $\mathcal{Z}$ were the shard complex with respect to which α was mosaic, this induces a new shard complex $\mathcal{Z}'$ for which α' is $O(L+1)$ -Lipschitz and mosaic. By our previous work, since β' is simplicial, there is a map $f_{\xi,\Delta}$ fitting into the diagonal of this diagram with the desired properties. By the homotopy lifting property, we attain a homotopy from $f_{\Delta,\xi}$ to some map f_ξ at $t = 0$ for which the following diagram commutes:

$$\begin{array}{ccc}
A & \xrightarrow{\alpha} & Y \\
\downarrow & \nearrow f_\xi & \downarrow p \\
X & \xrightarrow{\beta} & B
\end{array}$$

In fact, by the identification $p^{-1}(\star(b)) \cong \star(b) \times F$ for each $b \in B^{(0)}$ and all $t \in I$ we may define $H_\alpha(t)|_{p^{-1}(\star(b))} = (H_\beta(t)|_{\star(b)}, \mathbb{1})$ on $p^{-1}(\star(b)) \times F$. That is, we give $y \in p^{-1}(\star(b))$ coordinates $y = (\bar{y}, y') \in p^{-1}(\star(b)) \times F$ and we define

$$H_\alpha(y, t) = (H_\beta(\bar{y}, t), y').$$

This homotopy is piecewise linear, and therefore $f_\xi := H_\alpha(0)$ has Lipschitz constant C_ξ depending only on the geometry of the local trivializations and on L , as well as on the initial shard complex $\mathcal{Z}$. $\square$

Corollary 4.7. *Let $\mathcal{Z}_0$ be a fixed shard complex on Y for which $\mathcal{Z}_0^{(0)}$ consists of basepoints over fibers of vertices in B . There is a $C = C(\mathcal{Z}_0, n-1, Y)$ such that, if $f : X \rightarrow Y$ is externally 1-Lipschitz, and $f|_A$ is mosaic, then $f|_{X^{(n-1)}}$ is internally equivalent to a C -Lipschitz map. In particular there is a shard complex $\mathcal{Z}_{n-1} = \mathcal{Z}_{n-1}(\mathcal{Z}_0, n-1, Y)$ with respect to which $f|_{X^{(n-1)}}$ is mosaic.*

Proof. We proceed by induction. Because Y is finite, there is a $C_0 = \text{diam}(Y)$ for which $f|_{X^{(0)}}$ is C_0 -Lipschitz. Inductively, for $0 < k < n$ assume that f is internally homotopic to a map $\tilde{f}_{k-1}$ for which $\tilde{f}_{k-1}|_{X^{(k-1)}}$ is C_{k-1} -Lipschitz. Apply [Proposition 4.6](#) to the diagram

$$\begin{array}{ccc}
A \cup X^{(k-1)} & \xrightarrow{\tilde{f}_{k-1}|_{A \cup X^{(k-1)}}} & Y \\
\downarrow & \nearrow \tilde{f}_{k-1}|_{X^{(k)}} & \downarrow p \\
A \cup X^{(k)} & \xrightarrow{p \circ f|_{A \cup X^{(k)}}} & B
\end{array}$$

to attain an (*a priori*) externally equivalent C_k -Lipschitz map $\tilde{f}_k|_{X^{(k)}} : X^{(k)} \rightarrow Y$ for which C_k depends only on C_{k-1} and on $p : Y \rightarrow B$. The obstruction to performing an

internal homotopy between these maps is an element of $H^n(A \cup X^{(k)}, A \cup X^{(k-1)}; G)$, which is trivial since $k \neq n$. In particular, $\tilde{f}_k|_{X^{(k)}}$ must in fact be *internally* equivalent to $f_{k-1}|_{X^{(k)}}$ over this diagram, and we extend $\tilde{f}_k$ over higher dimensional cells to a map $\tilde{f}_k$ on all of X which is internally equivalent (relative to A) to f itself. Take $\mathcal{Z}_k = \mathcal{Z}_{k-1} \cup \left\{ \tilde{f}_k|_{\Delta} \mid \Delta \subseteq X \text{ is a } k\text{-simplex} \right\}$. Induction ceases at $k = n - 1$. $\square$

Of primary importance for our purposes is the situation where the domain is the standard n -simplex, $(\Delta^n, \partial\Delta^n)$.

Corollary 4.8. *Let $p : Y \rightarrow B$ be a principal $K(G, n)$ fibration of simplicial sets. Let $\xi \in G \cong H^n(\Delta, \partial\Delta; G)$. There is a constant $C_\xi = C_\xi(p)$ such that if $f : \Delta \rightarrow Y$ is externally 1-Lipschitz, then there is a C_ξ -Lipschitz map f_ξ externally equivalent to f . Moreover, the obstruction to performing an internal homotopy from f_0 to f_ξ is ξ (where f_0 is the map applied to $\xi = 0$).*

The following lemmata from [23, §2.2] are key for providing quantitative bounds on integration over simplices. As in that paper, we adapt the notation $\Omega^*(X)$ to denote the algebra of simplex-wise smooth forms on a simplicial complex X . Given a simplicial pair (X, A) , we use $\Omega^*(X, A)$ to denote the forms whose restriction to A is trivial. Context should always make clear when this notation is not to be confused (for example) with the De Rham algebra of genuinely smooth forms. The proofs of these are based on the proof of the usual Poincaré Lemma for compact support as found in [43, §1.4], replacing compactly supported forms with forms on manifolds with boundary whose restriction to the boundary is trivial. We here fill in some of the detail absent from [23].

Given a compact manifold N with boundary, we let $\Omega^*(N, \partial N)$ denote the smooth forms whose restriction to ∂N is trivial. Let M be a compact Riemannian manifold with boundary, and let t denote the $I = [0, 1]$ coordinate of $M \times I$. Let $\pi : M \times I \rightarrow M$ denote the obvious projection. A k -form $\omega \in \Omega^k(M \times I, \partial(M \times I))$ is a linear combination of

two types of forms:

$$(I) \quad \pi^* \varphi \cdot f(m, t)$$

$$(II) \quad \pi^* \varphi \cdot f(m, t) dt$$

where $\varphi \in \Omega^k(M)$ in case (I) and $\varphi \in \Omega^{k-1}(M)$ in case (II), and where $dt \in \Omega^1(I)$ is the obvious nontrivial 1-form, and where $f: M \times I \rightarrow \mathbb{R}$ is a smooth function for which $f|_{\partial(M \times I)} = 0$. Note in particular that the form φ and dt are not required to be trivial on the boundary components.

Definition 4.9. The **fiberwise integral on $M \times I$ in the direction of t** is the linear map $\int_0^1 : \Omega^*(M \times I, \partial(M \times I)) \rightarrow \Omega^{*-1}(M, \partial M)$ defined on the above cases by

$$(I) \quad \int_0^1 \pi^* \varphi \cdot f(m, t) = 0$$

$$(II) \quad \int_0^1 \pi^* \varphi \cdot f(m, t) dt = \varphi \int_0^1 f(m, t) dt$$

We also define $\int_0^t : \Omega^*(M \times I, \partial(M \times I)) \rightarrow \Omega^{*-1}(M \times I, \partial(M \times I))$ by

$$(I) \quad \int_0^t \pi^* \varphi \cdot f(m, s) = 0$$

$$(II) \quad \int_0^t \pi^* \varphi \cdot f(m, s) ds = \pi^* \varphi \int_0^t f(m, s) ds$$

Lemma 4.10. *Fiberwise integration is a chain map. That is, $d \int_0^1 \omega = \int_0^1 d\omega$ for any $\omega \in \Omega^*(M \times I, \partial(M \times I))$.*

Proof. If ω is an in (I), then so is $d\omega$ since $d\pi^* = \pi^*d$ commutes. Clearly then, in this

case, $d \int_0^1 \omega = d0 = 0 = \int_0^1 d\omega$. On the other hand, if ω is as in (II) then

$$\begin{aligned} d \int_0^1 \omega &= d \left(\varphi \int_0^1 f(m, t) dt \right) \\ &= (d\varphi) \int_0^1 f(m, t) dt \\ &= \int_0^1 \pi^*(d\varphi) f(m, t) dt \\ &= \int_0^1 d\omega. \end{aligned}$$

□

Following the notation of [43] let $\varphi \cdot f := \pi^* \varphi \cdot f(m, t)$ and $\varphi \cdot f dt := \pi^* \varphi \cdot f(m, t) dt$. Fix $\varepsilon : I \rightarrow I$ a smooth bump function which is 0 near 0 and 1 near 1. Denote $d\varepsilon := \varepsilon(t)dt$, and define an operator

$$\begin{aligned} K : \Omega^*(M \times I, \partial(M \times I)) &\rightarrow \Omega^{*-1}(M \times I, \partial(M \times I)) \\ \omega &\mapsto \int_0^t \omega - \pi^* \int_0^t d\varepsilon \int_0^1 \omega. \end{aligned}$$

Basic computations show that

Proposition 4.11 ([43, Proposition 1.4.6]). *Let $\omega \in \Omega^k(M \times I, \partial(M \times I))$. Then*

$$\omega = (-1)^{k-1} \left((dK - Kd)(\omega) - d\varepsilon \wedge \pi^* \int_0^1 \omega \right).$$

Corollary 4.12. *Suppose that $\omega \in \Omega^k(M \times I, \partial(M \times I))$ is a closed k -form. Then*

$$\omega = \pm(K\omega - d\varepsilon \wedge \pi^* \int_0^1 \omega).$$

Lemma 4.13 (First Quantitative Poincaré Lemma). *For every $0 < k < n$ there is a constant $C_{n,k}$ such that, whenever $\omega \in \Omega^k(\Delta^n, \partial\Delta^n)$ is a smooth k -form, then there is an $\alpha \in \Omega^{k-1}(\Delta^n, \partial\Delta^n)$ such that $d\alpha = \omega$ and for which $\|\alpha\|_\infty \leq C_{n,k} \|\omega\|_\infty$. In the event*

$k = n$, the same holds if additionally ω satisfies $\int_{\Delta^n} \omega = 0$.

Proof. We replace Δ^n with a diffeomorphic unit n -cube Q^n , which we further view as $Q^n = Q^{n-1} \times I$. We proceed by induction on k and n simultaneously, while keeping $n - k$ fixed. The case $k = 0$ is trivial for all n , therefore serving as the base case for all n .

Therefore, let $k, n \geq 1$ and suppose that the result holds for $k - 1, n - 1$. Define

$$C_{n,k} := 2 + C_{k-1,n-1} \|d\varepsilon\|_\infty.$$

Let ω be a k -form in $\Omega^k(Q^n, \partial Q^n) = \Omega^k(Q^{n-1} \times I, \partial(Q^{n-1} \times I))$. Since $\int_0^1 \omega \in \Omega^{k-1}(Q^{n-1}, \partial Q^{n-1})$, the inductive hypothesis indicates there is a form $\eta \in \Omega^{k-2}(Q^{n-1}, \partial Q^{n-1})$ for which $d\eta = \int_0^1 \omega$, such that

$$\|\eta\|_\infty \leq C_{n-1,k-1} \left\| \int_0^1 \omega \right\|_\infty \leq C_{n-1,k-1} \|\omega\|_\infty.$$

Let $\alpha := \int_0^t \omega - \varepsilon(t) \pi^* \int_0^1 \omega - d\varepsilon(t) \wedge \pi^* \eta$, where we write $\varepsilon(t) = \int_0^t d\varepsilon = \int_0^t \varepsilon(s) ds$ as a slight abuse of notation. In particular, note that

$$\|\alpha\|_\infty \leq \|\omega\|_\infty + \|\omega\|_\infty + C_{n-1,k-1} \|d\varepsilon\|_\infty \|\omega\|_\infty = C_{n,k} \|\omega\|_\infty.$$

Moreover, since the restrictions of $\int_0^t \omega$ and $\pi^* \int_0^1 \omega$ and $d\varepsilon \wedge \pi^* \eta$ to ∂Q^n are all trivial, we have $\alpha|_{\partial Q^n}$ is trivial as well. Finally, since ω is closed, [Corollary 4.12](#) implies that up

to a change of signs

$$\begin{aligned}
\omega &= dK\omega - d\varepsilon \wedge \int_0^1 \omega \\
&= dK\omega - d\varepsilon \wedge \pi^* d\eta \\
&= d(K\omega - d\varepsilon \wedge \pi^* \eta) \\
&= d\left(\int_0^t \omega - \varepsilon(t)\pi^* \int_0^1 \omega - d\varepsilon(t) \wedge \pi^* \eta\right) \\
&= d\alpha.
\end{aligned}$$

□

In the event that the restriction to $\partial\Delta$ is nonzero, one may still produce a bounded antiderivation of ω by careful analysis away from the boundary, together with bounded data on a collar of the boundary.

Lemma 4.14 (Second Quantitative Poincaré Lemma). *For every $0 < k < n$ there is a constant $\tilde{C}_{n,k}$ such that, whenever $\omega \in \Omega^k(\Delta^n)$ is a closed k -form, and $\alpha_\partial \in \Omega^{k-1}(\partial\Delta^n)$ is a $(k-1)$ -form with $d\alpha_\partial = \omega|_{\partial\Delta^n}$, then there is a $(k-1)$ -form $\alpha \in \Omega^{k-1}(\Delta^n)$ extending α_∂ , such that $d\alpha = \omega$, and for which $\|\alpha\|_\infty \leq \tilde{C}_{n,k}(\|\omega\|_\infty + \|\alpha_\partial\|_\infty)$. In the event $k = n$, the same holds if the pair additionally $(\omega, \alpha_\partial)$ satisfies Stokes' Theorem, so that $\int_{\Delta^n} \omega = \int_{\partial\Delta^n} \alpha_\partial$.*

Proof. Let $U \subseteq \Delta^n$ be the a neighborhood of $\partial\Delta^n$ of width $\frac{1}{2n}$. Let $\pi: U \rightarrow \partial\Delta^n$ be a smooth projection of minimal Lipschitz constant, say $\text{Lip } \pi \leq L_\pi$. Let $\rho: \Delta^n \rightarrow I$ be a smooth bump for which $\rho|_{\partial\Delta^n} = 1$ and $\rho|_{\Delta^n \setminus U} = 0$. Moreover, let ρ be of minimal Lipschitz constant, say $\text{Lip } \rho \leq L_\rho$. Let $C_{n,k}$ be as in [Lemma 4.13](#), and define

$$\tilde{C}_{n,k} := \max \left\{ (C_{n,k} + L_\rho)L_\pi^{k-1}, C_{n,k}(L_\pi^k + 1) \right\}.$$

Note that $\rho\pi^*\alpha_\partial \in \Omega^{k-1}(\Delta^n)$ and that $(\rho\pi^*\alpha_\partial)|_{\partial\Delta^n} = \alpha_\partial$. Define $\omega' := \omega - d(\rho\pi^*\alpha_\partial)$ and

observe that $\omega' \in \Omega^k(\Delta^n, \partial\Delta^n)$. By [Lemma 4.13](#) there is an $\alpha' \in \Omega^k(\Delta^n, \partial\Delta^n)$ such that $\|\alpha'\|_\infty \leq C_{n,k}\|\omega'\|_\infty$ and $d\alpha' = \omega'$. Define $\alpha := \alpha' + \rho\pi^*\alpha_\partial$. Then

$$d\alpha = d\alpha' + d(\rho\pi^*\alpha_\partial) = \omega' + d(\rho\pi^*\alpha_\partial) = \omega.$$

Moreover, we see that

$$\begin{aligned} \|d(\rho\pi^*\alpha_\partial)\|_\infty &\leq \|d\rho \wedge \pi^*\alpha_\partial\|_\infty + \|\rho\pi^*d\alpha_\partial\|_\infty \\ &\leq L_\rho L_\pi^{k-1}\|\alpha_\partial\|_\infty + L_\pi^k\|d\alpha_\partial\|_\infty \\ &\leq L_\rho L_\pi^{k-1}\|\alpha_\partial\|_\infty + L_\pi^k\|\omega\|_\infty \end{aligned}$$

so that

$$\|\omega'\|_\infty \leq L_\rho L_\pi^{k-1}\|\alpha_\partial\|_\infty + (L_\pi^k + 1)\|\omega\|_\infty.$$

Furthermore,

$$\|\rho\pi^*\alpha_\partial\|_\infty \leq L_\rho L_\pi^{k-1}\|\alpha_\partial\|_\infty$$

so that

$$\begin{aligned} \|\alpha\|_\infty &\leq \|\alpha'\|_\infty + \|\rho\pi^*\alpha_\partial\|_\infty \\ &\leq C_{n,k}\|\omega'\|_\infty + L_\rho L_\pi^{k-1}\|\alpha_\partial\|_\infty \\ &\leq C_{n,k}(L_\pi^{k-1}\|\alpha_\partial\|_\infty + (L_\pi^k + 1)\|\omega\|_\infty) + L_\rho L_\pi^{k-1}\|\alpha_\partial\|_\infty \\ &\leq \tilde{C}_{n,k}(\|\alpha_\partial\|_\infty + \|\omega\|_\infty) \end{aligned}$$

which completes the proof. $\square$

Together, these imply the following general quantitative version of the Poincaré lemma, providing quantitative bounds on the L^∞ norm of antidifferentials of forms.

Though we will not need this result in what follows, it demonstrates the general theme of induction and “stitching together” bounded simplicial maps to produce a globally bounded one.

Lemma 4.15 ([23, Lemma 2.2]). *Let (X, A) be a finite simplicial pair with the standard simplex-wise metric. For every k there is a constant $C(k, X, A)$ such that for every exact k -form $\omega \in d\Omega^{k-1}(X, A)$ there is a $(k-1)$ -form $\alpha \in \Omega^{k-1}(X, A)$ such that $d\alpha = \omega$ and for which $\|\alpha\|_\infty \leq C(k, X, A)\|\omega\|_\infty$.*

Proof. Let $w \in C^k(X, A)$ be a simplicial k -cochain given by integrating ω over simplices. That is, $w : \Delta \mapsto \int_\Delta \omega$. Since ω is exact, the De Rham theorem implies that there is a form $a \in C^{k-1}(X, A)$ for which $\delta a = w$. Moreover, since the space of coboundaries is finite-dimensional (since X is finite) there is a constant $c_0(k, X, A)$ called the isoperimetric constant such that

$$\begin{aligned} \|a\|_\infty &\leq c_0(k, X, A) \|w\|_\infty \\ &\leq c_0(k, X, A) \text{vol}(\Delta^k) \|\omega\|_\infty. \end{aligned}$$

To define $\alpha \in \Omega^k(X, A)$, we proceed on skeleta. Fix a bump function $\varphi : X^{(k-1)} \rightarrow I$ whose integral is 1. Define $\alpha|_{X^{(k-1)}}$ by $\alpha = a\varphi \, \text{dvol}_{X^{(k-1)}}$. By Lemma 4.14 we may extend to higher skeleta inductively. Since $c_0(k, X, A)$ is multiplied by a constant depending only on the dimension at each stage, we end with a constant $C(k, X, A)$ and a form $\alpha \in \Omega^{k-1}(X, A)$ for which $\|\alpha\|_\infty \leq C(k, X, A)\|\omega\|_\infty$ as desired. $\square$

4.2 The Shadowing Principles

We have almost arrived in a position to state and prove the Shadowing Principle for lifts of Lipschitz maps, as well as the relevant corollaries. This will in particular allow us

to expand the original principle to nilpotent (rather than simply connected) spaces.

For the rest of this chapter we fix G a finitely generated abelian group, we fix a principal $K(G, n)$ fibration $Y \xrightarrow{p} B$ of simplicial sets Y and B . Moreover, fix norms on $\mathcal{M}_B^*$ and $V = G \otimes \mathbb{Q}$. Define F_b to be the fiber over a given point $b \in B$, and let F denote a generic fiber. Assume moreover that the skeleta of Y and B are finite in every dimension (but **not** necessarily that Y and B are finite themselves). For example this is the case at each stage of the Postnikov tower of a finite type nilpotent space.

Recall that $\mathcal{D} = \mathcal{D}(X, A, Y, B)$ is the set of all diagrams of the form

$$\begin{array}{ccc} A & \longrightarrow & Y \\ \downarrow \iota & & \downarrow p \\ X & \longrightarrow & B \end{array}$$

whose obstruction class vanishes, for some simplicial pair (X, A) . Recall also that we may use $\mathcal{D}_\Delta = \mathcal{D}(\Delta, \partial\Delta, Y, B)$. Recall furthermore that $f: X \rightarrow Y$ is **internally L -Lipschitz** if f is internally equivalent to an L -Lipschitz map, relative to $\mathcal{D}_f \in \mathcal{D}(X, A, Y, B)$. That is, there is an L -Lipschitz map $g: X \rightarrow Y$ and an internal homotopy from f to g relative to $\mathcal{D}_f$.

Our goal, in some sense, is to produce a constant C depending only on $p: Y \rightarrow B$ and $N = \dim X$ such that every map f which is externally 1-Lipschitz will be internally C -Lipschitz. However, this will only be possible if the corresponding algebraic data is bounded in the right ways. Recall that the following diagram commutes up to homotopy, where the vertical maps are quasi-isomorphisms:

$$\begin{array}{ccccc} (\mathcal{M}_B^*, d) & \xrightarrow{1 \otimes 1} & (\mathcal{M}_B^* \otimes \wedge V, d) & \longrightarrow & (\wedge V, \bar{d}) \\ \downarrow m_B & & \downarrow m_Y & & \downarrow m_F \\ \Omega^*(B) & \xrightarrow{p^*} & \Omega^*(Y) & \longrightarrow & \Omega^*(F) \end{array}$$

The top row is a minimal Sullivan model for the morphism $p^*: \Omega^*(B) \rightarrow \Omega^*(Y)$. Fix now a map $f: X \rightarrow Y$, and define $\bar{f} := p \circ f$.

A **quadruple** for $(\bar{f}, f)$ as above consists of DGA morphisms $(\bar{\varphi}, \varphi, \bar{\Phi}, \Phi)$ where

$$(Q1) \quad \overline{\varphi} : \mathcal{M}_B^* \rightarrow \Omega^*(X)$$

$$(Q2) \quad \varphi : \mathcal{M}_B^* \otimes \wedge V \rightarrow \Omega^*(X)$$

$$(Q3) \quad \overline{\Phi} : \mathcal{M}_B^* \rightarrow \Omega^*(X) \otimes \mathbb{R} \langle t, dt \rangle$$

$$(Q4) \quad \Phi : \mathcal{M}_B^* \otimes \wedge V \rightarrow \Omega^*(X) \otimes \mathbb{R} \langle t, dt \rangle$$

such that

$$(H5) \quad \overline{\varphi} \simeq \overline{f}^* m_B \text{ via } \overline{\Phi}$$

$$(H6) \quad \varphi \simeq f^* m_Y \text{ via } \Phi$$

$$(H7) \quad \Phi|_A \text{ is constant (in particular, } \varphi|_A = f^* m_Y|_A).$$

We say that (φ, Φ) **extends** $(\overline{\varphi}, \overline{\Phi})$ if in addition

$$(E1) \quad \varphi|_{\mathcal{M}_B^*} := \varphi \circ (\mathbb{1} \otimes 1) = \overline{\varphi}$$

$$(E2) \quad \Phi|_{\mathcal{M}_B^*} := \Phi \circ (\mathbb{1} \otimes 1) = \overline{\Phi}.$$

Unless otherwise stated, all quadruples will be assumed to be two pairs that extend each other. Ignoring the complex A , quadruples fit into the diagram of [Figure 4.1](#), where the top and bottom triangles commute exactly, while the left and right triangles commute up to homotopy through $\overline{\Phi}$ and Φ respectively.

We say that this quadruple $(\overline{\varphi}, \varphi, \overline{\Phi}, \Phi)$ is **bounded by L downstairs** if

$$(B1) \quad \text{Dil}(\overline{\varphi}) \leq L$$

$$(B2) \quad \text{Dil}(\varphi) \leq L$$

$$(B3) \quad \text{Dil}_{1/L}(\overline{\Phi}) \leq L.$$

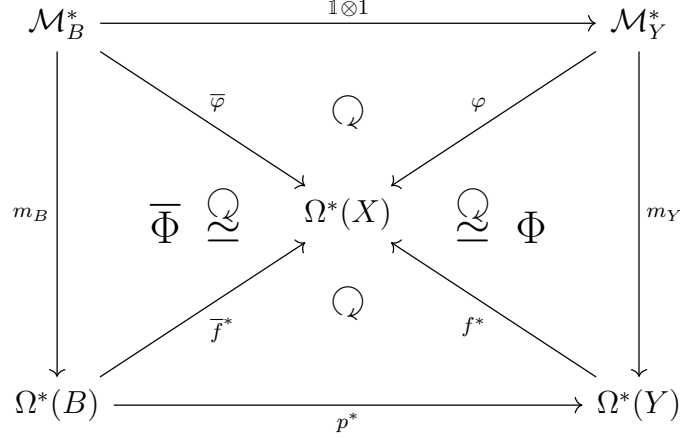


Figure 4.1: Quadruples

The quadruple is **totally bounded by L** (or a **totally L -bounded quadruple**) if in addition to being bounded by L downstairs we have

$$(B4) \quad \text{Dil}_{1/L}(\Phi) \leq L.$$

For a subcomplex $X' \subseteq X$ we say that $(\bar{\varphi}, \varphi, \bar{\Phi}, \Phi)|_{X'}$ is a totally L -bounded quadruple for $(\bar{f}, f)|_{X'}$ if $(\bar{\varphi}, \varphi, \bar{\Phi}, \Phi)$ is a quadruple for $(\bar{f}, f)$ whose appropriate restrictions to X' are all L -bounded. In this situation we may say more concisely that **the restriction of the quadruple to X'** is bounded.

The remainder of this chapter has strong roots in the original paper [23], but the current approach (induction up the Postnikov tower) allows the subsequent ideas to be presented somewhat independent of one another. Still, almost all the ideas present here can be found in the original paper. Before moving into the relevant lemmata, we set up notation which will help streamline some of the proofs.

Definition 4.16. Given maps $f, g: \Delta \rightarrow Y$ for which $f|_{\partial\Delta} = g|_{\partial\Delta}$, define $\hat{b}_f: \mathbf{S}^k \rightarrow Y$ to be the map given by $\hat{b}_f|_+ = g$ on the northern hemisphere and $\hat{b}_f|_- = f$ on the southern hemisphere, glued along their common boundary.

Recall from [Corollary 2.16](#) that since $p: Y \rightarrow B$ is principal, if f and g are externally equivalent then the obstruction to performing an internal homotopy from f to g (i.e., a homotopy which always fits in the diagonal of their common external diagram) defines a class

$$\begin{bmatrix} \bar{b} \\ f \end{bmatrix} \in H^k(\Delta, \partial\Delta; \pi_n(F)) = H^k(\Delta, \partial\Delta; G) \cong \begin{cases} G & k = n, \\ 0 & k \neq n. \end{cases}$$

We may write simply $[f]$ for $\begin{bmatrix} \bar{b} \\ f \end{bmatrix}$ if g is fixed ahead of time.

Definition 4.17. Let $\mathcal{D} \in \mathcal{D}_\Delta$ be an externally 1-Lipschitz diagram. The map f_ξ as given by [Corollary 4.8](#) is called the **elect representative of $\xi \in G$ in $\mathcal{D}$** . Note that this representative is defined relative to the diagram, as long as the diagram allows some diagonal filling. If $f: \Delta \rightarrow Y$ is a specific map for which $\mathcal{D}_f = \mathcal{D}$, we may also say that f_ξ is the **elect representative of ξ for f** , though in reality f_ξ is completely agnostic about the map f itself.

In what follows we let $\square \otimes 1: G \rightarrow G \otimes \mathbb{Q} \cong V^*$ be the obvious map, noting that in case $n = 1$ this is a slight abuse of notation. We may similarly denote $(\square \otimes 1)^*: G \rightarrow V$.

Lemma 4.18. *There is a $\tilde{C} > 0$ so that the following holds. Suppose that $f: \Delta \rightarrow Y$ is externally 1-Lipschitz, and let $\bar{f} := f \circ p$. Let $(\bar{\varphi}, \varphi, \bar{\Phi}, \Phi)$ be a quadruple for $(\bar{f}, f)$ which is 1-bounded downstairs, and whose restriction to $\partial\Delta$ is totally 1-bounded.*

Let f_0 be the elect representative of $0 \in G$ relative to f , and let $\mathcal{R} := \overset{0}{f}$. Then $[\mathcal{R}] \otimes 1$ is contained in a $\tilde{C}$ -ball around

$$\int_{\Delta} \left(\varphi|_V - f^* m_Y|_V - \int_0^1 \Phi d|_V \right) \in V^*.$$

Proof. Let C_0 be given by [Corollary 4.8](#) applied to $\xi = 0$, let $C(n-1, d)$ be the constant from [Proposition 2.48](#), and define $\tilde{C} = C(n-1, d) + (C_0)^n + 1$. This constant depends

only on n and the norms of V .

The class $[\mathcal{R}] \otimes 1$ is the obstruction in $\mathcal{M}_B^* \otimes \wedge V$ to homotoping $\mathcal{R}^* m_Y$ to 0, or equivalently, to another nullhomotopic map, such as ϕ , the map which is φ on each hemisphere. First, define $\Psi|_{\mathcal{M}_B^*} := \Phi|_{\mathcal{M}_B^*}$ on each hemisphere. We now evaluate the obstruction to extending Ψ to V on the northern hemisphere. The obstruction to extending Ψ to V is given according to [Proposition 2.47](#) by

$$\left[-\phi + \mathcal{R}^* m_Y + \int_0^1 \Psi d|_V \right] \in V^*.$$

Now because $\text{Dil}_1(\Phi|_{\mathcal{M}_B^*}) = \text{Dil}_1(\bar{\Phi}) \leq 1$, by [Proposition 2.48](#) the obstruction on the northern hemisphere is represented by a class $\mathcal{O}$ of norm

$$\begin{aligned} \|\mathcal{O}\|_{op} &\leq C(n-1, d) \text{Dil}_1(\Psi|_{\mathcal{M}_B^*})^{n+1} + \text{Dil}(f_0^* m_Y)^n + \text{Dil}(\varphi)^n \\ &= C(n-1, d) + (C_0)^n + 1 \\ &= \tilde{C}. \end{aligned}$$

while on the southern hemisphere, this class sends $v \in V$ to

$$\int_{\Delta} (\varphi(v) - f^* m_Y(v) - \int_0^1 \Phi(dv)).$$

and therefore, $[\mathcal{R}] \otimes 1$ is within distance $\tilde{C}$ of the term as claimed. $\square$

Lemma 4.19. *There is a constant C such that the following holds. Suppose the following diagram commutes:*

$$\begin{array}{ccc} A & \xrightarrow{f|_A} & Y \\ \downarrow \iota & \nearrow f & \downarrow p \\ X & \xrightarrow{\bar{f}} & B \end{array}$$

Suppose that $f|_A$ and $\bar{f}$ are 1-Lipschitz. Let $(\bar{\varphi}, \varphi, \bar{\Phi}, \Phi)$ be a quadruple for $(\bar{f}, f)$ which is 1-bounded downstairs, and whose restriction to A is totally 1-bounded. Suppose moreover that $\text{Lip } f|_{X^{(n-1)}} \leq 1$ and $\text{Dil}_1(\Phi|_{X^{(n-1)}}) \leq 1$.

Then there is a homotopy H from f to a map $\tilde{f}: X \rightarrow Y$ such that

$$(1) \ H|_{(X^{(n-2)} \cup A) \times \{t\}} = f|_{X^{(n-2)} \cup A} \text{ for all } t \in I \text{ (i.e., } H \text{ is a homotopy rel } X^{(n-2)} \cup A)$$

$$(2) \ p \circ H_t = \bar{f} \text{ for all } t \in I \text{ (i.e., } H \text{ is a fiberwise homotopy)}$$

$$(3) \ \tilde{f}|_{X^{(n-1)}} = f|_{X^{(n-1)}}$$

$$(4) \ \tilde{f} \text{ is } C\text{-Lipschitz.}$$

Proof. We assume without loss of generality that the star of A retracts to A after further subdivision if necessary. Let $\Delta \subseteq X$ be an n -simplex. For each $\zeta \in G$, let C_ζ be as in the consequence of [Corollary 4.8](#). Let $\tilde{C}$ be as in the result of [Lemma 4.18](#). Define

$$G_{\tilde{C}} := \left\{ \zeta \in G \mid \|\zeta \otimes 1\|_{V^*} \leq \tilde{C} + 1 \right\}.$$

Since $\ker(\square \otimes 1)$ is torsion, $\#G_{\tilde{C}} < \infty$ is a constant depending only on the same data as $\tilde{C}$, and so we may define a constant C_n depending again only on the same data as $\tilde{C}$ by

$$C_n := \max \{C_\zeta \mid \zeta \in G_{\tilde{C}}\}.$$

Now, let the map f be given. For each $\zeta \in G_{\tilde{C}}$, let $f_\zeta: \Delta \rightarrow Y$ be the elect representative of ζ relative to $\mathcal{D}_f$, as given by [Corollary 4.8](#). We may define a shard complex on Y by

$$\mathcal{Z} = \left\{ f_\zeta|_\Delta \mid f_\zeta|_{\partial\Delta} = f|_{\partial\Delta}, \zeta \in G_{\tilde{C}} \right\} \cup \mathcal{A}$$

where $\mathcal{A}$ is a given shard complex for which $f|_{A \cup X^{(n-1)}}$ is mosaic.

We show that there is an element $\xi \in G_{\tilde{C}}$ such that (the restriction to Δ of) the elect representative $\tilde{f} := f_\xi$ of ξ relative to f is mosaic with respect to this complex, and that this shard complex has bounded Lipschitz constant. The process goes by building a sufficiently nice homotopy on the boundary of each $\Delta^n \subseteq X^{(n)}$, so that the obstruction to extending to Δ is as desired.

Let $\beta = \int_0^1 \Phi$, and observe that for $v \in V$ we have $d\beta(v) = \varphi(v) - f^*m_Y(v) - \int_0^1 \Phi(dv)$. Let $b \in C^{n-1}(X, A; G)$ with the property that the image of $\langle b, \sigma \rangle \in G$ inside V^* is as close in norm as possible (say, within distance 1) to $\|\int_\sigma \beta|_V\|_{V^*}$ for any $(n-1)$ -simplex $\sigma \subseteq X^{(n-1)}$, but otherwise arbitrary.

Let $\Sigma \subseteq X^{(n)}$ be a simplex. Set $\xi := \langle \delta b, \Sigma \rangle - [f|_\Sigma]$. Let $\tilde{f}|_\Sigma := (f|_\Sigma)_\xi$ be the elect representative of ξ relative to $f|_\Sigma$. In particular, f_ξ is externally equivalent to $f|_\Sigma$.

Claim 4.19.1

The map $\tilde{f}|_\Sigma$ has $\text{Lip } \tilde{f}|_\Sigma \leq C_n$.

Proof of Claim: Since $\tilde{C}$ was the result of [Lemma 4.18](#), observe that $[f|_\Sigma] \otimes 1$ is contained in a $\tilde{C}$ -ball around

$$\int_\Sigma (\varphi|_V - f^*m_Y|_V - \int_0^1 \Phi d|_V) = \int_\Sigma d\beta(v).$$

As a result, $[f|_\Sigma] \otimes 1$ is contained within a $\tilde{C} + 1$ ball around $\langle \delta b, \Sigma \rangle \otimes 1$ by Stokes' Theorem. In particular,

$$\|\xi \otimes 1\| = \left\| \left([f|_\Sigma] - \langle \delta b, \Sigma \rangle \right) \otimes 1 \right\| \leq \tilde{C} + 1$$

Therefore, $\xi \in G_{\tilde{C}}$. Since $\tilde{f}|_\Sigma$ was elect, we have $\text{Lip } \tilde{f} \leq C_\xi \leq C_n$. ■

It remains to construct the desired homotopy H from f to $\tilde{f}$. For all $(n-2)$ -cells τ

Define $H|_{\tau \times \{t\}} = f|_{\tau}$ for all $t \in I := [0, 1]$. Let $\sigma \subseteq \Sigma$ be an $(n-1)$ -face. Define $H|_{\sigma \times \{1\}} := H|_{\sigma \times \{0\}} = f|_{\sigma}$. This provides a complete definition for $H|_{\partial(\sigma \times I)}$, which makes the following diagram commute for each $(n-1)$ -face $\sigma \subseteq \Sigma$:

$$\begin{array}{ccc} \partial(\sigma \times I) & \xrightarrow{H|_{\partial(\sigma \times I)}} & Y \\ \downarrow & \nearrow \text{dashed} & \downarrow \\ \sigma \times I & \xrightarrow{(\bar{f} \circ \pi)|_{\sigma}} & B \end{array}$$

By [Corollary 2.16](#) the obstruction to extending this to a map which fits in the dashed diagonal defines an element of G . There is well-defined extension $H|_{\sigma \times I}$ whose degree is $\langle b, \sigma \rangle \in G$. That is, precomposing H with a linear retract of σ induces a canonical nullhomotopy of this map, and one can find a filling $H|_{\sigma \times I}$ whose degree is as desired. This gives a complete definition of H on $\partial(\Sigma \times I) = (\Sigma \times \{0\}) \cup (\partial\Sigma \times I) \cup (\Sigma \times \{1\})$. It remains to show that H extends to all of $\Sigma \times I$.

Claim 4.19.2

There is an extension of $H|_{\Sigma}$ which defines a homotopy from $f|_{\Sigma}$ to $\tilde{f}|_{\Sigma}$.

Proof of Claim: Consider the following diagram:

$$\begin{array}{ccc} \mathbf{S}^n \cong (\Sigma \times \{0\}) \cup (\partial\Sigma \times I) \cup (\Sigma \times \{1\}) & \xrightarrow{f \cup H|_{\partial\Sigma \times I} \cup \tilde{f}} & Y \\ \downarrow & \nearrow \text{dashed} & \downarrow p \\ \mathbf{D}^{n+1} \cong \Sigma \times I & \xrightarrow{\bar{f} \circ \pi} & B \end{array}$$

By [Proposition 2.12](#), an extension of H will fit in this diagram if and only if the obstruction class $[H] \in G$ vanishes. By Stokes' Theorem this means that such a homotopy exists if and only if

$$[H|_{\Sigma \times \{1\}}] - [H|_{\Sigma \times \{0\}}] = [H|_{(\partial\Sigma) \times I}].$$

In particular, this happens if and only if

$$[\tilde{f}] - [f] = \langle \delta b, \Sigma \rangle$$

which is true by definition of $[\tilde{f}] = \xi = \langle \delta b, \Sigma \rangle - [f]$. Therefore H extends to an internal homotopy from f to $\tilde{f}$. ■

Therefore, $\tilde{f}|_{X^{(n)}}$ is C_n -Lipschitz. In fact, $\tilde{f}|_{X^{(n)}}$ is mosaic with respect to $\mathcal{Z}_n$. We redefine $\tilde{f}_n = \tilde{f}$ on higher skeleta of X using [Proposition 4.6](#) as in the proof of [Corollary 4.7](#).

Namely, inductively for $n < k \leq N = \dim X$ the diagram

$$\begin{array}{ccc} A \cup X^{(k-1)} & \xrightarrow{\tilde{f}_{k-1}|_{A \cup X^{(k-1)}}} & Y \\ \downarrow & \nearrow \tilde{f}_{k-1}|_{X^{(k)}} & \downarrow p \\ A \cup X^{(k)} & \xrightarrow{p \circ f|_{A \cup X^{(k)}}} & B \end{array}$$

is externally C_{k-1} -Lipschitz. We attain an (*a priori*) externally equivalent C_k -Lipschitz map $\tilde{f}_k|_{X^{(k)}} : X^{(k)} \rightarrow Y$ for which C_k depends only on C_{k-1} and on $p : Y \rightarrow B$. As before, the obstruction to performing an internal homotopy between these maps is an element of $H^n(A \cup X^{(k)}, A \cup X^{(k-1)}; G)$, which is trivial since $k \neq n$. In particular, $\tilde{f}_k|_{X^{(k)}}$ must in fact be *internally* equivalent to $f_{k-1}|_{X^{(k)}}$ over this diagram, and we extend $\tilde{f}_k$ over higher dimensional cells to a map $\tilde{f}_k$ on all of X which is internally equivalent to f itself, and whose restriction to lower skeleta is as desired. This extends the shard complex as well. When $k = N$ we attain (noting slight equivocation on the symbol $\tilde{f}$) the desired map $\tilde{f} := \tilde{f}_N$. □

Lemma 4.20. *Let $(\overline{\varphi}, \varphi, \overline{\Phi}, \Phi)$ and $(\overline{f}, f)$ be as in the setup to [Lemma 4.19](#) and let $\tilde{f}$ and H be as in the conclusion. In particular, assume that $\text{Dil}_1(\Phi|_{X^{(n-1)}}) \leq 1$.*

Let $\pi : X \times I \rightarrow X$ be the obvious projection. Then there is a homotopy

$$\Psi : \mathcal{M}_B^* \otimes \wedge V \rightarrow \Omega^*(X \times I) \otimes \mathbb{R} \langle s, ds \rangle$$

from $\pi^*\varphi$ to H^*m_Y for which $\Psi|_{t=0} = \Phi$, and such that $\tilde{\Phi} := \Psi|_{t=1}$ is a homotopy from φ to $\tilde{f}^*m_Y \text{ rel } A$ for which $\text{Dil}_1(\tilde{\Phi}) \lesssim 1$. In particular, $(\bar{\varphi}, \varphi, \bar{\Phi}, \tilde{\Phi})$ is a quadruple for $(\bar{f}, \tilde{f})$ which is totally $(\lesssim 1)$ -bounded.

Proof. First, define $\Psi|_{\mathcal{M}_B^*} := \pi^*\Phi|_{\mathcal{M}_B^*} = \pi^*(\bar{\Phi} \otimes 1)$. Observe that $\Psi|_{\mathcal{M}_B^*}$ is a homotopy from $\pi^*\varphi|_{\mathcal{M}_B^*}$ to $H^*m_Y|_{\mathcal{M}_B^*}$, since

$$(\Psi|_{\mathcal{M}_B^*})|_{X \times \{0\}} = \pi^*(\Phi|_{t=0})|_{\mathcal{M}_B^*} = \pi^*\varphi|_{\mathcal{M}_B^*}$$

and (because $p \circ H = p \circ f \circ \pi$) we furthermore have

$$(\Psi|_{\mathcal{M}_B^*})|_{X \times \{1\}} = \pi^*(\bar{\Phi}|_{t=1} \otimes 1)|_{\mathcal{M}_B^*} = \pi^*\bar{f}^*m_B = (p \circ f \circ \pi)^*m_B = H^*p^*m_B = H^*m_Y|_{\mathcal{M}_B^*}.$$

This defines Ψ (hence, $\tilde{\Phi}$) on $\mathcal{M}_B^*$, and in particular on $\text{im}(d|_V)$. We now seek to extend Ψ to V .

Recall that $V = \text{Hom}(G, \mathbb{Q})$ is of degree n , and that $\beta = \int_0^1 \Phi$. The primary goal is to define a bounded $\tilde{\beta}|_{X^{(n)}}$ on V for which $\tilde{\beta}(v) = \varphi(v) - \tilde{f}^*m_Y(v) - \int_0^1 \Phi(dv)$, and hence to define

$$\tilde{\Phi}(v) := \tilde{f}^*m_Y(v) + d(\tilde{\beta}(v) \otimes s) + \int_0^s \Phi(dv).$$

On $X^{(n-2)}$ we may define $\Psi(v)$ constantly equal to $\Phi(v)$, but on $X^{(n-1)}$ we will want to be somewhat more careful. The obstruction to defining such a homotopy Ψ from Φ to such a map $\tilde{\Phi}$ is (by [Proposition 2.46](#)) a class $\mathcal{O}(v) \in H^n(X \times I, X \times \{0, 1\}; V)$, defined

by

$$\mathcal{O}(v) = (\pi^*\varphi(v) - H^*m_Y(v) - \int_0^1 \Psi(dv); \beta(v) \oplus \tilde{\beta}(v)).$$

We analyze this obstruction class in order to determine a proper definition for $\tilde{\beta}$. To aid in this, we denote $\mathfrak{D}(v) := \pi^*\varphi(v) - H^*m_Y(v) - \int_0^1 \Psi(dv)$. First, note that $\mathfrak{D}(v)|_{X^{(n-2)} \times I} = 0$, and therefore for any $(n-1)$ -simplex $\sigma \subseteq X^{(n-1)}$ we have $\mathfrak{D}(v)|_{(\partial\sigma) \times I} = 0$ as well. Hence, by the Poincaré Lemma, there is a form $\mathcal{B}(v)$ for which $d\mathcal{B}(v) = \mathcal{O}(v)$, and whose restriction to $X^{(n-2)} \times I$ vanishes.

Therefore, it is enough to analyze $\mathfrak{B}(v)|_{X \times \{0,1\}}$ and thus to determine the class of $\tilde{\beta}(v)$. Let $b \in C^{n-1}(X, A; G)$ be as in the previous lemma. In particular, recall that H was defined so that $[H|_{\sigma \times I}] = \langle b, \sigma \rangle$.

Claim 4.20.1

For any $(n-1)$ -simplex σ we have

$$\int_{\sigma \times I} d\mathfrak{B}(v) = -\langle b, \sigma \rangle^*(v).$$

Proof of Claim: Identifying $\sigma \times \{0\} \sim \sigma \times \{1\}$ and flattening $\partial\sigma \times I$ to a single copy $\partial\sigma$, we form a quotient $Q : \sigma \times I \rightarrow \mathbf{S}^n$. Observe that $\mathfrak{D}(v)|_{\sigma \times \{0\}} = \mathfrak{D}(v)|_{\sigma \times \{1\}}$, and that $\mathfrak{D}(v)|_{\partial\sigma \times I}$ is constant. Similarly, by [Lemma 4.19](#) we have $H|_{\sigma \times \{0\}} = H|_{\sigma \times \{1\}}$ and constant $H|_{\partial\sigma \times I}$. Therefore $\mathfrak{D}(v)$ and H each factor through Q . In particular, we view $\int_{\sigma \times I} \mathfrak{D}(v)$ as a map $V \rightarrow \Omega^n(\mathbf{S}^n) \cong \mathbb{R}$ (and therefore as an element of $V^* \cong V$ itself) which is the obstruction to performing a homotopy from H^*m_Y to $\pi^*(\varphi|_\sigma)$ within $\mathbf{S}^n$. However, $\pi^*(\varphi|_\sigma)$ factors through $\Omega^*(\sigma) \cong \Omega^*(\mathbf{D}^{n-1})$ and is therefore algebraically nullhomotopic.

Hence, $\int_{\sigma \times I} d\mathfrak{B}$ is equivalently seen as the obstruction to nullhomoping H^*m_Y in

the quotient. By (2.40), the obstruction to an algebraic nullhomotopy of H^*m_Y is the same as the rationalization of the obstruction to a geometric nullhomotopy of H in this quotient. But this obstruction is exactly measured by the homotopy class of the torsor element $-[H|_{\sigma \times I}] = [H|_{\sigma \times \{1\}}] - [H|_{\sigma \times \{0\}}] \in G$, and so as required we get

$$\int_{\sigma \times I} d\mathfrak{B}(v) = -[H|_{\sigma \times I}]^*(v) = -\langle b, \sigma \rangle^*(v). \quad \blacksquare$$

Therefore, it is enough that $\tilde{\beta}$ satisfy

$$\int_{\sigma} \tilde{\beta}(v) = \int_{\sigma} \beta(v) - \langle b, \sigma \rangle^*(v) \quad (4.21)$$

since then

$$\begin{aligned} \int_{\sigma \times \{0\}} \mathfrak{B}(v) - \int_{\sigma \times \{1\}} \mathfrak{B}(v) &= \int_{\partial(\sigma \times I)} \mathfrak{B}(v) && (\text{since } \mathfrak{B}(v)|_{(\partial\sigma) \times I} = 0) \\ &= \int_{\sigma \times I} d\mathfrak{B}(v) && (\text{Stokes' Theorem}) \\ &= -\langle b, \sigma \rangle^*(v) && (\text{Claim 4.20.1}) \\ &= -\langle b, \sigma \rangle^*(v) + \int_{\sigma} \beta(v) - \int_{\sigma} \beta(v) \\ &= \int_{\sigma} \tilde{\beta}(v) - \int_{\sigma} \beta(v). && (\text{Property (4.21)}) \end{aligned}$$

Concretely, define $\tilde{\beta}|_{\sigma}$ be the volume form of σ times a bump function, scaled so that (4.21) is satisfied. This completes the definition of $\Psi(v)|_{X^{(n-1)} \times I}$, and it remains to define $\Psi(v)|_{X^{(n)}}$. Let $\Sigma \subseteq X^{(n)}$ be an n -simplex. By Lemma 4.14, there is an extension $\tilde{\beta}(v)|_{\Sigma}$ of $\tilde{\beta}(v)|_{\partial\Sigma}$ to all of Σ for which $\|\tilde{\beta}\|_{op} \leq O(1)$, and such that $d\tilde{\beta} = \varphi(v) - \tilde{f}^*m_Y(v) - \int_0^1 \Phi(dv)$.

Define

$$\tilde{\Phi}(v) := \tilde{f}^*m_Y(v) + d(\tilde{\beta}(v) \otimes s) + \int_0^s \Phi(dv).$$

Therefore, we have

$$\text{Dil}_1 \left(\tilde{\Phi}|_{X^{(n)}} \right) \leq \text{Lip } \tilde{f} + O(1) + 1 \lesssim 1,$$

as required.

Finally, we determine $\tilde{\Phi}$ on higher dimensional skeleta. Let Δ be a k -simplex for some $k > n$. We inductively apply [Lemma 4.14](#) (swapping the roles of n and k) to $\alpha_{\partial} = \tilde{\beta}(v)|_{\partial\Delta}$ and $\omega = \varphi(v) - \tilde{f}^*m_Y(v) - \int_0^1 \Phi(dv)$ to produce an extension $\tilde{\beta}(v)|_{\Delta}$ of $\tilde{\beta}(v)|_{\partial\Delta}$, for which $d\tilde{\beta}(v) = \omega(v)$ with $\|\tilde{\beta}(v)|_{\Delta}\|_{\infty} \leq O(1)$. Since v was an arbitrary unit vector this induces a definition of $\tilde{\beta} : V \rightarrow \Omega^*(\Delta)$ for which $\|\tilde{\beta}\|_{op} \leq O(1)$. Define $\tilde{\Phi}(v) = \tilde{f}^*m_Y(v) + d(\tilde{\beta}(v) \otimes s) + \int_0^s \Phi(dv)$ as before. Since $k > n$ there are no obstructions to extending Ψ to a homotopy from Φ to $\tilde{\Phi}$ over these cells, completing the proof. $\square$

We put all of these pieces together to get the following theorem, which we state for simplicial sets, and in particular can be applied to compact Riemannian manifolds:

Theorem 4.22 (The Shadowing Principle for Lifts of Lipschitz Maps). *Let the following diagram commute:*

$$\begin{array}{ccc} A & \xrightarrow{f|_A} & Y \\ \downarrow \iota & \nearrow f & \downarrow p \\ X & \xrightarrow{\bar{f}} & B \end{array}$$

where p is a principal $F = K(G, n)$ fibration of compact Riemannian manifolds with boundary, and where (X, A) is a finite N -dimensional simplicial pair. Assume that $\text{Lip } f|_A \leq L$ and that $\text{Lip } \bar{f} \leq L$. Moreover, suppose that there is a DGA map $\varphi : \mathcal{M}_B^* \otimes \wedge V \rightarrow \Omega^*(X)$ with $\varphi|_{\mathcal{M}_B^*} = \bar{f}^*m_B$, and $\text{Dil}(\varphi) \leq L$ such that $\varphi \simeq f^*m_Y$ by a homotopy Φ with $\Phi|_{\mathcal{M}_B^*} = \bar{f}^*m_B \otimes 1$.

Then f is homotopic rel A to a $C(L+1)$ -Lipschitz map $\tilde{f}$ for which $p \circ \tilde{f} = \bar{f}$, and such that $\varphi \simeq \tilde{f}^*m_Y$ by a homotopy $\tilde{\Phi} : \mathcal{M}_B^* \otimes \wedge V \rightarrow \Omega^*(X) \otimes \mathbb{R}\langle t, dt \rangle$ with $\text{Dil}_{1/L}(\tilde{\Phi}) \leq C(L+1)$. Here, C is a constant depending only on $p : Y \rightarrow B$, on the minimal model of

the fibration, on the norms of V , and on N (but otherwise is independent of X).

Proof. Let $\bar{\varphi} = \varphi|_{\mathcal{M}_B^*} = \bar{f}^* m_B$ and $\bar{\Phi} = \bar{\varphi} \otimes 1$. After taking a (regular) subdivision of (X, A) at scale $1/L$, we may assume without loss of generality that $L = 1$, and that each simplex of X is uniformly bilipschitz to the standard simplex. At the cost of slightly increasing the Lipschitz constant, and a slight adjustment to the DGA map $\varphi|_{\star(A)}$, we may assume without loss of generality (see the beginning of [23, Theorem 4.1]) that α is mosaic with respect to some fixed shard complex $\mathcal{Z}$ on Y for which $\mathcal{Z}^{(0)}$ consists of basepoints over fibers of vertices in B . By Corollary 4.7, f is internally equivalent to a map f' for which $f'|_{X^{(n-1)}}$ is C_{n-1} -Lipschitz. Moreover, by an inductive application of Lemma 4.14 (as in the end of Lemma 4.20) we can assume without loss of generality that $\text{Dil}_1(\Phi|_{X^{(n-1)}}) \leq C_{n-1}$. By Lemma 4.19, there is an internal homotopy H to a C_N -Lipschitz map $\tilde{f}$ for which H and $\tilde{f}$ meet the assumptions of Lemma 4.20. This yields a quadruple $(\bar{\varphi}, \varphi, \bar{\Phi}, \tilde{\Phi})$ for $(\bar{f}, \tilde{f})$ which is totally $\lesssim C := C_N$ bounded, as desired. $\square$

An inductive argument allows us to replace this single fibration with fibrations which admits a refinement of principal $K(G, n)$ fibrations. In particular, we recover the original shadowing principle for simply connected targets, but more generally this applies to nilpotent spaces.

Theorem 4.23 (The Shadowing Principle for Nilpotent Targets). *Let (X, A) be an N -dimensional finite simplicial pair, and let Y be a compact nilpotent space of finite type. Let $f : X \rightarrow Y$ be a map and let $\varphi : \mathcal{M}_Y^* \rightarrow \Omega^*(X)$ be such that*

$$(1) \ f^* m_Y|_A = \varphi|_A,$$

$$(2) \ f^* m_Y \simeq \varphi \text{ rel } A,$$

$$(3) \ \text{Lip } f|_A \leq L, \text{ and}$$

(4) $\text{Dil}(\varphi) \leq L$.

Then f is homotopic rel A to a $(\lesssim L)$ -Lipschitz map $\tilde{f} : X \rightarrow Y$ such that $\tilde{f}^*m_Y \simeq \varphi$ by a homotopy $\tilde{\Phi} : \mathcal{M}_Y^* \rightarrow \Omega^*(X) \otimes \mathbb{R} \langle t, dt \rangle$ such that $\tilde{\Phi}|_A$ is constant, and $\text{Dil}_{1/L}(\tilde{\Phi}) \lesssim L$.

Proof. Write $f_k^{(j)} := p_k^{(j)} \circ f$ and $\varphi_k^{(j)} := \varphi|_{Y_k^j}$ for each k, j where $p_k^{(j)} : Y \rightarrow Y_k^j$ is the map to the relevant Postnikov stage. Let c be such that Y is c -step nilpotent up through degree N . Without loss of generality we may assume that $\text{Lip}(p_k^{(j)}) \leq 1$ for all $(k, j) \leq (N, c)$. Observe that for all such (k, j) we therefore have

$$(1) \quad (f_k^{(j)})^*m_Y|_A = (\varphi_k^{(j)})|_A,$$

$$(2) \quad (f_k^{(j)})^*m_Y \simeq (\varphi_k^{(j)}) \text{ rel } A,$$

$$(3) \quad \text{Lip}(f_k^{(j)})|_A \leq L, \text{ and}$$

$$(4) \quad \text{Dil}(\varphi_k^{(j)}) \leq L.$$

In particular there is some quadruple $(\varphi_k^{(j)}, \varphi_k^{(j+1)}, \Phi_k^{(j)}, \Phi_k^{(j+1)})$ for $(f_k^{(j)}, f_k^{(j+1)})$ whose restriction to A is totally L -bounded for all $(k, j) \leq (N, c)$. Here, we note that if $j = c$, then we may replace (k, j) with $(k+1, 0)$ and interpret the pair $(k, j+1)$ to mean $(k+1, 1)$, so that this notation still makes sense. The proof is then a simple application of [Theorem 4.22](#) inductively up the principal $K(G, n)$ refinement of the Postnikov tower $\varprojlim Y_n$ of Y .

The inductive process begins trivially with a map to the base space $Y_0 \simeq *$ which is constant. The only serious adjustment we must make is that, in order to continue induction, we require the homotopies at each stage in the Postnikov tower to extend into all of Y both geometrically, and algebraically. Concretely, assume inductively that there is a map $g : X \rightarrow Y$ such that f is homotopic rel A to g , and $g^*m_Y \simeq \varphi \text{ rel } A$ by a homotopy Ψ such that $g_K^{(J)}$ is $(\lesssim L)$ -Lipschitz, and such that $(\varphi_K^{(J)}, \varphi_K^{(J+1)}, \Psi_K^{(J)}, \Psi_K^{(J+1)})$ is

a quadruple for $(g_K^{(J)}, g_K^{(J+1)})$ which is $(\lesssim L)$ -bounded downstairs. In particular, observe that

$$(\varphi_K^{(J)}, \varphi_K^{(J+1)}, \Psi_K^{(J)}, \Psi_K^{(J+1)})|_A = (\varphi_K^{(J)}, \varphi_K^{(J+1)}, \Phi_K^{(J)}, \Phi_K^{(J+1)})|_A$$

and therefore this restriction to A is totally $(\lesssim L)$ -bounded. Since the skeleta of each Y_k^j are finite, we may apply [Theorem 4.22](#) to obtain

- a map $\tilde{g}_K^{(J+1)}: X \rightarrow Y_K^{J+1}$
- a homotopy $\tilde{\Psi}_K^{(J+1)}$ from $\varphi_K^{(J+1)}$ to $(\tilde{g}_K^{(J+1)})^* m_{Y_K^{J+1}}$

for which

- $(\varphi_K^{(J)}, \varphi_K^{(J+1)}, \Psi_K^{(J)}, \tilde{\Psi}_K^{(J+1)})$ is a totally $(\lesssim L)$ -bounded quadruple for $(g_K^{(J)}, \tilde{g}_K^{(J+1)})$
- the diagram corresponding to $\tilde{g}_K^{(J+1)}$ is internally $(\lesssim L)$ -Lipschitz.

There is no topological obstruction to lifting $\tilde{g}_K^{(J+1)}$ up the remainder of the tower to produce a genuine map $g_K^{J+1}: X \rightarrow Y$. To get a homotopy $\Psi_K^{(J+1)}: \mathcal{M}_Y^* \rightarrow \Omega(X) \otimes \mathbb{R}\langle t, dt \rangle$ out of $\tilde{\Psi}_K^{(J+1)}: \mathcal{M}_{Y_K^{J+1}}^* \rightarrow \Omega(X) \otimes \mathbb{R}\langle t, dt \rangle$, we apply [Proposition 2.47](#) to the diagram

$$\begin{array}{ccccc} \mathcal{M}_{Y_K^{J+1}}^* & \xrightarrow{\tilde{\Psi}_K^{(J+1)}} & \Omega^*(X \times I) \otimes \mathbb{R}\langle s, ds \rangle & & \\ \downarrow & \nearrow \text{dashed} & \downarrow t=0 & \searrow & \\ \mathcal{M}_Y^* & \xrightarrow{\Psi_K^J} & \Omega^*(X) \otimes \mathbb{R}\langle s, ds \rangle & \xlongequal{\quad} & \Omega^*(X) \otimes \mathbb{R}\langle s, ds \rangle \end{array}$$

in which the middle vertical map is a quasi-isomorphism.

The result is a map $\tilde{g}: X \rightarrow Y$ and a homotopy $\tilde{\Psi}$ such that $f \simeq \tilde{g} \text{ rel } A$, and $\tilde{g}^* m_Y \simeq \varphi \text{ rel } A$ via $\tilde{\Psi}$. In particular $\tilde{g}_K^{(J+1)}$ is $(\lesssim L)$ -Lipschitz, and $(\varphi_K^{(J)}, \varphi_K^{(J+1)}, \tilde{\Psi}_K^{(J)}, \tilde{\Psi}_K^{(J+1)})$ is a quadruple for $(\tilde{g}_K^{(J+1)}, g_K^{(J+2)})$ which is $(\lesssim L)$ -bounded downstairs. This completes the inductive step. Once $n = N+1$, the map $g_{N+1}^{(0)}$ may therefore be identified with $g: X \rightarrow Y$ itself, and so we take $\tilde{f} = g_{N+1}^{(0)}$ and $\tilde{\Phi} = \Psi_{N+1}^{(0)}$, and the result follows. $\square$

One of the main challenges for applying the Shadowing Principle is that of finding DGA morphisms which satisfy the bounded conditions of the theorem. While there are some algorithmic processes for constructing such maps, the process does not in general produce optimal bounds. Even in the case of maps $\mathbf{S}^3 \rightarrow \mathbf{S}^2$ the bounds attained in this way are not sharp. Regardless, these algorithmic processes still provide some insight into general bounds, which we explore next.

Chapter 5

Upper Bounds on Volumes of Nullhomotopies

Throughout this chapter, we primarily work with the minimal model $\mathcal{M}_Y^* = \wedge V$ of a space Y which is c -step nilpotent (up through some finite stage), where $V = \{V_n\}_{n \geq 1}$. Since Y is c -step nilpotent, we use our notation in [Chapter 2](#). Recall then that $V_n = V_n^{\leq c}$, where $V_n^{\leq J} = \bigoplus_{1 \leq j \leq J} \Lambda_n^{(j)}$. Moreover, we recall that, using the notation of [\(2.17\)](#) for the principal refinement of the Postnikov tower of Y , we have a sequence of elementary extensions

$$\mathcal{M}_{Y_n^{J-1}}^* \subseteq \mathcal{M}_{Y_n^{J-1}}^* \langle \Lambda_n^{(J)} \rangle = \mathcal{M}_{Y_n^J}^*$$

dual to $p_n^{(J)} : Y_n^J \rightarrow Y_n^{J-1}$, and therefore that in particular the extension $\mathcal{M}_{Y_{n-1}}^* \subseteq \mathcal{M}_{Y_n}^*$ passes through a refinement of c -many elementary extensions.

Armed with the Nilpotent Shadowing Principle, the same proof as in [\[23, §5\]](#) leads to the following theorem. We state and prove only for the scenario pertaining to nullhomotopic maps, but note that the original statement and proof can be used more generally to bound homotopies between general maps as well.

Theorem 5.1 ([23, Thm. 5.7]). *Let X be a finite n -dimensional simplicial complex with the standard metric. Let $f : X \rightarrow Y$ be a nullhomotopic Lipschitz map and let*

$$\Phi : \mathcal{M}_Y^* \rightarrow \Omega^*(X) \otimes \mathbb{R} \langle t, dt \rangle$$

*be an algebraic nullhomotopy of f^*m_Y with $\text{Dil}_\tau(\Phi) \leq \sigma$ for some $\tau, \sigma \geq 0$. There are constants $\kappa = \kappa(X, Y)$ depending only on X and Y , and $C = C(Y, m_Y, n)$ depending in addition on m_Y such that there is a nullhomotopy of f with length $C(L_\lambda + 1)$ where*

$$L_\lambda = \sigma\tau + 1$$

and thickness $C(L_\theta + 1)$ where

$$L_\theta = \max \{ \sigma, \kappa \}.$$

Proof. Since Theorem 4.23 proves the Shadowing Principle for nilpotent targets, the proof as in [23] carries over practically verbatim. The main idea is to apply the Shadowing Principle to the pair

$$(L_\theta X \times [0, L_\lambda], L_\theta X \times \{0, L_\lambda\}).$$

In other words, we would like to take an “uncontrolled” homotopy $\mathfrak{H}$ from f to a constant map, and homotope $\mathfrak{H}$ so that its pullback is close to the realization

$$\rho\Phi : \mathcal{M}_Y^* \rightarrow \Omega^*(L_\theta X \times [0, L_\lambda]).$$

However, in order to apply the principle, $\mathfrak{H}^*m_Y$ must be homotopic rel $L_\theta X \times \{0, L_\lambda\}$ to $\rho\Phi$, which may not be the case. To resolve this, we will simultaneously concatenate $\mathfrak{H}$ with a self-homotopy of f while concatenating $\rho\Phi$ with an algebraic self-homotopy of f^*m_Y , while keeping the dilatation bounded, in order to produce homotopies which are

in the same relative DGA homotopy class.

To do this, let $\Psi: \mathcal{M}_Y^* \rightarrow \Omega^*(X \times \mathbf{S}^1)$ be formed by gluing $\mathfrak{H}^*m_Y$ and Φ along their common ends. That is, Ψ is $\mathfrak{H}^*m_Y$ on one half of the circle and Φ on the other half. Inductively apply [Proposition 2.46](#) to the following diagram, whose vertical map is a quasi-isomorphism

$$\begin{array}{ccccc} \mathcal{M}_{Y_K}^* & \xrightarrow{\tilde{\Psi}_K^J} & \Omega^*(X) \otimes \mathbb{R} \langle e^{(1)} \rangle & \xrightarrow{e \mapsto 0} & \Omega^*(X) \\ \downarrow & & \downarrow e \mapsto d\theta & \nearrow |_{\theta=0} & \\ \mathcal{M}_{Y_K^{J+1}}^* & \xrightarrow{\Psi|_{Y_K^{J+1}}} & \Omega^*(X \times \mathbf{S}^1) & & \end{array}$$

to attain a lift

$$\begin{array}{ccc} & \Omega^*(X) \otimes \mathbb{R} \langle e^{(1)} \rangle & \xrightarrow{e \mapsto 0} \Omega^*(X) \\ & \downarrow e \mapsto d\theta & \nearrow |_{\theta=0} \\ \mathcal{M}_Y^* & \xrightarrow{\Psi} \Omega^*(X \times \mathbf{S}^1) & \end{array}$$

This will allow us to define the desired algebraic self-homotopy.

By [\[23, Proposition 3.7\]](#), there is a group structure on $[\mathcal{M}_Y^*, \Omega^*(X) \otimes \mathbb{R} \langle e \rangle]_{f^*m_Y}$ with addition $\boxplus$ acting essentially as concatenation. By [\[10, Lemma 5.2\]](#) the realization map

$$\pi_1(Y^X; f) \rightarrow [\mathcal{M}_Y^*, \Omega^*(X) \otimes \mathbb{R} \langle e \rangle]_{f^*m_Y}$$

is $P(\text{Lip } f)$ -surjective for some polynomial P depending only on X and Y . (Note that a morphism from a finitely generated group to a normed vector space is called C -surjective if the image is C -coarsely dense.) In particular we may define

$$\Xi = f^*m_Y + \eta \otimes e: \mathcal{M}_Y^* \rightarrow \Omega^*(X) \otimes \mathbb{R} \langle e \rangle$$

in such a way that $\|\eta|_{V_k}\|_{op} \leq P(\|[f]\|_{Lip}) = P(0) =: \kappa(X, Y)$ is a constant depending

only on X and Y , since f is nullhomotopic. Moreover, such a Ξ may be found for which

$$[\Xi \boxplus \widetilde{\Psi}] \in [\mathcal{M}_Y^*, \Omega^*(X) \otimes \mathbb{R} \langle e \rangle]_{f^*m_Y}$$

is the rationalization of a class in $\pi_1(Y^X; f)$, i.e., of the class of a self-homotopy of $f \in Y^X$. Let F be any self-homotopy of f representing that class. Observe that rescaling $f^*m_Y + \eta \otimes t$ produces a map

$$\rho(f^*m_Y + \eta \otimes t) : \mathcal{M}_Y^* \rightarrow \Omega^*(L_\theta X \times [0, 1]).$$

with dilatation ≤ 1 . Define

$$\widetilde{\Phi} : \mathcal{M}_Y^* \rightarrow \Omega^*(L_\theta X \times [0, \sigma\tau + 1]) = \Omega^*(L_\theta X \times [0, L_\lambda])$$

to be the morphism obtained by concatenating $\rho(f^*m_Y + \eta \otimes t)$ and $\rho\Phi$, after rescaling as needed. Moreover, define

$$\widetilde{\mathfrak{H}} : L_\theta X \times [0, L_\lambda] \rightarrow Y$$

to be the map attained by concatenation of rescaled versions of F and $\mathfrak{H}$. Observe now that concatenation of $\widetilde{\Phi}$ with $\widetilde{\mathfrak{H}}^*m_Y$ on each side creates a map $\mathcal{M}_Y^* \rightarrow \Omega^*(X \times \mathbf{S}^1)$ which is homotopic rel $X \times \{0\}$ to $\rho(\Xi \boxplus \widetilde{\Psi})$. By the choice of F , the same is true of $\widetilde{\mathfrak{H}}^*m_Y$. Hence, $\widetilde{\Phi}$ and $\widetilde{\mathfrak{H}}^*m_Y$ are homotopic relative to their ends. Moreover, $\widetilde{\Phi}$ has dilatation bounded by 1. By [Theorem 4.23](#) applied to $\widetilde{\Phi}$ and $\widetilde{\mathfrak{H}}$, we attain the controlled nullhomotopy of f we were looking for. $\square$

Using this, we will get upper bounds on the volumes of nullhomotopies in nilpotent spaces. In particular, the goal of [Section 5.2](#) will be to prove the following proposition:

Proposition 5.2. *Let X be a finite N -dimensional simplicial complex with the standard*

metric. Let $\varphi : \mathcal{M}_Y^* \rightarrow \Omega^*(X) \otimes \mathbb{R} \langle t, dt \rangle$ have $\text{Dil}(\varphi) \leq L$, and let Φ be a nullhomotopy which can be built formally up through degree N without encountering any nonzero obstructions that make extendability dependent on choices made in lower degrees. Then writing Φ_i^j for the $t^i dt^j$ term of Φ , for all $1 < n \leq N$ we have

$$\|\Phi_i^j|_{V_n}\|_{op} \leq C(c, n, Y) L^{(n-1)(4c-1)-c+1} \lesssim L^{n(4c-1)-4c+1}$$

while for $n = 1$ we have

$$\|\Phi_i^j|_{V_1}\|_{op} \leq C(c, Y) L^c$$

In particular, $\text{Dil}_{L^{-4c+1}}(\Phi) \leq CL^{4c-1}$ for $N > 1$ and $\text{Dil}_{L^{c-1}}(\Phi) \leq CL$ for $N = 1$.

This fact is the analogue of [23, eq. 3.10], and therefore once we have proved this fact the proof of [23, Thm. 5.5] yields the following:

Theorem 5.3. *Let Y be a c -step nilpotent space up through dimension n , and let X be a finite simply connected complex of dimension n . Then there is a constant $C = C(X, Y)$ such that any nullhomotopic L -Lipschitz map $f : X \rightarrow Y$ is nullhomotopic via a C -length nullhomotopy of thickness $O(L^{4c-1})$.*

Proof (Assuming Proposition 5.2). It suffices to show that we can build a nullhomotopy Φ by formal extension in each degree without encountering any nontrivial obstructions, thus putting us in the setting of Proposition 5.2. The proof of this is identical to that of [23, Thm. 5.5], and we omit the details for brevity. The crux of the argument is that at each stage of the minimal model the morphism

$$[\mathcal{M}_Y^*(r+1), \Omega^*(X) \otimes \langle e \rangle]_0 \rightarrow [\mathcal{M}_Y^*(r), \Omega^*(X) \otimes \langle e \rangle]_0$$

is surjective so that obstructions to formally extending nullhomotopies vanish. That is, we

can take $\Phi : \mathbb{R} = \mathcal{M}_Y^*(0) \rightarrow \Omega^*(X) \otimes \mathbb{R} \langle t, dt \rangle$ to be the trivial map, and inductively apply [Proposition 2.48](#) to formally extend Φ from $\mathcal{M}_Y^*(r)$ to $\mathcal{M}_Y^*(r+1)$ without obstruction. We do this in such a way as to construct Φ as in [Proposition 5.2](#) for which

$$\text{Dil}_{L^{-4c+1}}(\Phi) = O(L^{4c-1}).$$

Applying [Theorem 5.1](#) the length of such a nullhomotopy is constant and has thickness bounded by $O(L^{4c-1})$. $\square$

Corollary 5.4. *If Y is c -step nilpotent up through dimension $n > 1$, then $\text{V}\delta_Y^n(L) = O(L^{(4c-1)n})$.*

Note in the application of [Proposition 5.2](#), we may have actually reduced the length of the homotopy by considering the slightly stronger bound $\|\Phi_i^j|_{V_n}\|_{op} \lesssim L^{(n-1)(4c-1)-c+1}$, but this would lead to a less symmetric result. This lack of symmetry suggests these bounds might be improved under more careful techniques.

For example, a more careful analysis of the situation when $c = 1$ (see [Section 5.3](#)) yields the following:

Proposition 5.5. *Let X be a finite N -dimensional simplicial complex with the standard metric. Let $c = 1$ so that Y is a simple space up through degree n . Let $\varphi : \mathcal{M}_Y^* \rightarrow \Omega^*(X) \otimes \mathbb{R} \langle t, dt \rangle$ have $\text{Dil}(\varphi) \leq L$, and let Φ be a nullhomotopy which can be built formally up through degree n without encountering any nonzero obstructions that make extendability dependent on choices made in lower degrees. Then writing Φ_i^j for the $t^i dt^j$ term of Φ , for all $1 < n \leq N$ we have*

$$\|\Phi_i^j|_{V_n}\|_{op} \leq C(n, Y)L^{2n-1}.$$

In particular, $\text{Dil}_{L^{-1}}(\Phi) \leq CL^2$.

Applying this in the same manner as above, we can construct a homotopy with length $O(L)$ and thickness $O(L^2)$, and therefore attain the following:

Corollary 5.6. *If Y is simple up through dimension n , then $V\delta_Y^n(L) = O(L^{2n+1})$.*

Similarly, when Y is coformal (i.e., the differential of the minimal model is quadratic) we can refine these bounds to be nearly sharp. In particular, the example from [Section 3.5](#) nearly meets the upper bounds of the following:

Proposition 5.7. *Let X be a finite N -dimensional simplicial complex with the standard metric. Let Y be a coformal, c -step nilpotent space up through degree n . Let $\varphi : \mathcal{M}_Y^* \rightarrow \Omega^*(X) \otimes \mathbb{R}\langle t, dt \rangle$ have $\text{Dil}(\varphi) \leq L$, and let Φ be a nullhomotopy which can be built formally up through degree n without encountering any nonzero obstructions that make extendability dependent on choices made in lower degrees. Then writing Φ_i^j for the $t^i dt^j$ term of Φ , for all $1 < n \leq N$ we have*

$$\|\Phi_i^j|_{V_n}\|_{op} \leq C(n, Y)L^{(c+1)(n-1)}$$

In particular, $\text{Dil}_{L^{-(c+1)}}(\Phi) \leq CL^{c+1}$.

Applying this in the same manner as above, we can construct a homotopy with length $O(1)$ and thickness $O(L^{c+1})$, and therefore attain the following:

Corollary 5.8. *If Y is c -step nilpotent and coformal up through dimension n , then $V\delta_Y^n(L) = O(L^{(c+1)n})$.*

We expect that [Corollary 5.4](#) can be improved somehow. However, proving this general refinement would likely benefit from a different approach. In particular, our use of the Sullivan model appears to be ill-suited for these proofs, as evidenced by many of the seemingly inelegant and repetitive methods we employ. The author expects the

Quillen model (or perhaps some other model) may provide more elegant structure absent in the current proofs. However, as with much of life, there is a certain “conservation of energy” that must be taken into account. It seems that translation from the language of Sullivan models into that of Quillen’s Lie model would take considerable effort, to the point that even if the proofs were simplified in that context, the translation itself could require too much effort to prove insightful.

The following sections are a largely a technical analysis of the structure of the differential on Sullivan models. This is purely algebraic information about the structure of Sullivan models which will provide the foundation of the quantitative results we are interested in.

The rest of this chapter is organized as follows: in [Section 5.1](#) we define a filtration on each V_n which will help with the analysis of the following sections. From this filtration we will see that any nilpotence in the Sullivan model is quantitatively recognized by the action of the abelianization of the fundamental group. In [Section 5.2](#), we will analyze the quantitative bounds to prove [Theorem 5.3](#). In [Section 5.3](#) we refine our approach when the target Y is a simple space, with more careful analysis of the differential. We use this to prove [Proposition 5.5](#) which results in [Corollary 5.6](#). Similar refinements in [Section 5.4](#) allow us to conclude with bounds for coformal c -step nilpotent spaces in [Proposition 5.7](#), which in turn yields [Corollary 5.8](#).

5.1 Filtrations

The main quantity we analyze is the weight of indecomposables of a minimal model. This will give us a measure of how the differential interacts with certain indecomposables.

Definition 5.9. Let $\omega \in V_n$ be indecomposable of degree n for a minimal Sullivan DGA $(\wedge V, d)$. Fix a basis $\{\bar{\mathbf{a}}_i\}$ on V . The **weight** of ω is defined by $\text{wt}(\omega) = n$ if $d\omega = 0$, and

otherwise, writing

$$d\omega = \sum_i r_i (\bar{\mathbf{a}}_{i_1} \wedge \bar{\mathbf{a}}_{i_2} \wedge \cdots \wedge \bar{\mathbf{a}}_{i_{R_i}})$$

as a linear combination of products of indecomposables, we inductively define

$$\text{wt}(\omega) = \max_i \left\{ \sum_k \text{wt}(\bar{\mathbf{a}}_{i_k}) \right\}.$$

We may further define the weight of a product of indecomposables to be the maximal sum of weights of the individual factors. Note that though we started with a basis for V , the weight is well-defined independent of the specific basis we started with. Later we will set particular bases and conventions of notation.

In order to perform analysis on the weights of indecomposables, we present some helpful filtrations on the Sullivan DGAs in terms of the differential d . The indexing of [29, §2] (from whom we borrow our notation) begins at $J = 0$. Unfortunately, this conflicts with the convention of initializing the indices of the central series at 1, to which we adhere. This collision of conventions is somewhat inevitable. We stick with convention—at the cost of this collision—and index starting at $J = 1$.

Definition 5.10. The **cautious filtration** on V_n is $W_n^J \subseteq W_n^{J+1}$ defined by

$$W_n^J := \begin{cases} V_n \cap d^{-1}(\wedge^{\geq 2} V_{<n}) & J = 1 \\ V_n \cap d^{-1}(\wedge^{\geq 2} V_{<n} \oplus (W_1^{J-1} \wedge W_n^{J-1})) & J > 1. \end{cases}$$

The **naive filtration** on V_n is $\widetilde{W}_n^J \subseteq \widetilde{W}_n^{J+1}$ defined by

$$\widetilde{W}_n^J := \begin{cases} V_n \cap d^{-1}(\wedge^{\geq 2} V_{<n}) & J = 1 \\ V_n \cap d^{-1}(\wedge^{\geq 2} V_{<n} \oplus (V_1 \wedge \widetilde{W}_n^{J-1})) & J > 1. \end{cases}$$

That is, one may always pick $V_n^{\leq J} = \widetilde{W}_n^J$, but it may be possible to find a more refined definition of $V_n^{\leq J}$. Moreover, having fixed a norm on V_n we define the M_n^J (resp. $\widetilde{M}_n^J$) the **J th cautious step** of the cautious filtration by

$$M_n^J := \begin{cases} W_n^1 & J = 1 \\ (W_n^{J-1})^\perp \cap W_n^J & J > 1. \end{cases}$$

and the **J th naive step** of the naive filtration by

$$\widetilde{M}_n^J := \begin{cases} \widetilde{W}_n^1 & J = 1 \\ (\widetilde{W}_n^{J-1})^\perp \cap \widetilde{W}_n^J & J > 1. \end{cases}$$

In particular observe that

$$M_n^{J+1} \oplus W_n^J = W_n^{J+1}$$

and

$$\widetilde{M}_n^{J+1} \oplus \widetilde{W}_n^J = \widetilde{W}_n^{J+1}.$$

Note. The distinction in the naive and cautious filtrations is subtle, but important. In the naive filtration, we essentially say that “we know π_1 acts on our space, and we don’t care *which* elements are doing the action”, and so we take all of the action all at once. In the cautious filtration, we move more slowly, considering only those pieces in the J th step of π_1 itself, when building the $(J+1)$ st stage of the filtration. Because of this, the cautious filtration might take twice as long as the naive one to terminate. The main result of this section is that in fact the two filtrations agree.

We begin with the easy direction:

Proposition 5.11. $V_n^{\leq J} \subseteq \widetilde{W}_n^J$ and $W_n^J \subseteq \widetilde{W}_n^J$ for all n and all J .

Proof. For $J = 1$, by definition we have $\text{im} \left(d|_{V_n^{\leq 1}} \right) \subseteq \wedge^{\geq 2} V_{<n}$, and so obviously

$$V_n^{\leq 1} \subseteq V_n \cap d^{-1} \left(\wedge^{\geq 2} V_{<n} \right) = W_n^1.$$

Moreover, clearly by definition we have

$$W_n^1 = \widetilde{W}_n^1.$$

Inductively then we have

$$\begin{aligned} \text{im} \left(d|_{V_n^{\leq J}} \right) &\subseteq \wedge^{\geq 2} V_{<n} \oplus (V_1 \wedge V_n^{\leq J-1}) \\ &\subseteq \wedge^{\geq 2} V_{<n} \oplus (V_1 \wedge \widetilde{W}_n^{J-1}) \end{aligned}$$

so that

$$V_n^{\leq J} \subseteq V_n \cap d^{-1} \left(\wedge^{\geq 2} V_{<n} \oplus (V_1 \wedge \widetilde{W}_n^{J-1}) \right) = \widetilde{W}_n^J.$$

Moreover, since $W_1^J \subseteq V_1$ for all J we have

$$\begin{aligned} W_n^J &= V_n \cap d^{-1} \left(\wedge^{\geq 2} V_{<n} \oplus (W_1^{J-1} \wedge W_n^{J-1}) \right) \\ &\subseteq V_n \cap d^{-1} \left(\wedge^{\geq 2} V_{<n} \oplus (V_1 \wedge W_n^{J-1}) \right) \\ &\subseteq V_n \cap d^{-1} \left(\wedge^{\geq 2} V_{<n} \oplus (V_1 \wedge \widetilde{W}_n^{J-1}) \right) \\ &= \widetilde{W}_n^J. \end{aligned}$$

□

Corollary 5.12. $\widetilde{W}_n^c = V_n$.

Proof. By the previous proposition $V_n = V_n^{\leq c} \subseteq \widetilde{W}_n^c \subseteq V_n$.

□

Now let $\omega \in M_n^{J+1}$. Then $d\omega \in \wedge^{\geq 2} V_{<n} \oplus (W_1^J \wedge W_n^J)$. Let $d = d_{sim} + d_{nil}$ where

$$d_{sim} : M_n^{J+1} \rightarrow \wedge^{\geq 2} V_{<n}$$

and

$$d_{nil} : M_n^{J+1} \rightarrow W_1^J \wedge W_n^J.$$

We call $d_{sim}(\omega)$ the **simple component**, and $d_{nil}(\omega)$ the **nilpotent component** of $d\omega$.

Moreover, for $n > 1$ we may write

$$W_1^J \wedge W_n^J = \bigoplus_{1 \leq i, j \leq J} M_1^i \wedge M_n^j$$

and we write

$$d_{nil}^{i,j} : M_n^{J+1} \rightarrow M_1^i \wedge M_n^j$$

as the (i, j) -nilpotent component of the differential.

Note. We may similarly define d_{sim} and d_{nil} on $V_n^{\leq J}$ in the obvious way.

Let $\omega \in M_n^J$. We define $\delta\omega = d_{nil}^{1,J-1}(\omega)$, and write $\delta = d_{nil}^{1,J-1}$, which we call the **abelian component** of the differential. Context should always make clear the domain of the abelian component. This terminology is inspired by [29], where the orthogonal subspaces of $\text{im}(d)$, and summands therein, are called “components.”

A number of the subsequent proofs will be based on certain maximality arguments. To aid with this, we define a lexicographical order on tuples of natural numbers. Many of the methods used here are essentially those found in [Lemma 5.15](#), but perhaps a bit streamlined by the use of this total order notation. The author expects there is a more concise argument lurking somewhere, but has been unable to find it.

Definition 5.13. For $K \in \mathbb{N}$, let $U_K = \{(m_1, \dots, m_K) \in \mathbb{N}^K \mid m_i \leq m_{i+1}\}$. For $K = 1$, define $m'_1 \leq_1 m_1 \iff m'_1 \leq m_1$. Inductively, define $(m'_1, \dots, m'_K) \leq_K (m_1, \dots, m_K)$ if and only if either $m'_K < m_K$ or

$$m'_K = m_K \text{ and } (m'_1, \dots, m'_{K-1}) \leq_{K-1} (m_1, \dots, m_{K-1}).$$

In the event that $(m'_1, \dots, m'_K) \leq_K (m_1, \dots, m_K)$ and $(m'_1, \dots, m'_K) \neq (m_1, \dots, m_K)$ we write $(m'_1, \dots, m'_K) <_K (m_1, \dots, m_K)$.

The following is a result of [29], but note that by our reindexing, the exact statements look slightly different.

Lemma 5.14 ([29, Lem 2.3]). *The nilpotent component of the differential on M_1^J satisfies*

$$d_{nil} = \sum_{i+j \leq J} d_{nil}^{i,j}.$$

That is, if $\omega \in M_1^J$ then $d\omega = \sum_{i+j \leq J} \Omega_{i,j}$ where $\Omega_{i,j} \in M_1^i \wedge M_1^j$. Moreover, δ is injective.

We generalize this to the following:

Lemma 5.15. *For all n and J , the nilpotent component of the differential on M_n^J satisfies*

$$d_{nil} = \sum_{i+j \leq J} d_{nil}^{i,j}.$$

In particular, if $\omega \in M_n^J$ is nonzero then

$$d\omega = d_{sim}\omega + d_{nil}\omega = d_{sim}\omega + \sum_{i+j \leq J} \Omega_{i,j}$$

where each $\Omega_{i,j} \in M_1^i \wedge M_n^j$. Moreover, the morphism $\delta : M_n^J \rightarrow M_1^1 \wedge M_n^{J-1}$ is injective for all $J \geq 2$.

Proof. When $n = 1$ this is the result of [Lemma 5.14](#), so let $n > 1$. First consider $J = 2$ and let $\omega \in M_n^2$. Note that in this case $\delta\omega = d_{nil}\omega$. Thus if $\delta\omega = 0$ then $d_{nil}\omega \in M_1^1 \wedge M_n^1$ is trivial. But by definition $\omega \in d^{-1}(\wedge^{\geq 2} V_{<n}) = M_n^1 = W_n^1$, and since M_n^2 is orthogonal to W_n^1 we have

$$\omega \in W_n^1 \cap M_n^2 = \{0\}.$$

Therefore the result holds for $J = 2$.

Now let $J > 2$ and assume the result for all $2 \leq j < J$. Let $\omega \in M_n^J$. Note that $0 = d^2\omega = (d \circ d_{sim})\omega + (d \circ d_{nil})\omega$. Moreover, note that

$$\text{im}(d \circ d_{sim}) \subseteq \wedge^{\geq 3} V_{<n}$$

and that

$$\text{im}(d \circ d_{nil}) \subseteq (\wedge^{\geq 3} V_{<n}) \oplus ((W_1^{J-2} \wedge W_1^{J-2} \wedge W_n^{J-1}) + (W_1^{J-1} \wedge W_1^{J-2} \wedge W_n^{J-2})).$$

In particular, the components of $(d \circ d_{nil})\omega$ outside of $\wedge^{\geq 3} V_{<n}$ must all be trivial, as they cannot cancel with components of $(d \circ d_{sim})\omega$. To this end, write $d_{nil}\omega = \sum_{i,j \leq J-1} \Omega_{i,j}$, such that $\Omega_{i,j} \in M_1^i \wedge M_n^j$. A priori, $i + j$ is unbounded. Let $\ell := \max \{i + j \mid \Omega_{i,j} \neq 0\}$. Moreover, fix $(i_0, j_0) := \max_{\leq 2} \{(i, j) \mid i + j = \ell, \Omega_{i,j} \neq 0\}$. In particular, j_0 is maximal over such pairs (i, j) for which $\Omega_{i,j} \neq 0$.

Suppose to the contrary that $d\Omega_{i_0, j_0} \neq 0$. Write $\Omega_{i_0, j_0} = \sum_r \bar{\mathbf{t}}_r \wedge \mathbf{v}_r \in M_1^{i_0} \wedge M_n^{j_0}$ such that the $\bar{\mathbf{t}}_r$ terms are linearly independent. By [\[29, Lem 2.3\]](#) (equivalently, by the base case $n = 1$) we know that $\delta|_{M_1^{i_0}}$ is injective. Hence the collection $\{\delta\bar{\mathbf{t}}_r\} \subseteq M_1^1 \wedge M_1^{i_0-1}$ is linearly independent. Consequently $\sum_r (\delta\bar{\mathbf{t}}_r) \wedge \mathbf{v}_r \in M_1^1 \wedge M_1^{i_0-1} \wedge M_n^{j_0}$ is itself nonzero.

Therefore, there is some term $\Omega_{i', j'} \in M_1^{i'} \wedge M_n^{j'}$ of $d_{nil}\omega$ for which $d\Omega_{i', j'}$ has a summand $\gamma \in M_1^1 \wedge M_1^{i_0-1} \wedge M_n^{j_0}$. Moreover, by induction (applying the differential to

$\Omega_{i',j'}$) we must have

$$\gamma \in \left(M_1^{a_i} \wedge M_1^{b_i} \wedge M_n^{j'} \right) \oplus \left(M_1^{i'} \wedge M_1^{a_j} \wedge M_n^{b_j} \right)$$

for which $a_i + b_i \leq i'$ and $a_j + b_j \leq j'$.

First, if

$$\gamma \in \left(M_1^{a_i} \wedge M_1^{b_i} \wedge M_n^{j'} \right) \cap \left(M_1^1 \wedge M_1^{i_0-1} \wedge M_n^{j_0} \right)$$

then (without loss of generality) we have $a_i = 1$ and $b_i = i_0 - 1$ and $j' = j_0$. But $i_0 = a_i + b_i \leq i'$ and $j' = j_0$. By maximality of (i_0, j_0) this implies that in fact $(i', j') = (i_0, j_0)$ and hence that $\Omega_{i',j'} = s\Omega_{i_0,j_0}$ for some $s \in \mathbb{R}$.

On the other hand, if

$$\gamma \in \left(M_1^{i'} \wedge M_1^{a_j} \wedge M_n^{b_j} \right) \cap \left(M_1^1 \wedge M_1^{i_0-1} \wedge M_n^{j_0} \right)$$

then $b_j = j_0$. But then, since j_0 is maximal and since $a_j > 0$, we have

$$j_0 < j_0 + a_j = b_j + a_j \leq j'$$

a contradiction to the maximality of j_0 .

Thus, $d\Omega_{i_0,j_0} = 0$. But this implies that $\sum_r (\delta \bar{\mathbf{t}}_r) \wedge \mathbf{v}_r = 0$. By linear independence of the $\delta \bar{\mathbf{t}}_r$ terms we must have $\delta \bar{\mathbf{t}}_r = 0$ for all r , which by injectivity of δ in degree 1 implies that $\bar{\mathbf{t}}_r = 0$ for all r . In particular we have $i_0 = 1$, and since $j_0 \leq J - 1$ we have $\ell = i_0 + j_0 = 1 + j_0 \leq J$. By the maximality of ℓ we see that d_{nil} is as desired.

Finally, if $\ell \neq J$, then each $\Omega_{i,j}$ actually belongs to $W_1^{J-2} \wedge W_n^{J-2}$ so that $\omega \in W_n^{J-1}$. Thus if $\omega \in M_n^J$ as we assumed, we would have had $\omega \in M_n^J \cap W_n^{J-1} = \{0\}$, a contradiction. Therefore, $\ell = J$. The above shows that the component $\Omega_{1,J-1}$ is

nontrivial, and thus that δ is injective as claimed. $\square$

This allows us to prove:

Lemma 5.16. $\widetilde{W}_n^J = W_n^J$ for all n and J .

Proof. For $J = 1$, this is clearly true by definition. By [Proposition 5.11](#) we know that $W_n^J \subseteq \widetilde{W}_n^J$. Inductively we can write

$$W_n^J = W_n^{J-1} \oplus M_n^J = \widetilde{W}_n^{J-1} \oplus M_n^J \subseteq \widetilde{W}_n^{J-1} \oplus \widetilde{M}_n^J = \widetilde{W}_n^J$$

where in fact $M_n^J \subseteq \widetilde{M}_n^J$ as they are both orthogonal to the same $W_n^{J-1} = \widetilde{W}_n^{J-1}$ by induction. More generally, we have M_n^K orthogonal to $W_n^{J-1} = \widetilde{W}_n^{J-1}$ for all $K > J - 1$. Let K be maximal so that there is $\omega \in \widetilde{M}_n^J$ for which the natural projection $p : \widetilde{M}_n^J \rightarrow M_n^K$ is nonzero. By injectivity of δ we have $\delta(p(\omega)) \neq 0$ as a component of $d\omega$ in $M_1^1 \wedge M_n^{K-1}$. But also, by definition, the differential sends the naive filtration down a degree in the filtration. Therefore, since $p(\omega) \in \widetilde{M}_n^J$, we have

$$\delta(p(\omega)) \in V_1 \wedge \widetilde{W}_n^{J-1} = V_1 \wedge W_n^{J-1}$$

by induction. Therefore

$$\delta(p(\omega)) \in (V_1 \wedge W_n^{J-1}) \cap (M_1^1 \wedge M_n^{K-1})$$

is nonzero which in particular implies that $K \leq J$. Since K was maximal for which this projection was nontrivial, every element of $\widetilde{W}_n^J$ projects trivially onto M_n^j for $j > J$ so that $W_n^J \subseteq \widetilde{W}_n^J \subseteq \bigoplus_{j=1}^J M_n^j = W_n^J$ as required. $\square$

Corollary 5.17. *For all J we have $V_n^{\leq J} \subseteq W_n^J$. In particular*

$$\operatorname{im} \left(d_{\text{nil}}|_{V_n^{\leq J}} \right) \subseteq \operatorname{im} \left(d_{\text{nil}}|_{W_n^J} \right) \subseteq \bigoplus_{i+j \leq J} M_1^i \wedge M_n^j.$$

This allows us to work exclusively with the cautious filtration from now on. We now establish conventions and terminology which we frequently use throughout the rest of the chapter. In particular, defining a basis of M_n^J for each n and J induces a basis of V and thus on $\wedge V$ in the obvious ways. Given such a basis we induce an inner product $\langle \cdot, \cdot \rangle$ for which this basis is orthonormal. Given such a basis and inner product, we try to adhere to the following conventions in what follows:

Notation.

- Standard *italic* Roman letters such as r, s and i, N denote scalars and indices.
- **Bold and barred** lower case Roman letters such as $\bar{\mathbf{x}}$ denote basic indecomposables, and in particular are elements of V_n for some degree n .
- The lower case Greek letter ω , and **bold** Roman lower case letters such as $\mathbf{x}$ denote generic indecomposables, and in particular are elements of V_n for some degree n .
- Greek letters such as $\Omega, \alpha, \beta, \xi$ (but **not** ω ; see above) denote scalar multiples of basis elements of $\wedge V$. In particular, these denote basis elements of $\wedge^k V$ for some k , and are products of basic indecomposables.
- Capital Roman letters such as A, B denote generic elements of $\wedge V$.

For example we might write $\mathbf{x} \wedge \bar{\mathbf{y}} = (\sum_i r_i \bar{\mathbf{x}}_i \wedge \bar{\mathbf{y}})$ to denote an element of $V_{|x|} \wedge \operatorname{span} \{\bar{\mathbf{y}}\}$. We will sometimes be concerned with specifying that certain summands live in complementary subspaces. Generally then if $A \in \wedge V$ (not necessarily basic) for some k , and $B \in \wedge V$ write $A + B = A \oplus B$ if A and B live in complementary subspaces.

Given $A \in \wedge V$ we will typically say that $\Omega \in \wedge^k V$ is a **(basic) summand** of A if $\Omega \in \text{span} \{ \tilde{\Omega} \}$ for some basic element $\tilde{\Omega}$ of $\wedge^k V$ for which $\langle A, \Omega \rangle > 0$. In particular, Ω is a nonzero scalar multiple of a basic element. (Note, that Ω is a basic summand of $d\omega$ if and only if $d\omega = \Omega \oplus B$ for some possibly trivial B .) A **summand** more generally is a linear combination of basic summands, whose coefficients are typically assumed to be nonzero, but we will often use the word “summand” when we really mean a basic summand. Context should make the meaning of the term clear. Finally, suppose that ξ is basic and $A \in \wedge V$. We say that ξ **cancels** with A if $\langle A, \xi \rangle < 0$; that is, A contains $r\xi$ as a basic summand for some $r < 0$.

Lemma 5.18. *Let ω be indecomposable, let $\langle d\omega, A \rangle \neq 0$, and suppose dA has a (nonzero) basic summand ξ . Then there is a (basic) summand Ω of $d\omega$ complementary to A for which $\langle d\Omega, \xi \rangle < 0$. In particular, $d\Omega$ has a (basic) summand which cancels with ξ .*

Proof. Write $d\omega = A \oplus B$. Since $dA + dB = d^2\omega = 0$ we have

$$0 = \langle 0, \xi \rangle = \langle d^2\omega, \xi \rangle = \langle dA, \xi \rangle + \langle dB, \xi \rangle$$

so that $\langle dB, \xi \rangle = -\langle dA, \xi \rangle$. Since ξ is a basic summand of dA we have $\langle dA, \xi \rangle > 0$ by definition. Since B is complementary to A , there must be a (basic) summand Ω of B for which $\langle d\Omega, \xi \rangle < 0$. □

Definition 5.19. If ξ is as in the above lemma we say that ξ is **cancelled externally** relative to A .

Definition 5.20. Let Ω be a basic summand of $d\omega$, and write $\Omega = r\bar{\mathbf{x}}_1 \wedge \cdots \wedge \bar{\mathbf{x}}_k$. Let ξ be a (possibly trivial) summand of $r\bar{\mathbf{x}}_1 \wedge \cdots \wedge (d\bar{\mathbf{x}}_i) \wedge \cdots \wedge \bar{\mathbf{x}}_k$ for some i . We say that ξ is **cancelled internally** relative to Ω if $\xi = 0$ or if ξ is a basic summand which cancels with a summand of $r\bar{\mathbf{x}}_1 \wedge \cdots \wedge (d\bar{\mathbf{x}}_j) \wedge \cdots \wedge \bar{\mathbf{x}}_k$ for some j .

Note. Let Ω be as above and let ξ be a basic summand of $r\bar{\mathbf{x}}_1 \wedge \cdots \wedge (d\bar{\mathbf{x}}_i) \wedge \cdots \wedge \bar{\mathbf{x}}_k$ for some i . If ξ is not cancelled externally (relative to Ω), then by contraposition of the previous lemma, ξ is **not** a nonzero summand of $d\Omega$. Hence, ξ must be cancelled internally (relative to Ω).

Lemma 5.21. *Let Ω be as above and let ξ be a summand of $r\bar{\mathbf{x}}_1 \wedge \cdots \wedge (d\bar{\mathbf{x}}_i) \wedge \cdots \wedge \bar{\mathbf{x}}_k$ for some i . If ξ is cancelled internally (relative to Ω) by a summand of $r\bar{\mathbf{x}}_1 \wedge \cdots \wedge (d\bar{\mathbf{x}}_j) \wedge \cdots \wedge \bar{\mathbf{x}}_k$ of $d\Omega$, then $\xi = 0$ or $\bar{\mathbf{x}}_i \neq \bar{\mathbf{x}}_j$.*

Proof. For contraposition, suppose $\bar{\mathbf{x}}_i = \bar{\mathbf{x}}_j$ and $\xi \neq 0$. If $i = j$ then $\xi = 0$, and so we may assume that $i \neq j$. If $|\bar{\mathbf{x}}_i|$ is odd, then $\Omega = 0$ since $\bar{\mathbf{x}}_i \wedge \bar{\mathbf{x}}_j = \bar{\mathbf{x}}_i^2 = 0$. Assume therefore that $|\bar{\mathbf{x}}_i|$ is even, and without loss of generality $\bar{\mathbf{x}}_i = \bar{\mathbf{x}}_1 = \bar{\mathbf{x}}_2 = \bar{\mathbf{x}}_j$. Then by graded commutativity we have

$$(d\bar{\mathbf{x}}_1) \wedge \bar{\mathbf{x}}_2 \wedge \cdots \wedge \bar{\mathbf{x}}_k = \bar{\mathbf{x}}_2 \wedge (d\bar{\mathbf{x}}_1) \wedge \cdots \wedge \bar{\mathbf{x}}_k = \bar{\mathbf{x}}_1 \wedge (d\bar{\mathbf{x}}_2) \wedge \cdots \wedge \bar{\mathbf{x}}_k$$

since $\bar{\mathbf{x}}_1 = \bar{\mathbf{x}}_2$. Therefore, by definition of ξ being a basic summand we have

$$0 < \langle (d\bar{\mathbf{x}}_1) \wedge \bar{\mathbf{x}}_2 \wedge \cdots \wedge \bar{\mathbf{x}}_k, \xi \rangle = \langle \bar{\mathbf{x}}_1 \wedge (d\bar{\mathbf{x}}_2) \wedge \cdots \wedge \bar{\mathbf{x}}_k, \xi \rangle$$

so that ξ is not cancelled by such a summand. □

Lemma 5.22. *Let N be given and let $\Omega = \alpha \wedge \bar{\mathbf{x}}^\ell \in (\wedge^{\geq 0} V) \wedge (\wedge^\ell M_N^J)$ be a basic summand of $d\omega$. In particular, $\bar{\mathbf{x}}$ is basic, and α is a product of basic indecomposables. Moreover, assume that ℓ is maximal in the sense that α does not contain $\bar{\mathbf{x}}$ as a factor. Finally, assume that $J > 1$. Let $\bar{\mathbf{t}} \wedge \zeta \in M_1^p \wedge (\wedge^k V)$ be a basic summand of $d\bar{\mathbf{x}}$, and let $\xi := \alpha \wedge \bar{\mathbf{x}}^{\ell-1} \wedge \bar{\mathbf{t}} \wedge \zeta$. Then either $\xi = 0$ (in which case $\bar{\mathbf{t}}$ is a factor of α , or ζ and α share a common factor) or ξ is cancelled externally relative to Ω .*

Proof. If ξ is not cancelled externally, then ξ is cancelled internally relative to Ω . Suppose to the contrary that $\xi \neq 0$. By the above lemma, internal cancellation does not happen within any of the $\bar{\mathbf{x}}$ factors, so we must have some basic summand β of $d\alpha$ for which $\langle \beta \wedge \bar{\mathbf{x}}^\ell, \alpha \wedge \bar{\mathbf{x}}^{\ell-1} \wedge \bar{\mathbf{t}} \wedge \zeta \rangle < 0$. There are $\geq \ell$ factors of $\bar{\mathbf{x}}$ in $\beta \wedge \bar{\mathbf{x}}^\ell$, but only $\ell - 1$ factors of $\bar{\mathbf{x}}$ in $\alpha \wedge \bar{\mathbf{x}}^{\ell-1} \wedge \bar{\mathbf{t}} \wedge \zeta$ by the maximality of ℓ and the fact that $\bar{\mathbf{x}}$ is not a factor of $\bar{\mathbf{t}} \wedge \zeta$ by degree/step considerations. Since $\langle \cdot, \cdot \rangle$ is defined to make our basis orthogonal, we must have $\langle \beta \wedge \bar{\mathbf{x}}^\ell, \alpha \wedge \bar{\mathbf{x}}^{\ell-1} \wedge \bar{\mathbf{t}} \wedge \zeta \rangle = 0$, a contradiction.

Hence, $0 = \xi = \alpha \wedge \bar{\mathbf{x}}^{\ell-1} \wedge \bar{\mathbf{t}} \wedge \zeta$. Since each factor is basic indecomposable, we must have a repeated factor of odd degree. Since Ω itself is nonzero, this repeated factor must be a factor of $\bar{\mathbf{t}} \wedge \zeta$. Since $\bar{\mathbf{t}} \wedge \zeta$ is a summand of $d\bar{\mathbf{x}}$ this repeated factor cannot be $\bar{\mathbf{x}}$ itself, which completes the proof. $\square$

5.2 Bounds on Nilpotent Spaces

We continue using our prior [notation](#). The main goal of this section is the proof of [Proposition 5.2](#). We now put some quantitative bounds in place on the weights of indecomposables. First we make an observation about the indecomposables in degree 1.

Proposition 5.23. *If $\omega \in W_1^J$ then $\text{wt}(\omega) \leq J$. In particular, $\text{wt}(\omega) = J$ if $\omega \in M_1^J$.*

Proof. When $J = 1$ by definition $d\omega = 0$ and so $\text{wt}(\omega) = 1$. Now [Lemma 5.14](#) implies that any basic summand of $d\omega$ is of the form $\Omega_{i,j} = \bar{\mathbf{x}}_i \wedge \bar{\mathbf{y}}_j \in M_1^i \wedge M_1^j$ where $i + j \leq J$, and inductively $\text{wt}(\bar{\mathbf{x}}_i) \leq i$ and $\text{wt}(\bar{\mathbf{y}}_j) \leq j$. Thus if $\Omega_{i,j}$ maximizes the weight of ω , then $\text{wt}(\omega) \leq i + j \leq J$. Because δ is injective $\delta\omega \in M_1^1 \wedge M_1^{J-1}$ is nonzero and therefore inductively has weight exactly J . $\square$

Naively, one might expect that the indecomposables in M_2^1 of a c -step nilpotent space might have weight at most $3c$, but this bound can be sharpened. In fact, we see

in [Corollary 5.28](#) that $3c$ ends up being the largest weight of V_2 overall. We expect the techniques in degree 2 carry through higher dimensions to produce sharper upper bounds on the weight of indecomposables. Here, we examine one such technique, with an eye towards [Conjecture 1.4](#).

Definition 5.24. If Ω is a summand of $d\omega$ and $\text{wt}(\omega) = \text{wt}(\Omega)$, we say that Ω is a **full weight**, or **weight maximizing** summand of ω .

Definition 5.25. If $\mathbf{x} \in M_n^1$ for some n , we say that $\mathbf{x}$ is **light**.

Lemma 5.26. *Let $\omega \in M_2^J$ and suppose that all full weight summands of ω are summands of $d_{\text{sim}}\omega$. That is, $\text{wt}(\omega) = \text{wt}(d_{\text{sim}}\omega) > \text{wt}(d_{\text{nil}}\omega)$. Then some full weight basic summand of $d\omega$ contains a light factor.*

Proof. Note that all basic summands of $d_{\text{sim}}\omega$ belong to $M_1^{J_1} \wedge M_1^{J_2} \wedge M_1^{J_3}$ for some $J_1 \leq J_2 \leq J_3$. Let $\Omega = \bar{\mathbf{x}} \wedge \bar{\mathbf{y}} \wedge \bar{\mathbf{z}}$ be a full weight summand of $d_{\text{sim}}\omega$ for which (J_1, J_2, J_3) is maximal under $\leq_3$. By [Proposition 5.23](#) we see in particular that $\text{wt}(\omega) = J_1 + J_2 + J_3$. That is, if $\Omega' \in M_1^{J'_1} \wedge M_1^{J'_2} \wedge M_1^{J'_3}$ with $J'_1 \leq J'_2 \leq J'_3$ is another full weight summand of $d_{\text{sim}}\omega$, then $(J'_1, J'_2, J'_3) \leq_3 (J_1, J_2, J_3)$. Furthermore, up to a change of basis we may assume that if $\bar{\mathbf{x}}' \wedge \bar{\mathbf{y}} \wedge \bar{\mathbf{z}}$ is a summand of $d\omega$ for which $(d\bar{\mathbf{x}}') \wedge \bar{\mathbf{y}} \wedge \bar{\mathbf{z}}$ has $(\delta\bar{\mathbf{x}}) \wedge \bar{\mathbf{y}} \wedge \bar{\mathbf{z}}$ as a summand, then $\bar{\mathbf{x}}' = \bar{\mathbf{x}}$.

Suppose to the contrary $J_1 > 1$, and let $\xi := (\delta\bar{\mathbf{x}}) \wedge \bar{\mathbf{y}} \wedge \bar{\mathbf{z}}$. Suppose then $\Omega' = \bar{\mathbf{x}}' \wedge \bar{\mathbf{y}}' \wedge \bar{\mathbf{z}}'$ is a summand of $d\omega$ for which $d\Omega'$ has a summand which cancels with ξ . By [Proposition 5.23](#) we also see that $\text{wt}(\xi) = \text{wt}(\Omega) = 1 + (J_1 - 1) + J_2 + J_3 = \text{wt}(\omega)$, and thus $\text{wt}(\Omega') \geq \text{wt}(\xi)$. In particular, Ω' is full weight. By maximality of (J_1, J_2, J_3) under full weight summands of $d\omega$, we see that $(J'_1, J'_2, J'_3) \leq_3 (J_1, J_2, J_3)$.

See first that $J'_3 \geq J_3$. Otherwise $d\Omega'$ contains no summands in $M_1^{J_1} \wedge M_1^{J_2} \wedge M_1^{J_3}$ by consideration step of the filtration. By maximality, $J'_3 = J_3$, and in fact $\bar{\mathbf{z}}' = \bar{\mathbf{z}}$. Similarly,

once $J'_3 = J_3$ we see that $J'_2 = J_2$ in the same way and $\bar{\mathbf{y}}' = \bar{\mathbf{y}}$, and hence that $J'_1 = J_1$ as well. In particular, we must then have $\Omega' = r\bar{\mathbf{x}}' \wedge \bar{\mathbf{y}} \wedge \bar{\mathbf{z}}$ for some $r \in \mathbb{R}$, where $(d\bar{\mathbf{x}}') \wedge \bar{\mathbf{y}} \wedge \bar{\mathbf{z}}$ has $(\delta\bar{\mathbf{x}}) \wedge \bar{\mathbf{y}} \wedge \bar{\mathbf{z}}$ as a summand. Hence, $\bar{\mathbf{x}}' = \bar{\mathbf{x}}$ by our initial assumptions. In particular, $\Omega' \in \text{span}\{\Omega\}$. Cancellation is thus not external so that [Lemma 5.22](#) implies $\xi = 0$. But $\xi \in M_1^1 \wedge M_1^{J_1-1} \wedge M_1^{J_2} \wedge M_1^{J_3}$ is trivial if and only if $\delta\bar{\mathbf{x}} = 0$ by consideration of step of the filtration. Since δ is injective, $\bar{\mathbf{x}} = 0$, contradicting the fact that $\bar{\mathbf{x}} \in M_1^{J_1}$ with $J_1 > 1$. $\square$

Lemma 5.27. *Let $\omega \in M_2^J$. Then $\text{wt}(\omega) \leq J + 2c$.*

Proof. Let $\omega \in M_2^J$. If $J = 1$ then $d\omega = d_{\text{sim}}\omega$. By [Lemma 5.26](#), there is a full weight summand with a light factor. Hence, [Proposition 5.23](#) implies that $\text{wt}(\omega) \leq 1 + 2c$.

Inductively, assume the result for all $1 \leq j < J$. If $\omega \in M_2^J$ has weight maximized only by terms of $d_{\text{sim}}\omega$ then [Lemma 5.26](#) again implies that $\text{wt}(\omega) \leq 1 + 2c < J + 2c$. On the other hand, if $\text{wt}(\omega)$ is maximized by a summand of $d_{\text{nil}}\omega$, then such a summand is of the form $\bar{\mathbf{t}} \wedge \bar{\mathbf{y}} \in M_1^{i_1} \wedge M_2^{i_2}$ for which $i_1 + i_2 \leq J$ by [Lemma 5.14](#). Inductively then $\text{wt}(\Omega) = \text{wt}(\bar{\mathbf{t}} \wedge \bar{\mathbf{y}}) \leq i_1 + (i_2 + 2c) \leq J + 2c$. $\square$

Corollary 5.28. *If $\omega \in V_2$ then $\text{wt}(\omega) \leq 3c$.*

Unfortunately, it is not clear that this process can be extended to degrees higher than 2. The following conjecture seems likely though:

Conjecture 5.29. *Suppose the weight of $\omega \in M_n^J$ is maximized only by summands of $d_{\text{sim}}\omega$ (e.g., $J = 1$). Then some weight-maximizing summand of $d\omega$ has a light factor.*

If this conjecture holds, then [Conjecture 1.4](#) holds as well. Unfortunately, we have been unable to verify [Conjecture 5.29](#) for $n > 2$. As explained in the introduction to this section, we expect that this sort of analysis of the simple component can be done more carefully to lead to sharper upper bounds in each degree than the ones that follow. We

provide such an analysis in the following section for the situation when $c = 1$ and the space is simple. For now we can prove the following general bounds:

Proposition 5.30. *Let $\omega \in V_N$ for some $N \geq 2$ where V is c -step nilpotent up through degree N . If $\text{wt}(\omega)$ is maximized by summands of $d_{\text{sim}}\omega$ then $\text{wt}(\omega) \leq N(4c-1) - 3(2c-1)$. If $\omega \in M_N^J$, then $\text{wt}(\omega) \leq N(4c-1) - 3(2c-1) + (J-1)$. In particular, $\text{wt}(\omega) \leq N(4c-1) - 5c + 2$.*

Proof. By [Corollary 5.28](#) this is true for $N = 2$. Inductively, assume the result for all $2 \leq n < N$. Suppose first $\omega \in V_N$ has weight maximized by a summand $\Omega = \bar{\mathbf{x}}_1 \wedge \cdots \wedge \bar{\mathbf{x}}_k \in M_{n_1}^{i_1} \wedge \cdots \wedge M_{n_k}^{i_k}$ of $d_{\text{sim}}\omega$. Let $r = \#\{j \mid n_j = 1\}$, and write

$$\text{wt}(\Omega) = \sum_j \text{wt}(\bar{\mathbf{x}}_j) = \left(\sum_{\{j \mid n_j=1\}} \text{wt}(\bar{\mathbf{x}}_j) \right) + \left(\sum_{\{j \mid n_j \neq 1\}} \text{wt}(\bar{\mathbf{x}}_j) \right).$$

Therefore,

$$\begin{aligned} \text{wt}(\Omega) &\leq \sum_{\{j \mid n_j=1\}} cn_j + \sum_{\{j \mid n_j \neq 1\}} (n_j(4c-1) - 5c + 2) \\ &= \left(\sum_{\{j \mid n_j=1\}} n_j(4c-1) \right) - r(3c-1) + \left(\sum_{\{j \mid n_j \neq 1\}} n_j(4c-1) \right) - (k-r)(5c-2) \\ &= \left(\sum_j n_j \right) (4c-1) - r(3c-1) - (k-r)(5c-2) \\ &= (N+1)(4c-1) - r(3c-1) - (k-r)(5c-2) \quad (\text{since } \sum n_j = N+1) \\ &= N(4c-1) + (4c-1) - r(3c-1) - (k-r)(5c-2). \end{aligned}$$

First, suppose $r = 0$. Since $k \geq 2$ and $5c - 2 > 0$ we have

$$\begin{aligned} \text{wt}(\Omega) &\leq N(4c - 1) + (4c - 1) - 2(5c - 2) \\ &= N(4c - 1) - 3(2c - 1). \end{aligned}$$

Next, suppose $r = 1$. Since Ω is a summand of $d_{sim}\omega$ we must have $k \geq 3$. Hence,

$$\begin{aligned} \text{wt}(\Omega) &\leq N(4c + 1) + (4c - 1) - r(3c - 1) - (k - r)(5c - 2) \\ &= N(4c + 1) + (4c - 1) - (3c - 1) - 2(5c - 2) \\ &= N(4c + 1) - 9c + 4 \\ &\leq N(4c + 1) - 8c + 3 && (\text{since } c \geq 1) \\ &< N(4c + 1) - 3(2c - 1). \end{aligned}$$

Now suppose $2 \leq r < k$. Then

$$\begin{aligned} \text{wt}(\Omega) &\leq N(4c + 1) + (4c - 1) - r(3c - 1) - (k - r)(5c - 2) \\ &\leq N(4c + 1) + (4c - 1) - (6c - 2) - (5c - 2) \\ &= N(4c + 1) - 7c + 3 \\ &< N(4c + 1) - 3(2c - 1). \end{aligned}$$

Finally if $r = k$ we must in fact have $r \geq 4$ since $N \geq 3$ so that

$$\begin{aligned}
 \text{wt}(\Omega) &\leq N(4c+1) + (4c-1) - r(3c-1) \\
 &\leq N(4c+1) + (4c-1) - 4(3c-1) \\
 &= N(4c+1) - 8c - 3 \\
 &< N(4c+1) - 3(2c-1).
 \end{aligned}$$

This completes the proof that if $\text{wt}(\omega)$ is maximized only by terms of $d_{sim}\omega$ then $\text{wt}(\omega) \leq N(4c+1) - 3(2c-1)$. In particular, this applies to $\omega \in M_N^1$.

Inductively on $1 \leq j < J$ (the result for $j = 1$ being part of what was just established) assume that if $\bar{\mathbf{x}} \in M_N^j$ then $\text{wt}(\bar{\mathbf{x}}) \leq N(4c+1) - 3(2c-1) + (j-1)$. Let $\omega \in M_N^J$, and assume $\text{wt}(\omega)$ is maximized by a component $\Omega = \bar{\mathbf{t}} \wedge \bar{\mathbf{x}} \in M_1^a \wedge M_N^b$ of $d_{nil}\omega$. By [Proposition 5.23](#) we have $a+b \leq J$ so that

$$\begin{aligned}
 \text{wt}(\omega) &\leq \text{wt}(\bar{\mathbf{t}}) + \text{wt}(\bar{\mathbf{x}}) \\
 &\leq a + N(4c+1) - 3(2c-1) + (b-1) && \text{(induction)} \\
 &\leq N(4c+1) - 3(2c-1) + J - 1 && (a+b \leq J)
 \end{aligned}$$

as required. When $J = c$, the final result is as claimed. $\square$

We thus have the following, which consequently yields [Corollary 5.4](#).

Proof of Proposition 5.2. By [Proposition 2.48](#), $\|\Phi|_{V_n}\|_{op}$ has an upper bound estimated in terms of $\|\Phi|_{dV_n}\|_{op} + O(L^n)$, and in particular we can see that for $\omega \in V_n$ we have $\|\Phi(v)\|_{op} \leq CL^{\text{wt}(\omega)}$ for some constant C depending only on n , on Y , on the norms of V_n , and the algebraic structure of the differential. But $\omega \in V_n$ implies that $\text{wt}(\omega) \leq n(4c-1) - 5c + 2$, which implies the result. $\square$

5.3 Bounds on Simple Spaces

We continue using our prior [notation](#). As an immediate corollary of [Proposition 5.30](#) we have the following naive observation:

Corollary 5.31. *If $\mathcal{M}_Y^*$ is simple, then $\text{wt}(\omega) \leq 3(n-1)$ for all indecomposables $\omega \in V_n$.*

However, we will see in the rest of this section that this upper bound can be improved. We expect these methods to generalize to higher step actions as well, but the proofs and statements even in the current context are cumbersome and inelegant. Hence, we expect other methods exist for refining these bounds.

Now, we refine the above arguments in the event that $c = 1$ (or equivalently, in the event that $d_{\text{nil}} = 0$ and $d = d_{\text{sim}}$). In particular, this means that we are assuming our space is simple. As in the above section, to prove [Proposition 5.5](#) it is enough to show that the weight of any indecomposable in V_n is bounded by $2n - 1$. Therefore, this section is dedicated to the proof of [Lemma 5.33](#).

Definition 5.32. An indecomposable $\omega \in V_n$ is **overweight** if $\text{wt}(\omega) \geq 2n - 1$.

Note that in the setting of a simple space $V_n = V_n^{\leq 1}$ for all n . Many of the arguments made in this next lemma are closely related to those of the claims of [Lemma 5.26](#), but with some modifications to accommodate for the fact that we are working with $c = 1$.

Lemma 5.33. *Suppose $\wedge V$ is simple, and let $\omega \in V_n$ be indecomposable. If ω is overweight, then $\text{wt}(\omega) = 2n - 1$, and the weight maximizing summands of $d\omega$ are of the form $\bar{\mathbf{p}} \wedge \bar{\mathbf{q}}$, or $\bar{\mathbf{a}} \wedge \bar{\mathbf{b}} \wedge \bar{\mathbf{c}}$. In particular, in the quadratic case, at most one of $\bar{\mathbf{p}}$ or $\bar{\mathbf{q}}$ is overweight. In the cubic case, each factor is overweight.*

Proof. The result is trivial for indecomposables of V_1 . If $\omega \in V_2$, then either $d\omega = 0$ or every summand of $d\omega$ belongs to $V_1 \wedge V_1 \wedge V_1$. As every element of V_1 is (by definition) overweight, the result holds practically trivially for $n = 2$ as well.

Let $n > 2$ and assume inductively the result is true for all elements of degree less than n . Let $\omega \in V_n$ be overweight, so that $\text{wt}(\omega) \geq 2n - 1$. In a similar approach to many of the previous propositions, the primary tool for analysis is the fact that $d^2\omega = 0$.

Suppose to the contrary that $d\omega$ contains a nonzero summand $\bar{\mathbf{x}} \wedge \bar{\mathbf{y}}$ for which both $\bar{\mathbf{x}}$ and $\bar{\mathbf{y}}$ are overweight. Among all quadratic terms with overweight factor $\bar{\mathbf{x}}$, take $\bar{\mathbf{x}}$ to be minimal degree. That is, if $\bar{\mathbf{x}}' \wedge \bar{\mathbf{y}}'$ is another summand with $\bar{\mathbf{x}}'$ overweight, then $|\bar{\mathbf{x}}'| \geq |\bar{\mathbf{x}}|$. Since $|\bar{\mathbf{y}}| + |\bar{\mathbf{x}}| = n + 1$, this implies that $|\bar{\mathbf{y}}'| \leq |\bar{\mathbf{y}}|$ as well, so that the degree of $\bar{\mathbf{y}}$ is maximal among such overweight factors. Moreover, we may assume without loss of generality there are no other summands $\bar{\mathbf{x}}' \wedge \bar{\mathbf{y}}$ of $d\omega$ for which $d\bar{\mathbf{x}}'$ contains ξ as a summand. Otherwise, replace $\bar{\mathbf{x}}$ with $\bar{\mathbf{x}} + \bar{\mathbf{x}}'$.

Since $\bar{\mathbf{x}}$ is overweight, by induction $d\bar{\mathbf{x}}$ contains an overweight summand of the form $\xi_2 = \bar{\mathbf{p}} \wedge \bar{\mathbf{q}}$ with $\bar{\mathbf{p}}$ overweight and $\bar{\mathbf{q}}$ not overweight, or of the form $\xi_3 = \bar{\mathbf{a}} \wedge \bar{\mathbf{b}} \wedge \bar{\mathbf{c}}$ where each factor is an overweight basic indecomposable. We assume that the coefficient of this summand is 1 by rescaling ω if necessary, and we will write $\xi \in \{\xi_2, \xi_3\}$ when dealing with both cases simultaneously.

We seek a contradiction to the fact that $d^2\omega = 0$. Consider the summand $\xi \wedge \bar{\mathbf{y}}$ of $(d\bar{\mathbf{x}}) \wedge \bar{\mathbf{y}}$. By [Lemma 5.21](#) either $\xi \wedge \bar{\mathbf{y}} = 0$ or $\xi \wedge \bar{\mathbf{y}}$ is canceled externally. If $\xi \wedge \bar{\mathbf{y}} = 0$ then $\bar{\mathbf{y}}$ must be a factor of ξ . But each factor of ξ has degree strictly less than $|\bar{\mathbf{x}}| \leq |\bar{\mathbf{y}}|$, a contradiction. Therefore, the only option is external cancellation.

To this end, suppose $\bar{\mathbf{x}}' \wedge \bar{\mathbf{y}}'$ is a quadratic summand of $d\omega$ for which $d(\bar{\mathbf{x}}' \wedge \bar{\mathbf{y}}')$ contains a cancelling summand of $\xi \wedge \bar{\mathbf{y}}$. Suppose without loss of generality that $|\bar{\mathbf{x}}'| \leq |\bar{\mathbf{y}}'|$. If $(d\bar{\mathbf{x}}') \wedge \bar{\mathbf{y}}'$ cancels with $\xi \wedge \bar{\mathbf{y}}$, then by consideration of degrees we must have $\bar{\mathbf{y}}' = \bar{\mathbf{y}}$. On the other hand, suppose $\bar{\mathbf{x}}' \wedge d\bar{\mathbf{y}}'$ cancels with $\xi \wedge \bar{\mathbf{y}}$. If $\bar{\mathbf{x}}'$ is overweight, then either $\bar{\mathbf{x}}'$ agrees with $\bar{\mathbf{y}}$ (in which case, we were actually considering internal cancellation), or else $\bar{\mathbf{x}}'$ is an overweight factor of ξ which contradicts the minimality assumptions on $|\bar{\mathbf{x}}|$. On the other hand, if $\bar{\mathbf{x}}'$ is not overweight, then $\xi = \xi_2$ must have $\bar{\mathbf{x}}'$ as a factor. That is,

$\xi_2 = \bar{\mathbf{p}} \wedge \bar{\mathbf{q}} = \bar{\mathbf{p}} \wedge \bar{\mathbf{x}}'$. But then $\langle \bar{\mathbf{x}}' \wedge d\bar{\mathbf{y}}', \xi_2 \wedge \bar{\mathbf{y}} \rangle = \langle \bar{\mathbf{x}}' \wedge d\bar{\mathbf{y}}', \bar{\mathbf{p}} \wedge \bar{\mathbf{x}}' \wedge \bar{\mathbf{y}} \rangle \neq 0$ implies that $\bar{\mathbf{p}} \wedge \bar{\mathbf{y}}$ is a nonzero summand of $d\bar{\mathbf{y}}'$. Since $\bar{\mathbf{y}}$ and $\bar{\mathbf{p}}$ are both overweight, this contradicts the inductive assumption.

Finally, suppose there is a cubic summand $\bar{\mathbf{x}}' \wedge \bar{\mathbf{y}}' \wedge \bar{\mathbf{z}}'$ in $d\omega$ whose differential contains (without loss of generality) a cancelling summand $\bar{\mathbf{x}}' \wedge \bar{\mathbf{y}}' \wedge \bar{\mathbf{p}}' \wedge \bar{\mathbf{q}}'$ which was produced by $\bar{\mathbf{x}}' \wedge \bar{\mathbf{y}}' \wedge d\bar{\mathbf{z}}'$. Inductively, since this term came from a quadratic differential on $\bar{\mathbf{z}}'$, we may assume that $\bar{\mathbf{q}}'$ is not overweight. If $\xi = \xi_2$, then $\langle \bar{\mathbf{x}}' \wedge \bar{\mathbf{y}}' \wedge \bar{\mathbf{p}}' \wedge \bar{\mathbf{q}}', \xi_2 \wedge \bar{\mathbf{y}} \rangle = 0$ by consideration of word lengths, and so it is enough to consider cancellation when $\xi = \xi_3$. But in this case, note that each factor of $\bar{\mathbf{a}} \wedge \bar{\mathbf{b}} \wedge \bar{\mathbf{c}} \wedge \bar{\mathbf{y}}$ is overweight by assumption. This implies that each factor of $\bar{\mathbf{x}}' \wedge \bar{\mathbf{y}}' \wedge \bar{\mathbf{p}}' \wedge \bar{\mathbf{q}}'$ is overweight as well, contradicting the weight of $\bar{\mathbf{q}}'$.

Therefore, if the weight of ω is maximized by a summand of the form $\bar{\mathbf{x}} \wedge \bar{\mathbf{y}}$, at most one factor is overweight, and so without loss of generality $\text{wt}(\bar{\mathbf{x}}) \leq 2|\bar{\mathbf{x}}| - 2$ and $\text{wt}(\bar{\mathbf{y}}) \leq 2|\bar{\mathbf{y}}| - 1$. In particular,

$$\text{wt}(\omega) = \text{wt}(\bar{\mathbf{x}}) + \text{wt}(\bar{\mathbf{y}}) \leq 2(|\bar{\mathbf{x}}| + |\bar{\mathbf{y}}|) - 3 = 2(|\omega| + 1) - 3 = 2|\omega| - 1.$$

On the other hand, if the weight of ω is maximized by a summand of the form $\bar{\mathbf{a}} \wedge \bar{\mathbf{b}} \wedge \bar{\mathbf{c}}$, each factor may be overweight, which inductively implies that

$$\text{wt}(\omega) \leq (2|\bar{\mathbf{a}}| - 1) + (2|\bar{\mathbf{b}}| - 1) + (2|\bar{\mathbf{c}}| - 1) = 2(|\omega| + 1) - 3 = 2|\omega| - 1,$$

and ω is overweight when $\text{wt}(\omega)$ is maximized in one of these ways. □

We record our final improvement as a separate result:

Corollary 5.34. *If $\mathcal{M}_Y^* = \wedge V$ is simple, then $\text{wt}(\omega) \leq 2n - 1$ for all $\omega \in V_n$.*

In particular the bounds in [Proposition 5.5](#) hold, and thus so does [Corollary 5.6](#).

5.4 Bounds on Coformal Spaces

We continue using our prior [notation](#). Finally, we prove some stronger bounds for quadratic differentials. Compare the proof structure (and indeed, the claim itself) with [Lemma 5.15](#) and [Conjecture 5.29](#).

Lemma 5.35. *If d is quadratic and $\omega \in M_n^J$, then*

$$d\omega \in \bigoplus_{\substack{n_1 \leq n_2 \\ n_1 + n_2 = n+1}} \left(\bigoplus_{i_1 + i_2 \leq J+c} M_{n_1}^{i_1} \wedge M_{n_2}^{i_2} \right).$$

Proof. Note that *a priori*, we know only that $i_1, i_2 \leq J$ in the above expression. That is, since d is quadratic, we can write

$$d\omega = \sum_{\substack{n_1 \leq n_2 \\ n_1 + n_2 = n+1}} \left(\sum_{i_1, i_2 \leq c} \Omega_{(i_1, i_2, n_1, n_2)} \right) \in \bigoplus_{\substack{n_1 \leq n_2 \\ n_1 + n_2 = n+1}} \left(\bigoplus_{i_1, i_2 \leq c} M_{n_1}^{i_1} \wedge M_{n_2}^{i_2} \right).$$

The result is obvious for $n \leq 2$. Inductively assume the result is true for all $k < n$ for some $n > 2$. By [Lemma 5.15](#), if $\omega \in M_n^J$, then $d_{nil}\omega \in \bigoplus_{i+j \leq J} M_1^i \wedge M_n^j$ so that it is enough to consider terms of $d_{sim}\omega$. To this end, let (N_1, N_2) be such that $\Omega_{(i_1, i_2, N_1, N_2)} \neq 0$ for some (i_1, i_2) . In particular, assume $1 < N_1 \leq N_2 < n$. Moreover, define

$$\ell := \max \{i_1 + i_2 \mid \Omega_{(i_1, i_2, N_1, N_2)} \neq 0\}$$

and fix

$$(I_1, I_2) := \max_{\leq 2} \{(i_1, i_2) \mid i_1 + i_2 = \ell, \Omega_{(i_1, i_2, N_1, N_2)} \neq 0\}.$$

Let $\Omega = \bar{\mathbf{x}} \wedge \bar{\mathbf{y}} \in M_{N_1}^{I_1} \wedge M_{N_2}^{I_2}$ be a nonzero basic summand of $d\omega$, and define $\xi := (\delta \bar{\mathbf{x}}) \wedge \bar{\mathbf{y}} \in M_1^1 \wedge M_{N_1}^{I_1-1} \wedge M_{N_2}^{I_2}$.

By [Lemma 5.22](#), either $\xi = 0$ or ξ is cancelled externally. However, injectivity of δ

and the fact that $(I_1 - 1, N_1) <_2 (I_2, N_2)$ implies nearly immediately that $\xi \neq 0$. Hence, there is some external summand $\Omega' = \bar{\mathbf{x}}' \wedge \bar{\mathbf{y}}' \in M_{N'_1}^{I'_1} \wedge M_{N'_2}^{I'_2}$ of $d\omega$ for which $d\Omega'$ has a summand ξ' that cancels with ξ . Here as before we order degrees so that $N'_1 \leq N'_2$.

Since ξ' is a summand of $d\Omega'$ and since ξ' cancels with ξ we must have

$$\xi' \in \left[\left(\text{im} \left(d|_{M_{N'_1}^{I'_1}} \right) \wedge M_{N'_2}^{I'_2} \right) + \left(M_{N'_1}^{I'_1} \wedge \text{im} \left(d|_{M_{N'_2}^{I'_2}} \right) \right) \right] \cap [M_1^1 \wedge M_{N_1}^{I_1-1} \wedge M_{N_2}^{I_2}].$$

In particular there are a_1, b_1, a_2, b_2 for which

$$\xi' \in \left(M_{n_{1,1}}^{a_1} \wedge M_{n_{1,2}}^{b_1} \wedge M_{N'_2}^{I'_2} \right) + \left(M_{N'_1}^{I'_1} \wedge M_{n_{2,1}}^{a_2} \wedge M_{n_{2,2}}^{b_2} \right)$$

where $n_{1,1} + n_{1,2} = N'_1 + 1$ and $n_{2,1} + n_{2,2} = N'_2 + 1$. Suppose first that

$$\xi \in \left(M_{n_{1,1}}^{a_1} \wedge M_{n_{1,2}}^{b_1} \wedge M_{N'_2}^{I'_2} \right) \cap \left(M_1^1 \wedge M_{N_1}^{I_1-1} \wedge M_{N_2}^{I_2} \right). \quad (\text{A1})$$

Claim 5.35.1

Assuming (A1), then $\Omega' \in (I_1, I_2, N_1, N_2)$.

Proof of Claim: By considering degrees, and the assumption that $N'_1 \leq N'_2$, we see that we must have $N'_2 = N_2$. Moreover, without loss of generality we have $n_{1,1} = 1$ and $N'_1 = n_{1,2} = N_1$, since $1 < N_1$. In particular, this implies that $a_1 = 1$. Furthermore, since

$$\xi' \in \text{im} \left(d|_{M_{N'_1}^{I'_1}} \right) \wedge M_{N'_2}^{I'_2},$$

this implies that in fact ξ' came from the nilpotent component of the differential. By definition we must then have $b_1 \leq I'_1 - 1$.

Suppose first that $N_1 < N_2$. Then $I'_2 = I_2$ and $b_1 = I_1 - 1$. But then $I_1 = 1 + b_1 \leq I'_1$. Thus since $(N'_1, N'_2) = (N_1, N_2)$, the maximality of (I_1, I_2) implies that in fact

$$(I'_1, I'_2) = (I_1, I_2).$$

If $N_1 = N_2$, there are two possibilities. If $b_1 = I_1 - 1$, then the above proof shows $(I'_1, I'_2, N'_1, N'_2) = (I_1, I_2, N_1, N_2)$. On the other hand, if $b_1 = I_2$ then $I'_1 \geq b_1 + 1 > I_2$, which contradicts the maximality of I_2 . $\blacksquare$

By injectivity of δ this implies that $\Omega' \in \text{span}\{\Omega\}$, which means that cancellation was internal, a contradiction.

Suppose next that

$$\xi' \in \left(M_{N'_1}^{I'_1} \wedge M_{n_{2,1}}^{a_2} \wedge M_{n_{2,2}}^{b_2} \right) \cap \left(M_1^1 \wedge M_{N_1}^{I_1-1} \wedge M_{N_2}^{I_2} \right). \quad (\text{A2})$$

Note that by induction we have $a_2 + b_2 \leq I'_2 + c$. In the previous case, we could essentially ignore the situation $N'_1 = 1$. Now however we have either $N'_1 = 1$ or (without loss of generality) $n_{2,1} = 1$. We consider these in the reverse order:

Claim 5.35.1

Assuming (A2), if $n_{2,1} = 1$ then $(I'_2, I'_1, N'_1, N'_2) = (I_1, I_2, N_1, N_2)$.

Proof of Claim: In this situation, we see by the same arguments as in the previous claim, that $N'_1 = N_1$ and $N'_2 = n_{2,2} = N_2$, and that furthermore $a_2 = 1$ so that $b_2 \leq I'_2 - 1$.

In the event that $N_1 < N_2$ we have therefore that $b_2 = I_2$ which means $I'_2 \geq 1 + b_2 > I_2$, contradicting the maximality of I_2 .

If $N_1 = N_2$ there are two possibilities. If $b_2 = I_2$ then as above we contradict the maximality of I_2 . On the other hand, if $b_2 = I_1 - 1$ then $I'_2 \geq b_2 + 1 = I_1$. Hence (noting the switch of position of I'_2 and I'_1 in the lefthand term) we must have $(I'_2, I'_1, N'_1, N'_2) = (I_1, I_2, N_1, N_2)$. $\blacksquare$

As before, in the above setting injectivity of δ implies then that $\Omega' \in \text{span}\{\Omega\}$.

Claim 5.35.2

Assuming (A2), if $N'_1 = 1$, then $I_1 + I_2 \leq J + c$.

Proof of Claim: In this case, because $1 < N_1$, we know that $N'_1 \neq N_1$. Therefore, without loss of generality, we have $n_{2,1} = N_1$ and $n_{2,2} = N_2$, which implies that $a_2 = I_1 - 1$ and $b_2 = I_2$. Moreover we must have $I'_1 = 1$, and thus Ω' is a term in $d_{\text{nil}}\omega$. In particular, $I'_2 = J - 1$. By induction, therefore, we see that $(I_1 - 1) + I_2 = a_2 + b_2 \leq (J - 1) + c$. ■

Hence $\ell = I_1 + I_2 \leq J + c$ as required. □

The above result indicates that we should be able to refine the bounds on the weights of indecomposables in the event that d is quadratic. Indeed, we can.

Lemma 5.36. *Let $\omega \in M_n^J$ for some c -step nilpotent DGA $\wedge V$ with quadratic differential. Then for $n \geq 2$ we have*

$$\text{wt}(\omega) \leq (c + 1)(n - 1) + J - c.$$

In particular any $\omega \in V_n$ has $\text{wt}(\omega) \leq (c + 1)(n - 1)$.

Proof. For $n = 2$ and $J = 1$ the result is trivial because $d|_{M_2^1} = 0$. Note that for $\bar{\mathbf{t}} \in M_1^J$ we have $\text{wt}(\bar{\mathbf{t}}) = J$. Since d is quadratic we have $d|_{M_2^J} = d_{\text{nil}}|_{M_2^J}$, so that Lemma 5.15 implies inductively that if $\omega \in M_2^J$ then $\text{wt}(\omega) \leq J + 1$ which is the desired form when $n = 2$. In particular, $\text{wt}(\omega) \leq c + 1$ for all $\omega \in V_2$.

Inductively assume the result holds for all $(j, k) <_2 (J, n)$ for some (J, n) . Suppose that $\bar{\mathbf{x}} \wedge \bar{\mathbf{y}} \in M_{n_1}^{i_1} \wedge M_{n_2}^{i_2}$ is a summand of $d\omega$. By Lemma 5.35 we have $i_1 + i_2 \leq J + c$.

By induction, we have

$$\begin{aligned}
\text{wt}(\bar{\mathbf{x}} \wedge \bar{\mathbf{y}}) &\leq (c+1)(n_1-1) + i_1 - c \\
&\quad + (c+1)(n_2-1) + i_2 - c \\
&= (c+1)(n_1+n_2-2) + (i_1+i_2) - 2c \\
&\leq (c+1)(n-1) + (J+c) - 2c \\
&= (c+1)(n-1) + J - c.
\end{aligned}$$

Since $\bar{\mathbf{x}} \wedge \bar{\mathbf{y}}$ was an arbitrary summand of $d\omega$ we have $\text{wt}(\omega) \leq (c+1)(n-1) + J - c$ as desired. $\square$

Corollary 5.37. *If Y is coformal then $\text{wt}(\omega) \leq (c+1)(n-1)$ for all $\omega \in V_n$.*

In particular the bounds in [Proposition 5.7](#) hold, and thus so does [Corollary 5.8](#).

Bibliography

- [1] M. Gromov, *Homotopical effects of dilatation*, *Journal of Differential Geometry* **13** (1978), no. 3 303–310.
- [2] M. Gromov, *Quantitative homotopy theory*, *Prospects in Mathematics: Invited Talks on the Occasion of the 250th Anniversary of Princeton University (H. Rossi, ed.)* (1999) 45–49.
- [3] A. Nabutovsky and S. Weinberger, *Variational problems for riemannian functionals and arithmetic groups*, *Publications Mathématiques de l’IHÉS* **92** (2000) 5–62.
- [4] A. Nabutovsky and R. Rotman, *The area of a minimal embedded 2-sphere in a manifold diffeomorphic to S^3* , *International Mathematics Research Notices* **2003** (2003), no. 39 2121–2129.
- [5] F. Manin, *Asymptotic invariants of homotopy groups*. The University of Chicago, 2015.
- [6] F. Manin, *Volume distortion in homotopy groups*, *Geometric and Functional Analysis* **26** (2016) 607–679.
- [7] L. Guth, *Volumes of balls in riemannian manifolds and uryson width*, *Journal of Topology and Analysis* **9** (2017), no. 02 195–219.
- [8] G. Chambers, D. Dotterer, F. Manin, and S. Weinberger, *Quantitative null-cobordism*, *Journal of the American Mathematical Society* **31** (2018), no. 4 1165–1203.
- [9] G. R. Chambers, F. Manin, and S. Weinberger, *Quantitative nullhomotopy and rational homotopy type*, *Geometric and Functional Analysis* **28** (Apr., 2018) 563–588.
- [10] F. Manin and S. Weinberger, *Integral and rational mapping classes*, *Duke Mathematical Journal* **169** (2020), no. 10 1943 – 1969.
- [11] V. Kapovitch, A. Petrunin, and W. Tuschmann, *Nilpotency, almost nonnegative curvature, and the gradient flow on alexandrov spaces*, *Annals of Mathematics* (2010) 343–373.

- [12] R. Young, *Homological and homotopical higher-order filling functions*, *Groups, Geometry, and Dynamics* **5** (2011), no. 3 683–690.
- [13] R. Young, *Filling inequalities for nilpotent groups through approximations*, *Groups, Geometry, and Dynamics* **7** (2013), no. 4 977–1011.
- [14] N. Brady, M. R. Bridson, M. Forester, and K. Shankar, *Snowflake groups, perron–frobenius eigenvalues and isoperimetric spectra*, *Geometry & Topology* **13** (2008), no. 1 141–187.
- [15] M. Gromov, *Filling riemannian manifolds*, *Journal of Differential Geometry* **18** (1983), no. 1 1–147.
- [16] B. White, *Mappings that minimize area in their homotopy classes*, *Journal of Differential Geometry* **20** (1984), no. 2 433–446.
- [17] M. Gromov, *Geometric Group Theory: Asymptotic invariants of infinite groups*, vol. 2. Cambridge University Press, 1993.
- [18] C. Pittet, *Isoperimetric inequalities in nilpotent groups*, *Journal of the London Mathematical Society* **55** (1997), no. 3 588–600.
- [19] S. Wenger, *Nilpotent groups without exactly polynomial dehn function*, *Journal of Topology* **4** (2011), no. 1 141–160.
- [20] P. Pansu, *An isoperimetric inequality on the heisenberg group*, *Rend. Sem. Mat. Univ. Politec. Torino* (1983), no. Special issue 159–174.
- [21] N. T. Varopoulos, *Analysis and geometry on groups*, (No Title) (1993).
- [22] R. Young, *High-dimensional fillings in heisenberg groups*, *The Journal of Geometric Analysis* **26** (2016), no. 2 1596–1616.
- [23] F. Manin, *Plato’s cave and differential forms*, *Geometry & Topology* **23** (Dec., 2019) 3141–3202.
- [24] A. Berdnikov, *Lipschitz homotopies of mappings from 3-sphere to 2-sphere*, *Journal of Topology* **15** (2022), no. 2 815–843.
- [25] A. Berdnikov and F. Manin, *Scalable spaces*, *Inventiones mathematicae* **229** (2022), no. 3 1055–1100.
- [26] M. Gromov, M. Katz, P. Pansu, and S. Semmes, *Metric structures for Riemannian and non-Riemannian spaces*, vol. 152. Springer, 1999.
- [27] A. Hatcher, *Algebraic Topology*. Algebraic Topology. Cambridge University Press, 2002.

- [28] Y. Félix, S. Halperin, and J.-C. Thomas, *Rational homotopy theory*, vol. 205. Springer Science & Business Media, 2012.
- [29] Y. Félix, S. Halperin, and J.-C. Thomas, *Rational homotopy theory II*. World Scientific, 2015.
- [30] P. Hilton, G. Mislin, and J. Roitberg, *Localization of nilpotent groups and spaces*. Elsevier, 2016.
- [31] P. G. Goerss and J. F. Jardine, *Simplicial homotopy theory*. Springer Science & Business Media, 2009.
- [32] P. Griffiths and J. Morgan, *Rational Homotopy Theory and Differential Forms*, vol. 16. Springer Science & Business Media, 2013.
- [33] M. Schlessinger and J. Stasheff, *Deformation theory and rational homotopy type*, *arXiv preprint arXiv:1211.1647* (2012).
- [34] Y. Félix, J. Oprea, and D. Tanré, *Algebraic models in geometry*. OUP Oxford, 2008.
- [35] P. J. Hilton, *On the homotopy groups of the union of spheres*, *J. London Math. Soc* **30** (1955), no. 154-172 71.
- [36] D. Sullivan, *Differential forms and the topology of manifolds*, *Manifolds Tokyo* (1973) 37–49.
- [37] J. Neisendorfer and T. Miller, *Formal and coformal spaces*, *Illinois Journal of Mathematics* **22** (1978), no. 4 565–580.
- [38] P. Andrews and M. Arkowitz, *Sullivan’s minimal models and higher order whitehead products*, *Canadian Journal of Mathematics* **30** (1978), no. 5 961–982.
- [39] M. Golasinski and T. de Melo, *On the higher whitehead product*, *Journal of Homotopy and Related Structures* **11** (2016), no. 4 825–845.
- [40] G. J. Porter, *Higher order whitehead products*, *Topology* **3** (1965), no. 2 123–135.
- [41] S. Ferry and S. Weinberger, *Quantitative algebraic topology and lipschitz homotopy*, *Proceedings of the National Academy of Sciences* **110** (2013), no. 48 19246–19250.
- [42] H. Edelsbrunner and D. R. Grayson, *Edgewise subdivision of a simplex*, in *Proceedings of the fifteenth annual symposium on Computational geometry*, pp. 24–30, 1999.
- [43] R. Bott and L. W. Tu, *Differential forms in algebraic topology*, vol. 82. Springer Science & Business Media, 2013.

- [44] A. Berdnikov, L. Guth, and F. Manin, *Degrees of maps and multiscale geometry*, *Forum of Mathematics, Pi* **12** (2024) e2.
- [45] D. Sullivan, *Infinitesimal computations in topology*, *Publications Mathématiques de l’IHÉS* **47** (1977) 269–331.
- [46] M. Gruber, *Filling invariants of stratified nilpotent lie groups*, *Mathematische Zeitschrift* **293** (2019) 39–79.
- [47] S. M. Gersten, D. F. Holt, and T. R. Riley, *Isoperimetric inequalities for nilpotent groups*, *Geometric & Functional Analysis GAFA* **13** (2003) 795–814.
- [48] C. Llosa Isenrich, G. Pallier, and R. Tessera, *Cone-equivalent nilpotent groups with different dehn functions*, *Proceedings of the London Mathematical Society* **126** (2023), no. 2 704–789.
- [49] B. H. Bowditch, *Bilipschitz triangulations of riemannian manifolds*, *Warwick* (2020).
- [50] L. Guth, *Recent progress in quantitative topology*, *Surveys in Differential Geometry* **22** (2017), no. 1 191–216.
- [51] M. Čadek, M. Krčál, J. Matoušek, L. Vokřínek, and U. Wagner, *Extendability of continuous maps is undecidable*, *Discrete & Computational Geometry* **51** (2014), no. 1 24–66.
- [52] J. Alonso, X. Wang, and S. Pride, *Higher-dimensional isoperimetric (or dehn) functions of groups*, *J. Group Theory* **2** (1999) 81–112.
- [53] H. Edelsbrunner and D. Grayson, *Edgewise subdivision of a simplex*, *Discrete & Computational Geometry* **4** (2000), no. 24 707–719.
- [54] S. Rigot and S. Wenger, *Lipschitz non-extension theorems into jet space carnot groups*, *International Mathematics Research Notices* **2010** (2010), no. 18 3633–3648.
- [55] S. Wenger and R. Young, *Lipschitz extensions into jet space carnot groups*, *arXiv preprint arXiv:0911.0919* (2009).